Phase 1: Planning & Design

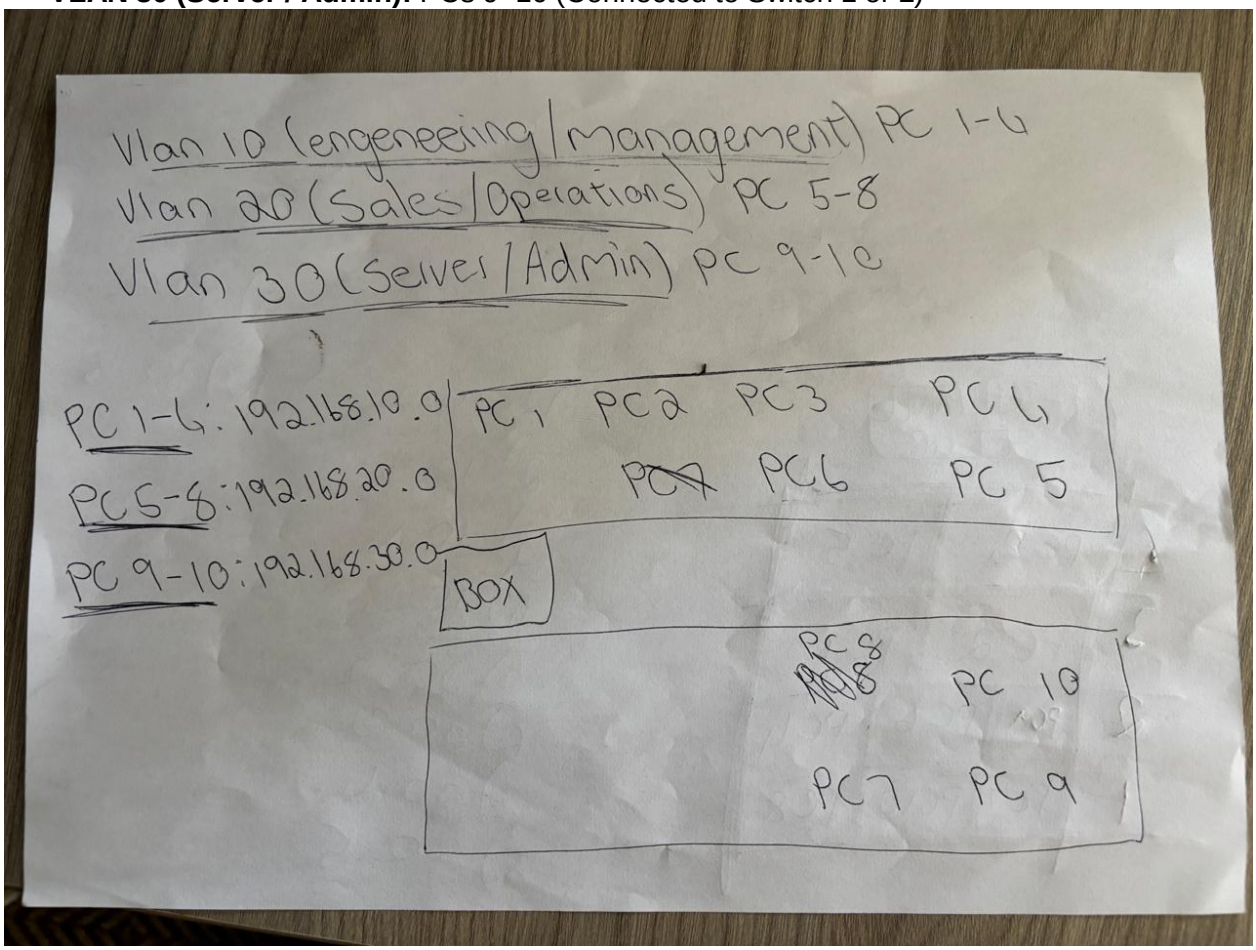
1. Physical Topology

- **The Single Switch Cabinet (Rack):**
  - **U1–U4 (Top):** 4 × D-Link Patch Panels (Grouped together at the top to aggregate incoming PC lines safely).
  - **U5–U6 (Middle):** 2 × Cisco Catalyst 3560 Switches (Placed directly under the patch panels for short, clean patch cord routing).
  - **U7–U8 (Bottom):** 2 × Cisco ISR 4331 Routers (Placed at the bottom due to heavier weight and separate power requirements).
- **The Device Connections:** 10 PCs sit externally. Their solid-core cables run directly into the back of the 4 patch panels inside the cabinet.

2. Logical Topology & VLAN Infrastructure

To accommodate 10 PCs across 2 switches and 2 routers, the network should be segmented logically into departmental VLANs to demonstrate routing capability.

- **VLAN 10 (Engineering / Management):** PCs 1–4 (Connected to Switch 1)
- **VLAN 20 (Sales / Operations):** PCs 5–8 (Connected to Switch 2)
- **VLAN 30 (Server / Admin):** PCs 9–10 (Connected to Switch 1 or 2)

3. 

- **Native/Management VLAN 99:** For Switches and Routers IP addresses.

4. IP Address Planning

Using a private Class C subnet space (**192.168.0.0/24**), divided using VLSM:

- **VLAN 10 Subnet:** 192.168.10.0/26 (Gateway: .1)
- **VLAN 20 Subnet:** 192.168.20.0/26 (Gateway: .1)
- **VLAN 30 Subnet:** 192.168.30.0/26 (Gateway: .1)
- **Management VLAN 99:** 192.168.99.0/24
- **Router-to-Router Link:** 10.0.0.0/30 (R1: .1, R2: .2)

Phase 2: Hardware Installation & Cabling

1. **Install Switch Cabinet:** Securely mount the rack enclosure. Ensure grounding cables are bonded to the cabinet chassis.
2. **Install Patch Panels:** Screw all 4 D-Link patch panels into the top rack units (U1–U4).
3. **Terminate Internal Cable Runs:**
    - Route the raw CAT6 cables coming from the 10 PC stations into the cabinet.
    - Pull them to the *back* side of the patch panels.
    - Use the **insertion (punch-down) tool** to map the wires directly to the **T568B** color-coded blocks.
    - *Note: Because outlets/faceplates were removed, the PC side of the cables must be*

punched down into loose CAT6 keystone jacks sitting at each desk.

4. **Test Cabling:** Use a hardware cable continuity tester on the punched-down lines to ensure no split pairs or shorts exist before powering up electronics.

Phase 3: Device Initialization & Layer 2 Config

1. **Physical Console Access:** Connect a PC running **PuTTY** to the Cisco devices using the provided **Console cables** (RJ45-to-DB9 or USB-to-RJ45) plugged into the device's Console port.
2. **Basic Configuration (Switches & Routers):**
   ```
   Router(config)# hostname R1
   R1(config)# enable secret YourStrongPassword
   R1(config)# line vty 0 4
   R1(config-line)# password YourVtyPassword
   R1(config-line)# login
   ```

3. **Switch Management Interfaces:**
   ```
   SW1(config)# vlan 99
   SW1(config-vlan)# name Management
   SW1(config)# interface vlan 99
   SW1(config-if)# ip address 192.168.99.10 255.255.255.0
   SW1(config-if)# no shutdown
   ```

4. **Configure VLANs & Access Ports:**
   ```
   SW1(config)# vlan 10
   SW1(config-vlan)# name Engineering
   SW1(config)# interface range fa0/1 - 4
   SW1(config-if-range)# switchport mode access
   SW1(config-if-range)# switchport access vlan 10
   ```

5. **Trunking between Switches:** Configure Inter-Switch Links (ISL/802.1Q) to allow VLAN traffic to traverse both Catalyst 3560s.

Phase 4: Layer 3 Configuration & Routing

1. **Configure Router Interfaces:** Assign IPs to the physical interfaces connecting the two Cisco ISR4331 routers together.
2. **Inter-VLAN Routing (Router-on-a-Stick):** Connect a trunk port from Switch 1 to Router 1's GigabitEthernet interface. Create sub-interfaces on the router to act as defaults gateways for the PCs:
   ```
   R1(config)# interface gi0/0/0.10
   R1(config-subif)# encapsulation dot1Q 10
   R1(config-subif)# ip address 192.168.10.1 255.255.255.192
   ```

3. **Configure Routing Protocols:** Set up static routing or OSPF between Router 1 and Router 2 so networks on Switch 1 can communicate with networks on Switch 2.
4. **Verify Routing Tables:** Run show ip route via PuTTY to ensure all subnets are visible.

Phase 5: Network Printers (Skipped / Simulated)

- **Status: Omitted per instructions.**
- *Alternative Action:* If the instructor requires verification of this phase despite missing hardware, assign an unused IP address (e.g., 192.168.30.50) to a loopback interface on a switch or router to act as a "Virtual Printer," then verify that all VLANs can successfully ping it.

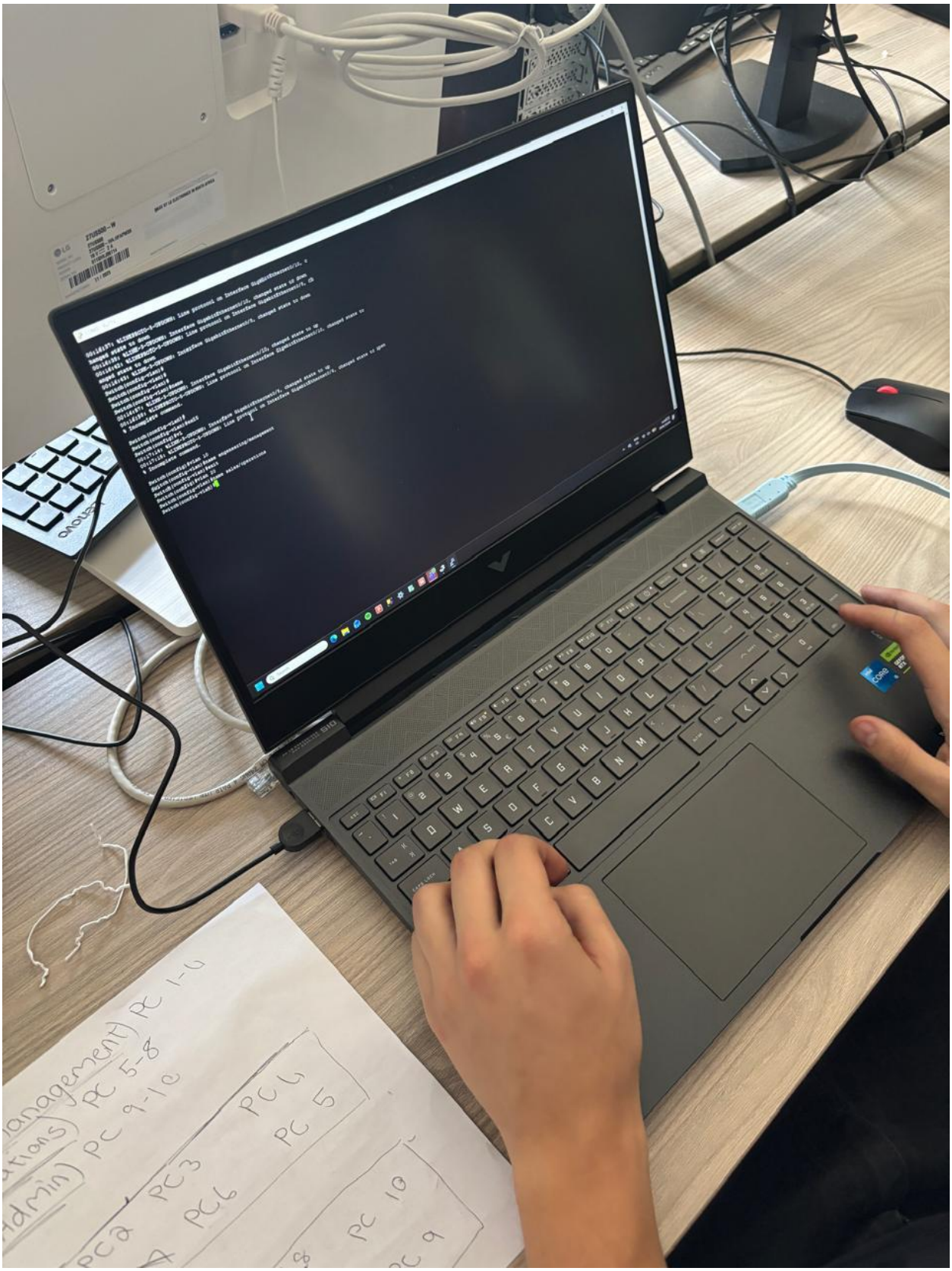
Phase 6: Testing & Troubleshooting

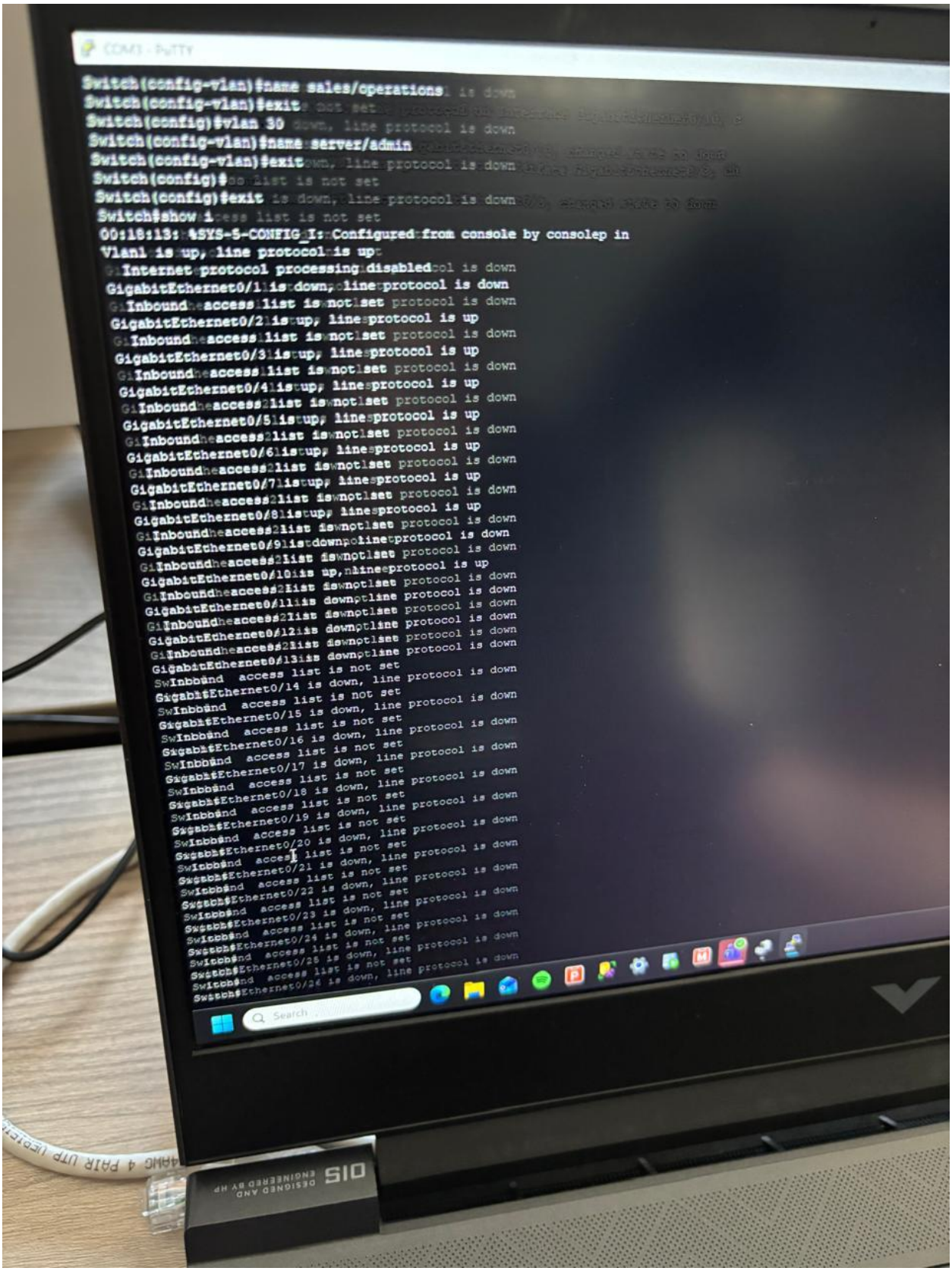
1. **Verify Connectivity:** Ping from a PC in VLAN 10 to a PC in VLAN 20.
2. **Verify VLAN Functionality:** Use show vlan brief on the switches to confirm ports are grouped correctly.
3. **Resolve Instructor Faults:** Common simulated faults to look out for via PuTTY command line:
    - **Wrong VLAN Assignment:** Fixed using switchport access vlan X.
    - **Native VLAN Mismatch:** Fixed by matching native IDs on both ends of a trunk link.
    - **Subnet Mask Mismatch:** Check PC IP configurations vs the Router sub-interface gateway.

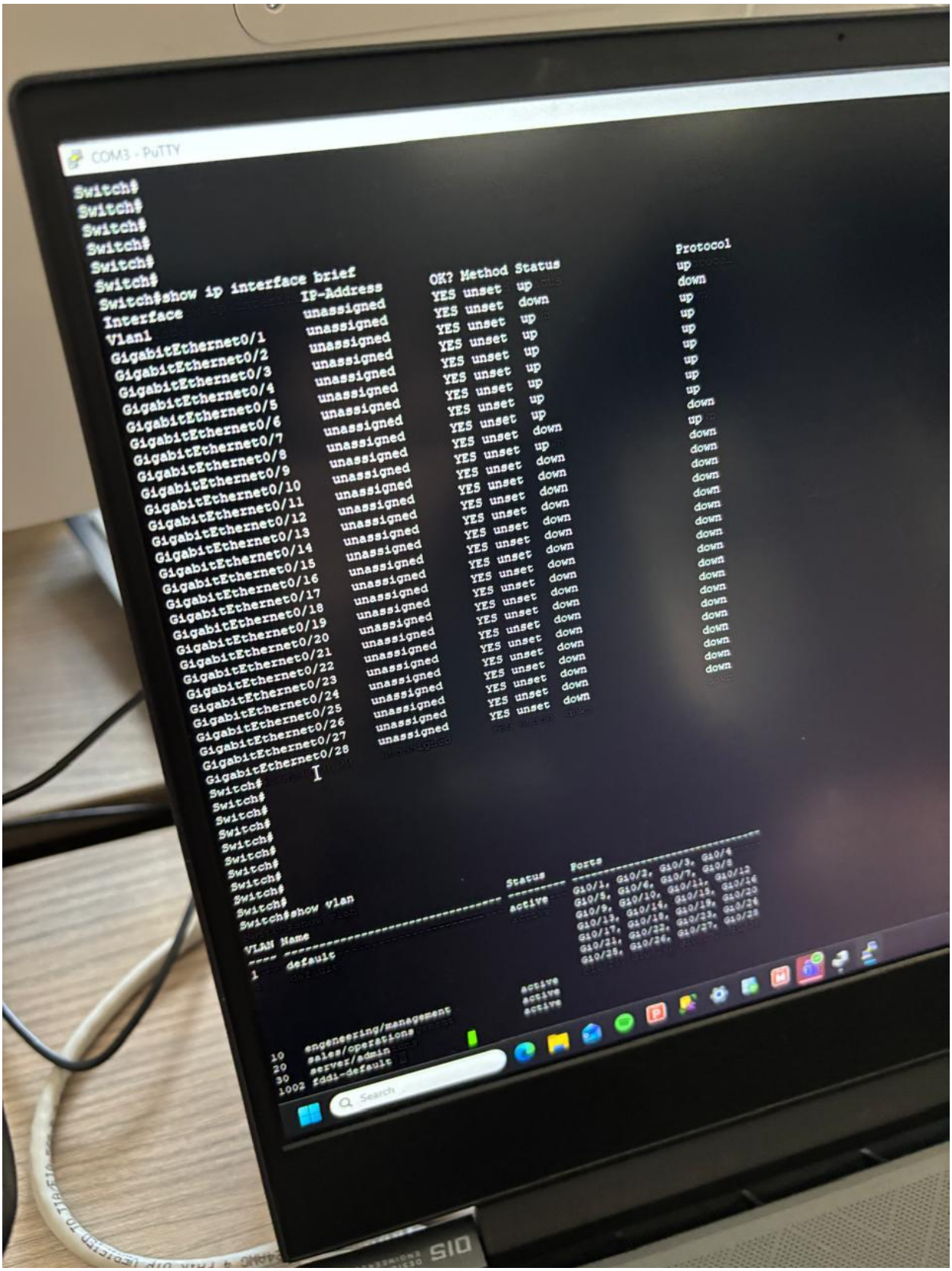
Phase 7: Documentation

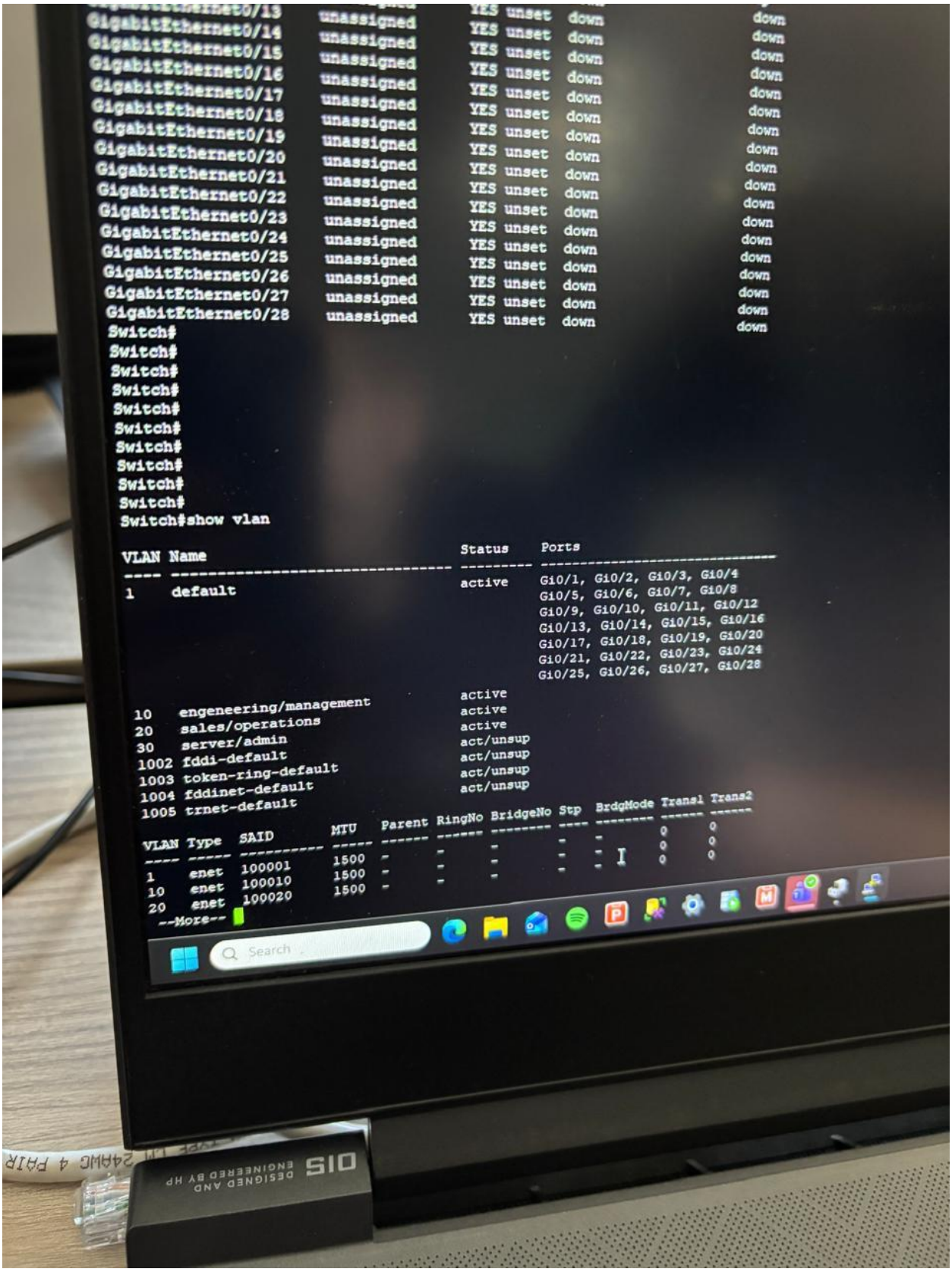
Produce a final network book containing:

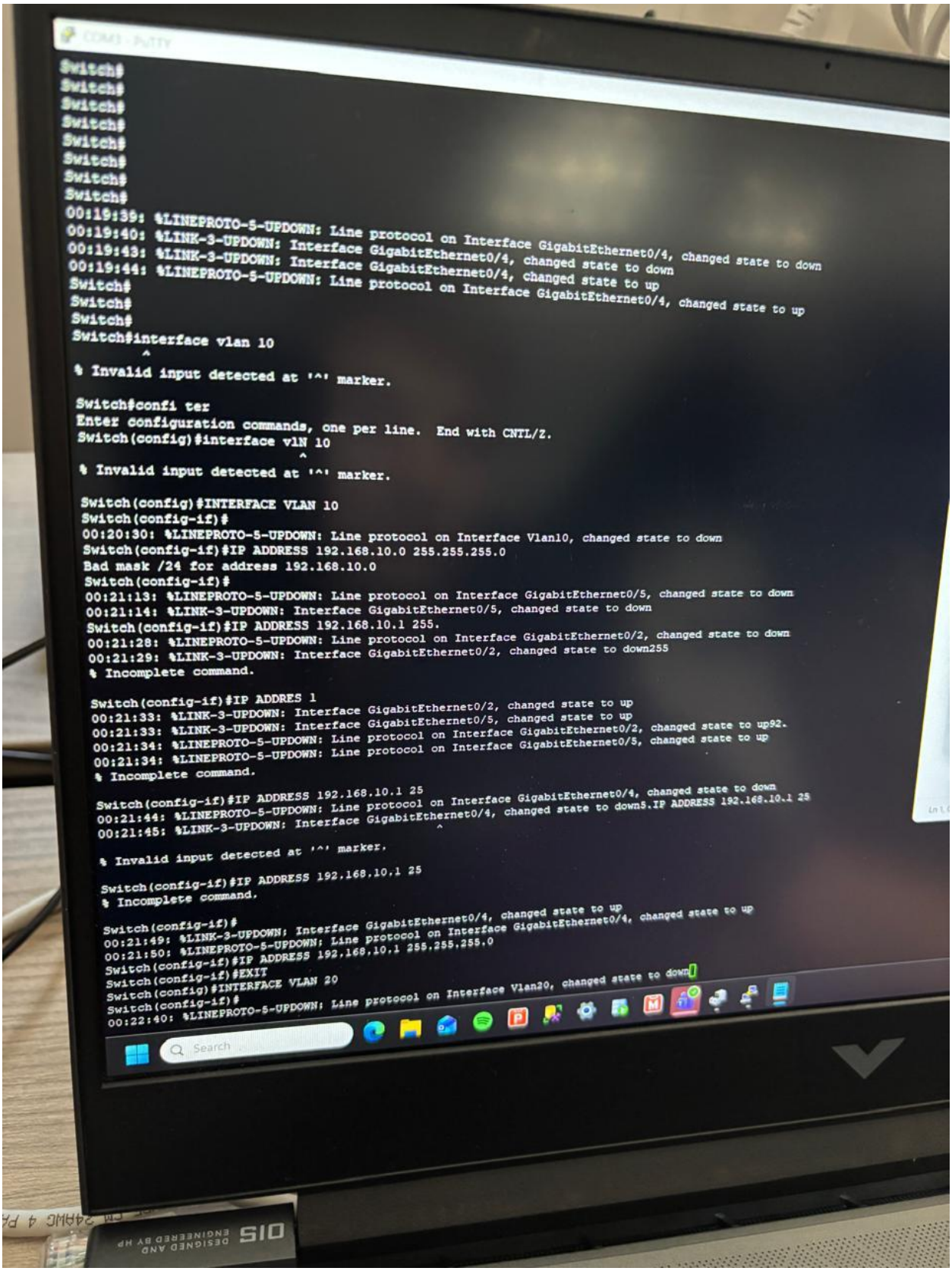
- Completed Visio/Packet Tracer **Logical Diagram**.
- High-level **Cabinet Rack Layout map** (Physical topology).
- Master **IP Addressing Matrix table** (listing every PC, Switch, Router interface, and functional gateway).
- Text file backups of the running configurations (show running-config) saved directly from the PuTTY session logs.

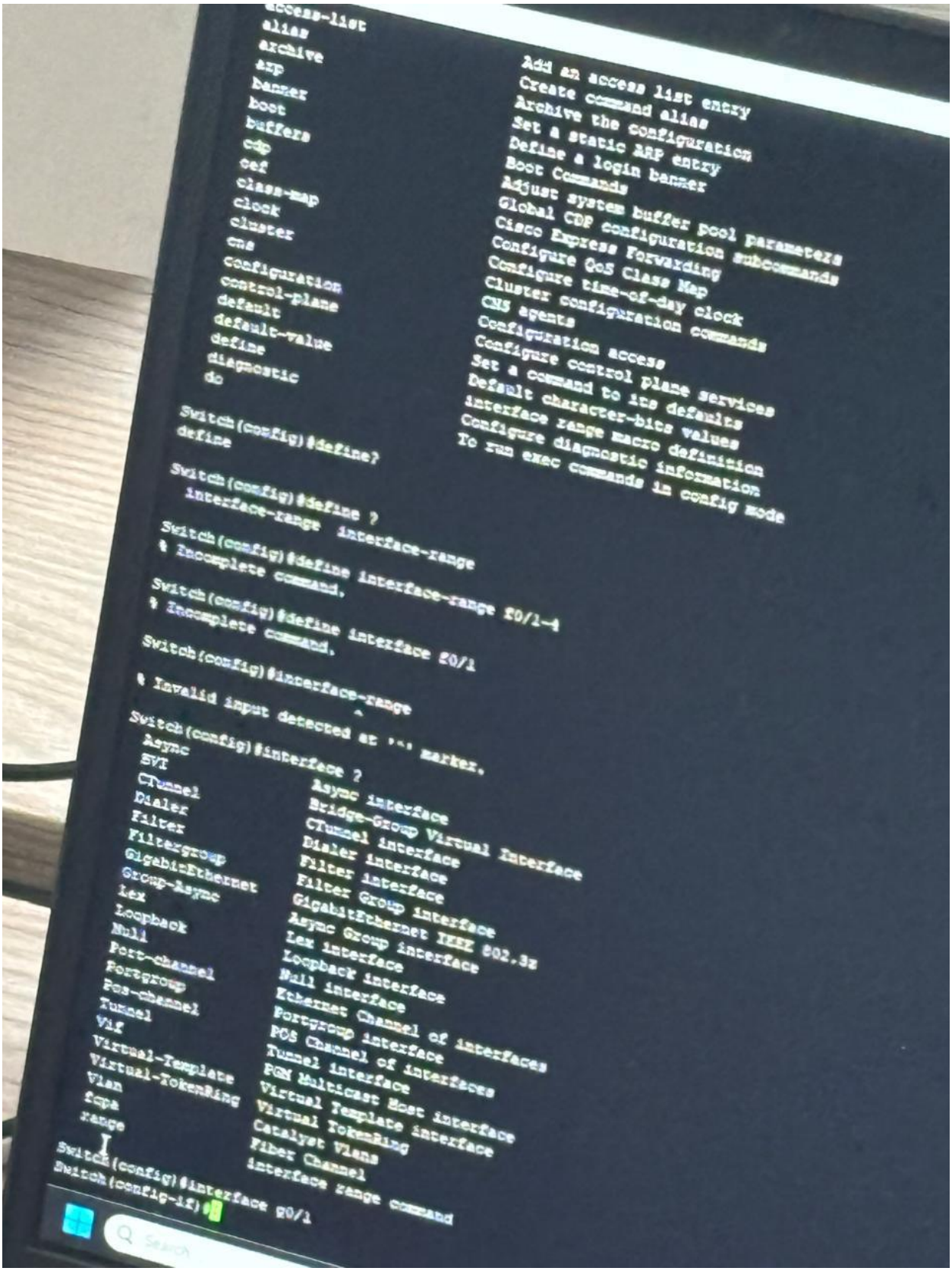

PC 1-4
PC 5-8
PC 9-10
PC3
PC4
PC5
PC6
PC 10
PC 9

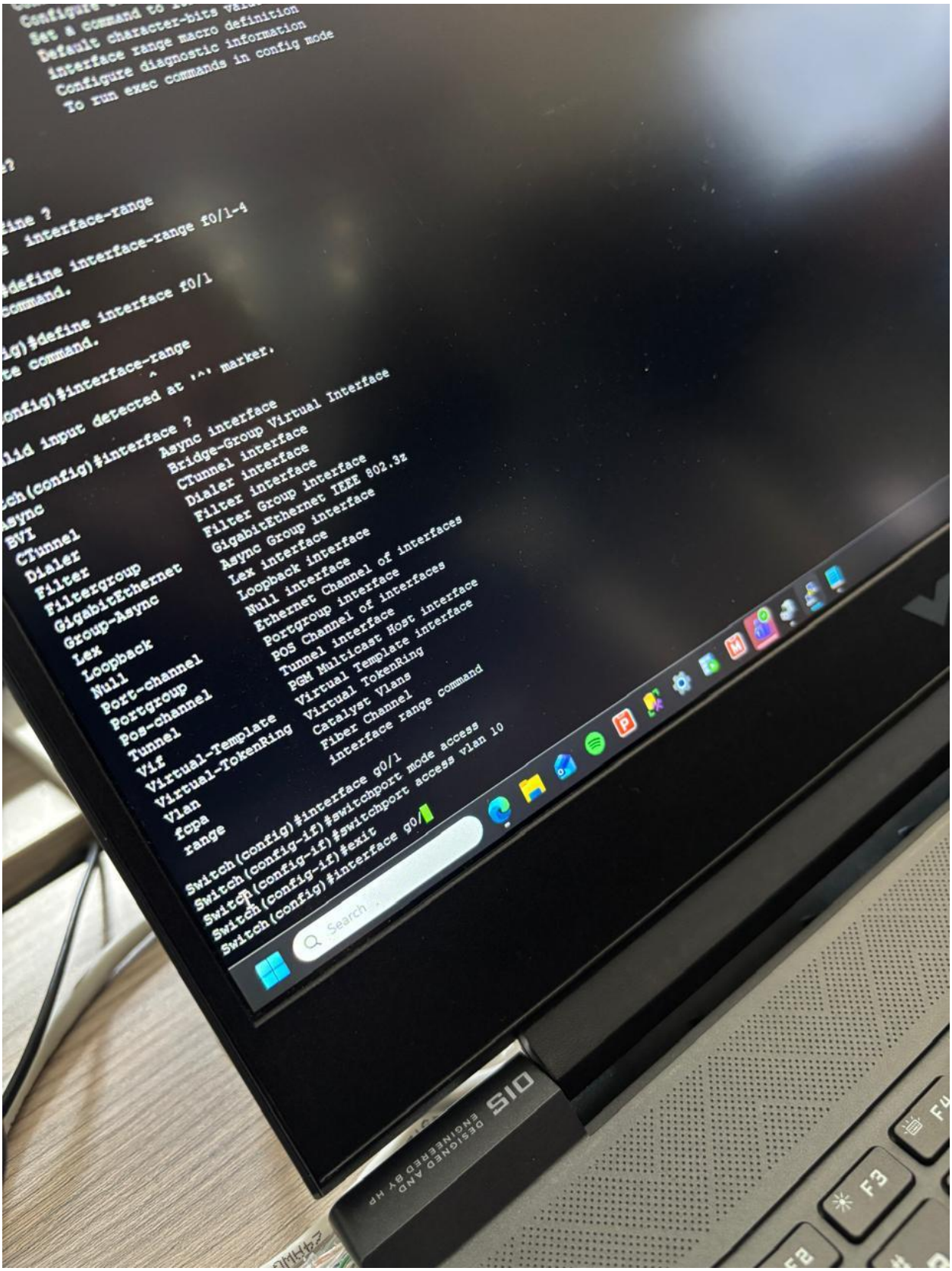

COM3 - PuTTY
Switch(config-vlan)#name sales/operations
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name server/admin
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#exit
Switch#show
00:18:13: %SYS-5-CONFIG_I: Configured from console by consolep in
Vlan1 is up, line protocol is up
Internet protocol processing disabled
GigabitEthernet0/1 is down, line protocol is down
Inbound access list is not set
GigabitEthernet0/2 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/3 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/4 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/5 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/6 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/7 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/8 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/9 is down, line protocol is down
Inbound access list is not set
GigabitEthernet0/10 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/11 is down, line protocol is down
Inbound access list is not set
GigabitEthernet0/12 is down, line protocol is down
Inbound access list is not set
GigabitEthernet0/13 is down, line protocol is down
SwInbound access list is not set
GigabitEthernet0/14 is down, line protocol is down
SwInbound access list is not set
GigabitEthernet0/15 is down, line protocol is down
SwInbound access list is not set
GigabitEthernet0/16 is down, line protocol is down
SwInbound access list is not set
GigabitEthernet0/17 is down, line protocol is down
SwInbound access list is not set
GigabitEthernet0/18 is down, line protocol is down
SwInbound access list is not set
GigabitEthernet0/19 is down, line protocol is down
SwInbound access list is not set
GigabitEthernet0/20 is down, line protocol is down
SwInbound access list is not set
GigabitEthernet0/21 is down, line protocol is down
SwItbb#nd access list is not set
SwItbb#Ethernet0/22 is down, line protocol is down
SwItbb#nd access list is not set
SwItbb#Ethernet0/23 is down, line protocol is down
SwItbb#nd access list is not set
SwItbb#Ethernet0/24 is down, line protocol is down
SwItbb#nd access list is not set
SwItbb#Ethernet0/25 is down, line protocol is down
SwItbb#nd access list is not set
SwItbb#Ethernet0/26 is down, line protocol is down
Search
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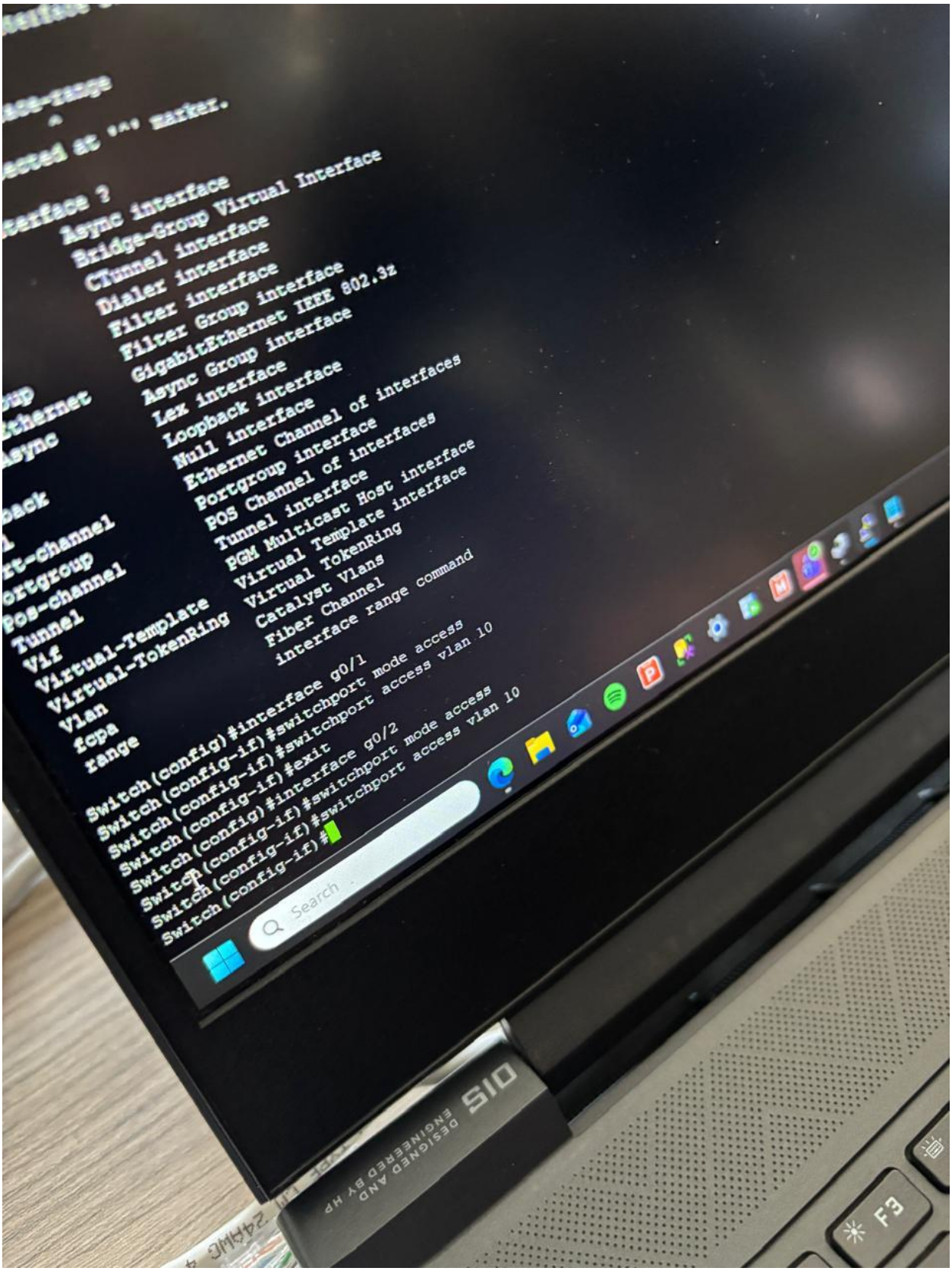

COM3 - PuTTY
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#show ip interface brief
Interface IP-Address OK? Method Status Protocol
Vlan1 unassigned YES unset up up
GigabitEthernet0/1 unassigned YES unset down down
GigabitEthernet0/2 unassigned YES unset up up
GigabitEthernet0/3 unassigned YES unset up up
GigabitEthernet0/4 unassigned YES unset up up
GigabitEthernet0/5 unassigned YES unset up up
GigabitEthernet0/6 unassigned YES unset up up
GigabitEthernet0/7 unassigned YES unset up down
GigabitEthernet0/8 unassigned YES unset down up
GigabitEthernet0/9 unassigned YES unset up down
GigabitEthernet0/10 unassigned YES unset down down
GigabitEthernet0/11 unassigned YES unset down down
GigabitEthernet0/12 unassigned YES unset down down
GigabitEthernet0/13 unassigned YES unset down down
GigabitEthernet0/14 unassigned YES unset down down
GigabitEthernet0/15 unassigned YES unset down down
GigabitEthernet0/16 unassigned YES unset down down
GigabitEthernet0/17 unassigned YES unset down down
GigabitEthernet0/18 unassigned YES unset down down
GigabitEthernet0/19 unassigned YES unset down down
GigabitEthernet0/20 unassigned YES unset down down
GigabitEthernet0/21 unassigned YES unset down down
GigabitEthernet0/22 unassigned YES unset down down
GigabitEthernet0/23 unassigned YES unset down down
GigabitEthernet0/24 unassigned YES unset down down
GigabitEthernet0/25 unassigned YES unset down down
GigabitEthernet0/26 unassigned YES unset down down
GigabitEthernet0/27 unassigned YES unset down down
GigabitEthernet0/28 unassigned YES unset down down
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#show vlan
VLAN Name Status Ports
1 default active Gi0/1, Gi0/2, Gi0/3, Gi0/4
Gi0/5, Gi0/6, Gi0/7, Gi0/8
Gi0/9, Gi0/10, Gi0/11, Gi0/12
Gi0/13, Gi0/14, Gi0/15, Gi0/16
Gi0/17, Gi0/18, Gi0/19, Gi0/20
Gi0/21, Gi0/22, Gi0/23, Gi0/24
Gi0/25, Gi0/26, Gi0/27, Gi0/28
10 engeneering/management active
20 sales/operations active
30 server/admin active
1002 fddi-default
Search

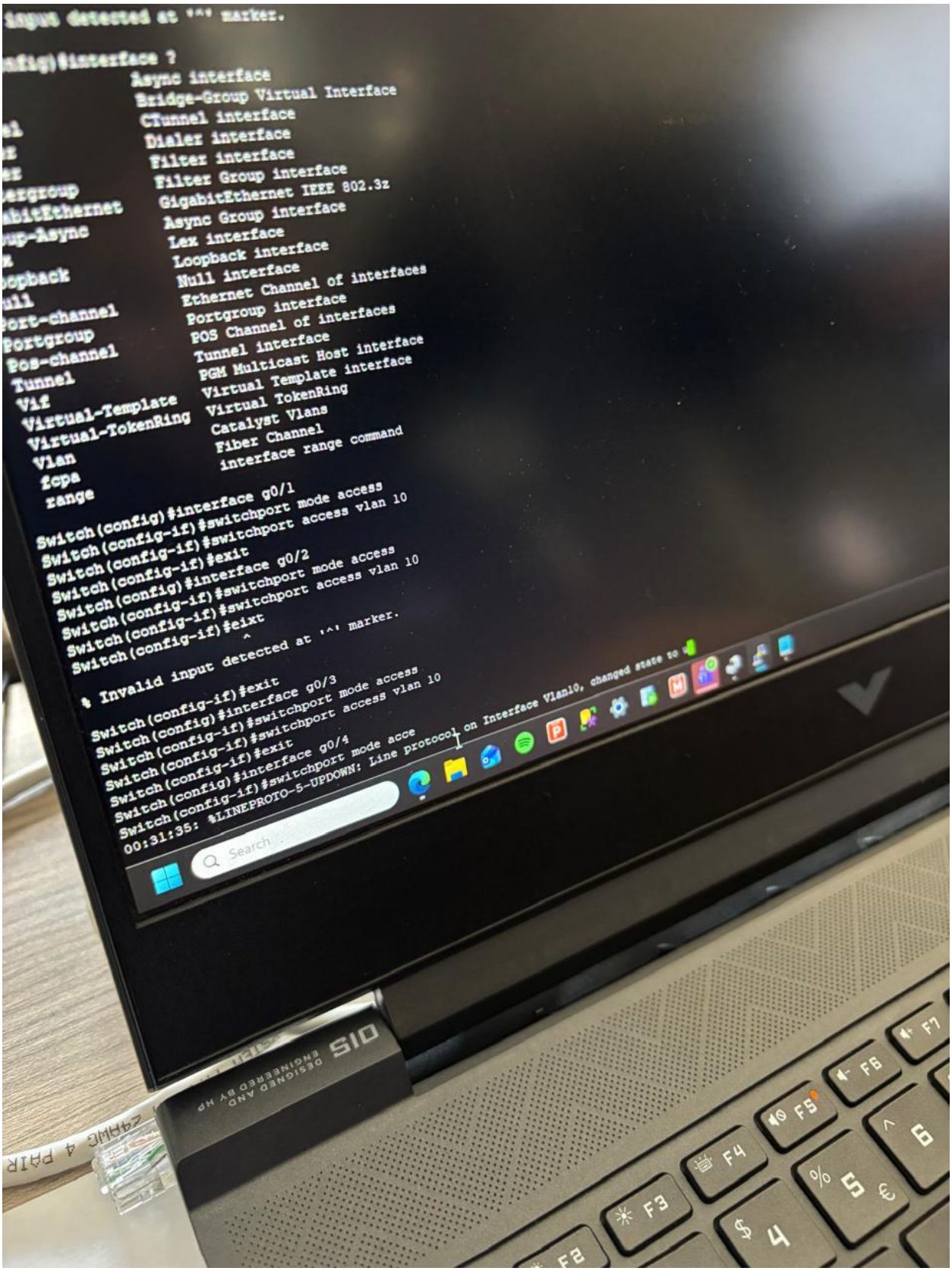

GigabitEthernet0/13 unassigned YES unset down down
GigabitEthernet0/14 unassigned YES unset down down
GigabitEthernet0/15 unassigned YES unset down down
GigabitEthernet0/16 unassigned YES unset down down
GigabitEthernet0/17 unassigned YES unset down down
GigabitEthernet0/18 unassigned YES unset down down
GigabitEthernet0/19 unassigned YES unset down down
GigabitEthernet0/20 unassigned YES unset down down
GigabitEthernet0/21 unassigned YES unset down down
GigabitEthernet0/22 unassigned YES unset down down
GigabitEthernet0/23 unassigned YES unset down down
GigabitEthernet0/24 unassigned YES unset down down
GigabitEthernet0/25 unassigned YES unset down down
GigabitEthernet0/26 unassigned YES unset down down
GigabitEthernet0/27 unassigned YES unset down down
GigabitEthernet0/28 unassigned YES unset down down
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#show vlan
VLAN Name Status Ports
1 default active Gi0/1, Gi0/2, Gi0/3, Gi0/4
Gi0/5, Gi0/6, Gi0/7, Gi0/8
Gi0/9, Gi0/10, Gi0/11, Gi0/12
Gi0/13, Gi0/14, Gi0/15, Gi0/16
Gi0/17, Gi0/18, Gi0/19, Gi0/20
Gi0/21, Gi0/22, Gi0/23, Gi0/24
Gi0/25, Gi0/26, Gi0/27, Gi0/28
10 engeneering/management active
20 sales/operations active
30 server/admin active
1002 fddi-default act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default act/unsup
1005 trnet-default act/unsup
VLAN Type SAID MTU Parent RingNo BridgeNo Stp BrdgMode Trans1 Trans2
1 enet 100001 1500 - - - - - 0 0
10 enet 100010 1500 - - - - - 0 0
20 enet 100020 1500 - - - - - 0 0
--More--
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24AWG 4 PAIR


COM3 - PuTTY
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
00:19:39: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/4, changed state to down
00:19:40: %LINK-3-UPDOWN: Interface GigabitEthernet0/4, changed state to down
00:19:43: %LINK-3-UPDOWN: Interface GigabitEthernet0/4, changed state to up
00:19:44: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/4, changed state to up
Switch#
Switch#
Switch#
Switch#interface vlan 10
% Invalid input detected at '^' marker.
Switch#confi ter
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlN 10
% Invalid input detected at '^' marker.
Switch(config)#INTERFACE VLAN 10
Switch(config-if)#
00:20:30: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to down
Switch(config-if)#IP ADDRESS 192.168.10.0 255.255.255.0
Bad mask /24 for address 192.168.10.0
Switch(config-if)#
00:21:13: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/5, changed state to down
00:21:14: %LINK-3-UPDOWN: Interface GigabitEthernet0/5, changed state to down
Switch(config-if)#IP ADDRESS 192.168.10.1 255.
00:21:28: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
00:21:29: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to down255
% Incomplete command.
Switch(config-if)#IP ADDRES 1
00:21:33: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up
00:21:33: %LINK-3-UPDOWN: Interface GigabitEthernet0/5, changed state to up
00:21:34: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up92.
00:21:34: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/5, changed state to up
% Incomplete command.
Switch(config-if)#IP ADDRESS 192.168.10.1 25
00:21:44: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/4, changed state to down
00:21:45: %LINK-3-UPDOWN: Interface GigabitEthernet0/4, changed state to down5.IP ADDRESS 192.168.10.1 25
% Invalid input detected at '^' marker.
Switch(config-if)#IP ADDRESS 192.168.10.1 25
% Incomplete command.
Switch(config-if)#
00:21:49: %LINK-3-UPDOWN: Interface GigabitEthernet0/4, changed state to up
00:21:50: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/4, changed state to up
Switch(config-if)#IP ADDRESS 192.168.10.1 255.255.255.0
Switch(config-if)#EXIT
Switch(config)#INTERFACE VLAN 20
Switch(config-if)#
00:22:40: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to down
Search


access-list
alias
archive
arp
banner
boot
buffers
cdp
cef
class-map
clock
cluster
cns
configuration
control-plane
default
default-value
define
diagnostic
do
Add an access list entry
Create command alias
Archive the configuration
Set a static ARP entry
Define a login banner
Boot Commands
Adjust system buffer pool parameters
Global CDP configuration subcommands
Cisco Express Forwarding
Configure QoS Class Map
Configure time-of-day clock
Cluster configuration commands
CNS agents
Configuration access
Configure control plane services
Set a command to its defaults
Default character-bits values
interface range macro definition
Configure diagnostic information
To run exec commands in config mode
Switch(config)#define?
define
Switch(config)#define ?
interface-range interface-range
Switch(config)#define interface-range f0/1-4
% Incomplete command.
Switch(config)#define interface f0/1
% Incomplete command.
Switch(config)#interface-range
^
% Invalid input detected at '^' marker.
Switch(config)#interface ?
Async
SVI
CTunnel
Dialer
Filter
Filtergroup
GigabitEthernet
Group-Async
Lex
Loopback
Null
Port-channel
Portgroup
Pos-channel
Tunnel
Vif
Virtual-Template
Virtual-TokenRing
Vlan
fcpa
range
Async interface
Bridge-Group Virtual Interface
CTunnel interface
Dialer interface
Filter interface
Filter Group interface
GigabitEthernet IEEE 802.3z
Async Group interface
Lex interface
Loopback interface
Null interface
Ethernet Channel of interfaces
Portgroup interface
POS Channel of interfaces
Tunnel interface
PGM Multicast Host interface
Virtual Template interface
Virtual TokenRing
Catalyst Vlans
Fiber Channel
interface range command
Switch(config)#interface f0/1
Switch(config-if)#

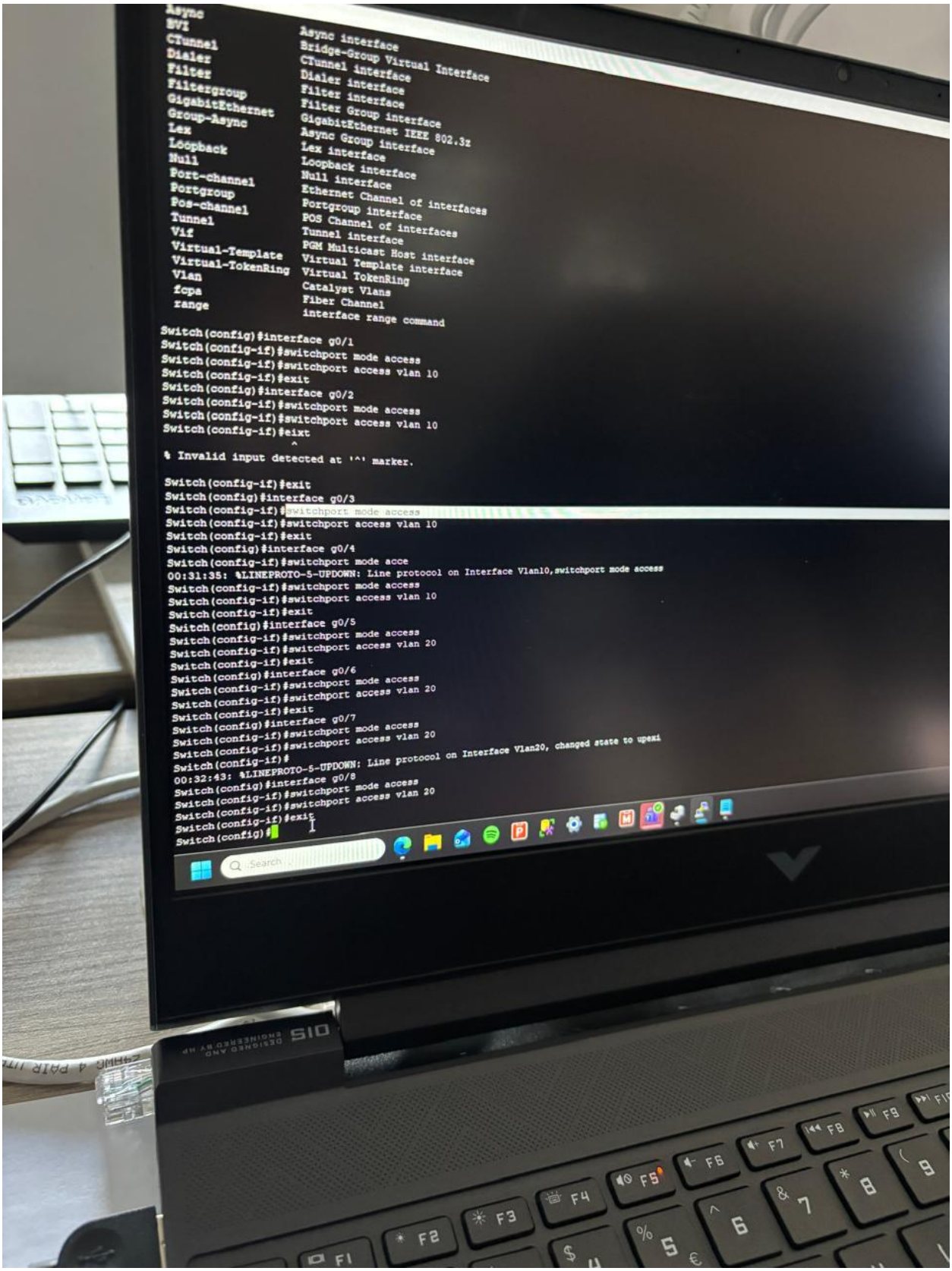

Set a command to its
Default character-bits
interface range macro definition
Configure diagnostic information
To run exec commands in config mode
interface-range
define interface-range f0/1-4
command.
g)$define interface f0/1
te command.
onfig)$interface-range
^
lid input detected at '^' marker.
ch(config)$interface ?
Async
BVI
CTunnel
Dialer
Filter
FilterGroup
GigabitEthernet
Group-Async
Lex
Loopback
Null
Port-channel
Portgroup
Pos-channel
Tunnel
Vif
Virtual-Template
Virtual-TokenRing
Vlan
fcpa
range
Async interface
Bridge-Group Virtual Interface
CTunnel interface
Dialer interface
Filter interface
Filter Group interface
GigabitEthernet IEEE 802.3z
Async Group interface
Lex interface
Loopback interface
Null interface
Ethernet Channel of interfaces
Portgroup interface
POS Channel of interfaces
Tunnel interface
PGM Multicast Host interface
Virtual Template interface
Virtual TokenRing
Catalyst Vlans
Fiber Channel
interface range command
Switch(config)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/
Search
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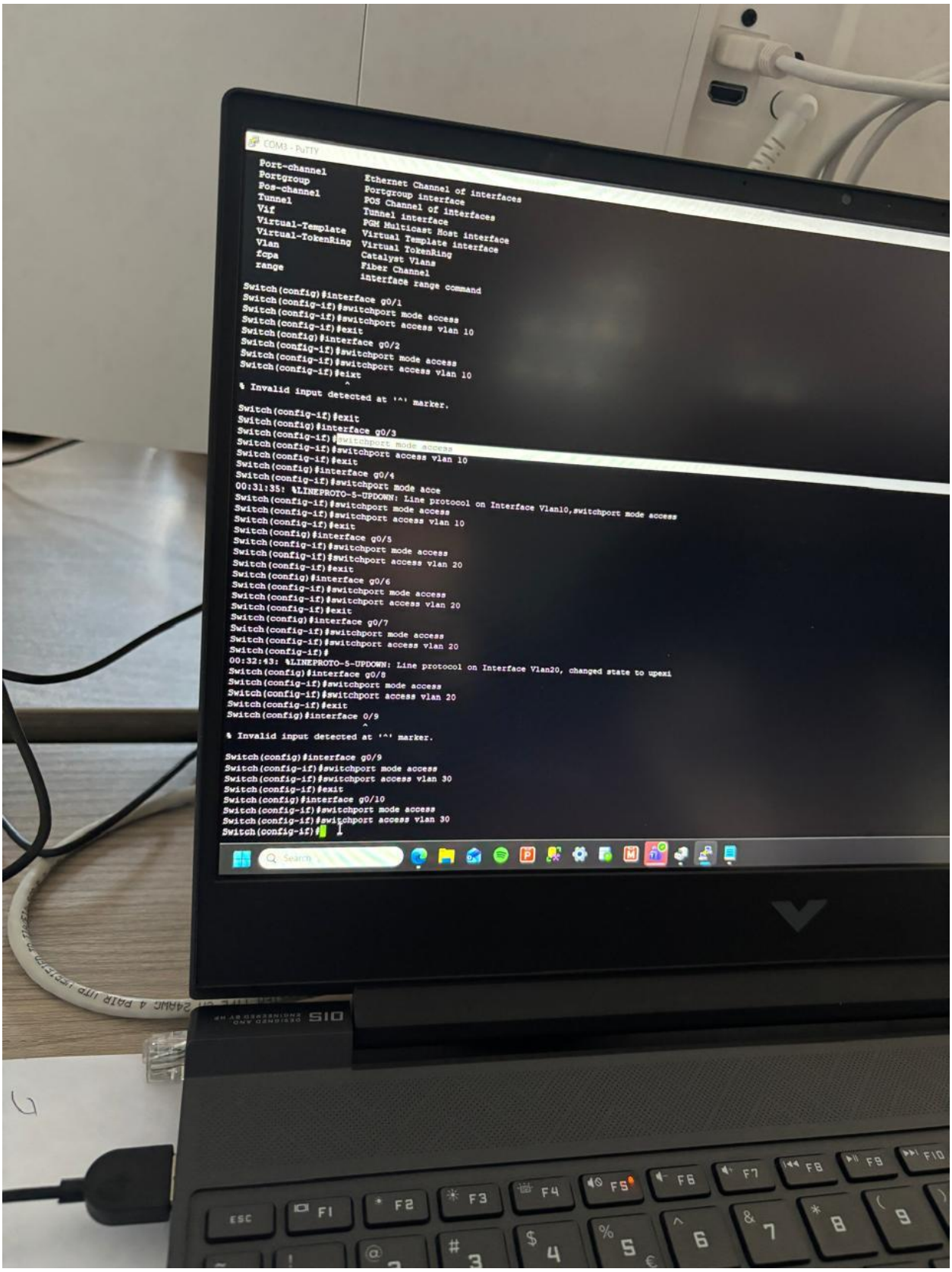

Async interface
Bridge-Group Virtual Interface
CTunnel interface
Dialer interface
Filter interface
Filter Group interface
GigabitEthernet IEEE 802.3z
Async Group interface
Lex interface
Loopback interface
Null interface
Ethernet Channel of interfaces
Portgroup interface
POS Channel of interfaces
Tunnel interface
PGM Multicast Host interface
Virtual Template interface
Virtual TokenRing
Catalyst Vlans
Fiber Channel
interface range command
Switch(config)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
Search
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F3

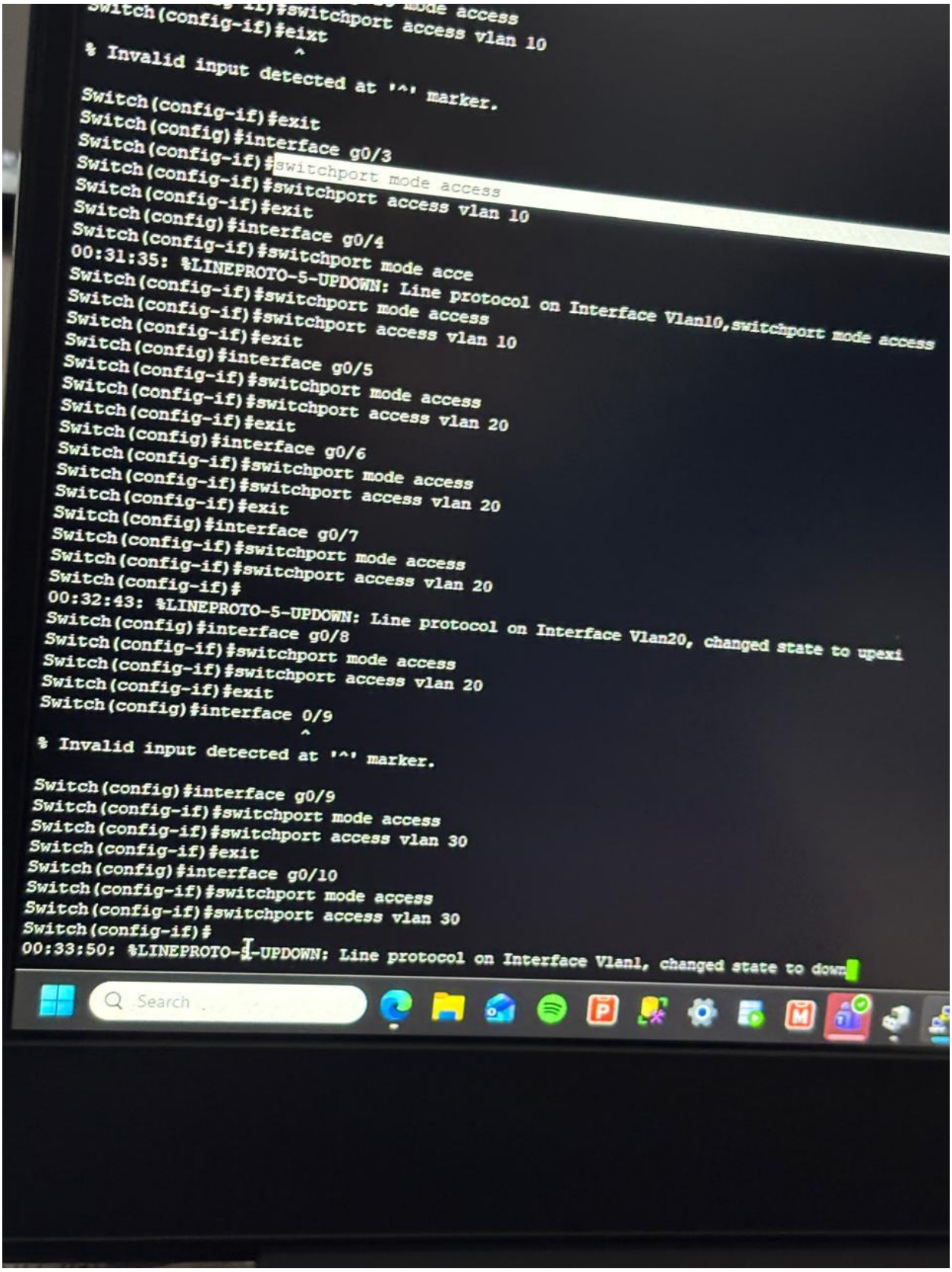

input detected at '^' marker.
nfig)#interface ?
Async interface
Bridge-Group Virtual Interface
CTunnel interface
Dialer interface
Filter interface
Filter Group interface
GigabitEthernet IEEE 802.3z
Async Group interface
Lex interface
Loopback interface
Null interface
Ethernet Channel of interfaces
Portgroup interface
POS Channel of interfaces
Tunnel interface
PGM Multicast Host interface
Virtual Template interface
Virtual TokenRing
Catalyst Vlans
Fiber Channel
interface range command
Switch(config)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#eixt
% Invalid input detected at '^' marker.
Switch(config-if)#exit
Switch(config)#interface g0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/4
Switch(config-if)#switchport mode acce
00:31:35: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to
Search

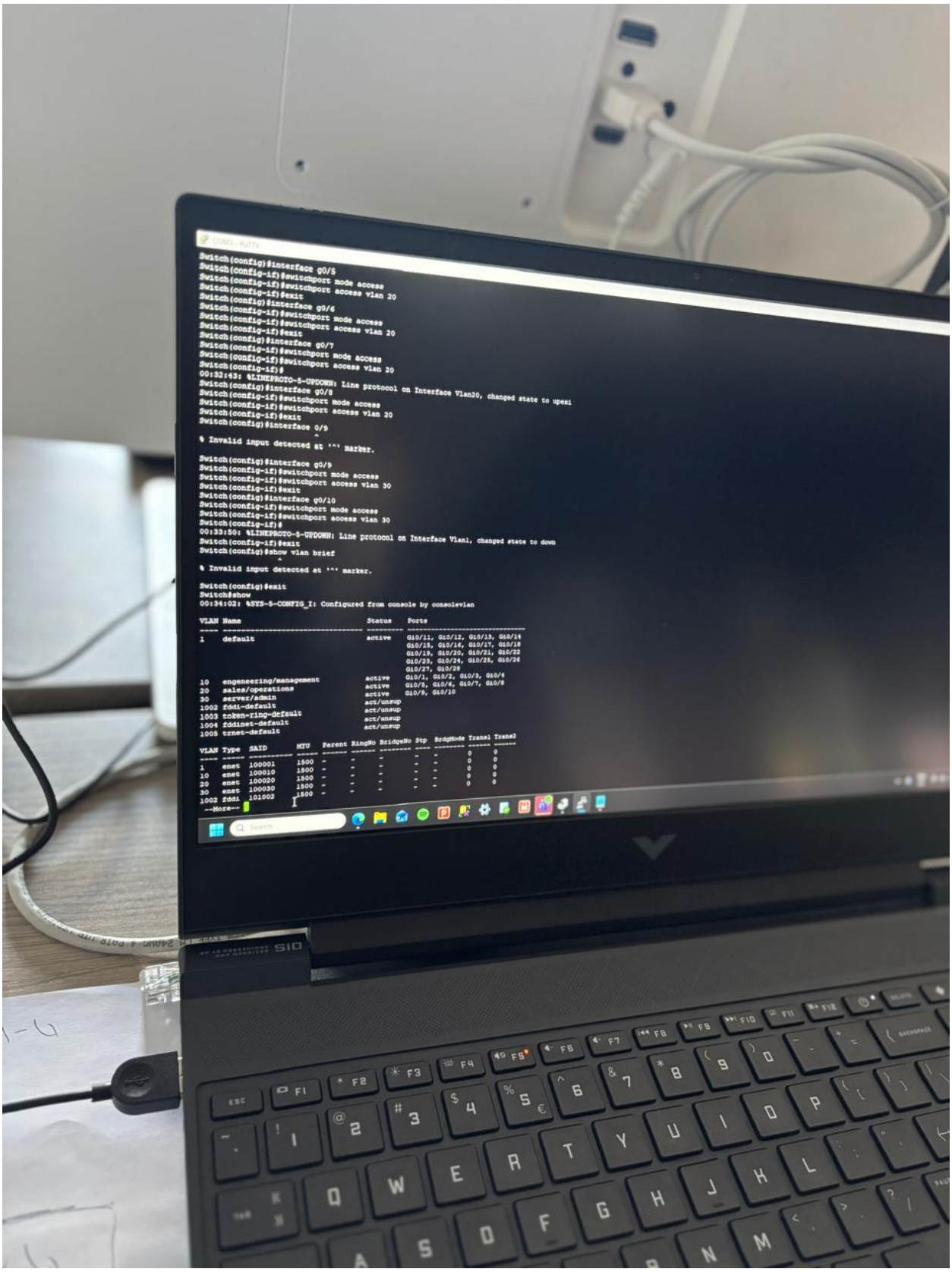
02.3z
nel of interfaces
terface
of interfaces
terface
ticast Host interface
ual Template interface
alyst Vlans
Fiber Channel
interface range command
nterface g0/1
(if)#switchport mode access
g-if)#exit
nterface g0/2
nfig-if)#switchport mode access
(config-if)#switchport access vlan 10
(config-if)#exit
^
% Invalid input detected at '^' marker.
Switch(config-if)#exit
Switch(config)#interface g0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/4
Switch(config-if)#switchport mode acce
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
00:31:35: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10,switchport mode access
Switch(config)#interface g0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#sw
switchport mode access
Search
DESIGNED AND
ENGINEERED BY HP
F1
F2
F3
F4
F5
%
4
3
2

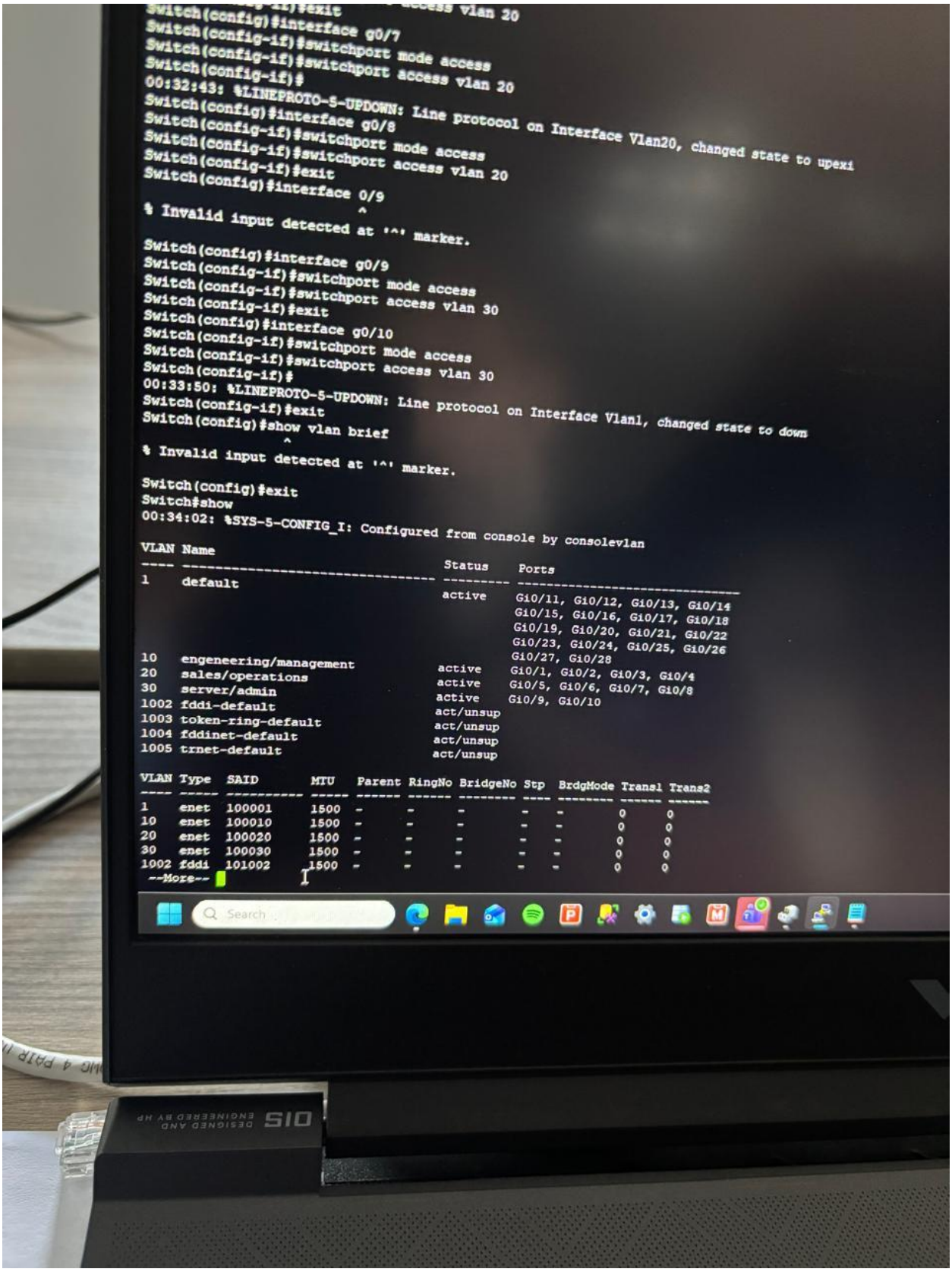



Async
BVI
CTunnel
Dialer
Filter
Filtergroup
GigabitEthernet
Group-Async
Lex
Loopback
Null
Port-channel
Portgroup
Pos-channel
Tunnel
Vif
Virtual-Template
Virtual-TokenRing
Vlan
fcpa
range
Async interface
Bridge-Group Virtual Interface
CTunnel interface
Dialer interface
Filter interface
Filter Group interface
GigabitEthernet IEEE 802.3z
Async Group interface
Lex interface
Loopback interface
Null interface
Ethernet Channel of interfaces
Portgroup interface
POS Channel of interfaces
Tunnel interface
PGM Multicast Host interface
Virtual Template interface
Virtual TokenRing
Catalyst Vlans
Fiber Channel
interface range command
Switch(config)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#eixt
^
% Invalid input detected at '^' marker.
Switch(config-if)#exit
Switch(config)#interface g0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/4
Switch(config-if)#switchport mode acce
00:31:35: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10,switchport mode access
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
Switch(config-if)#
00:32:43: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to upexi
Switch(config)#interface g0/8
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
Search

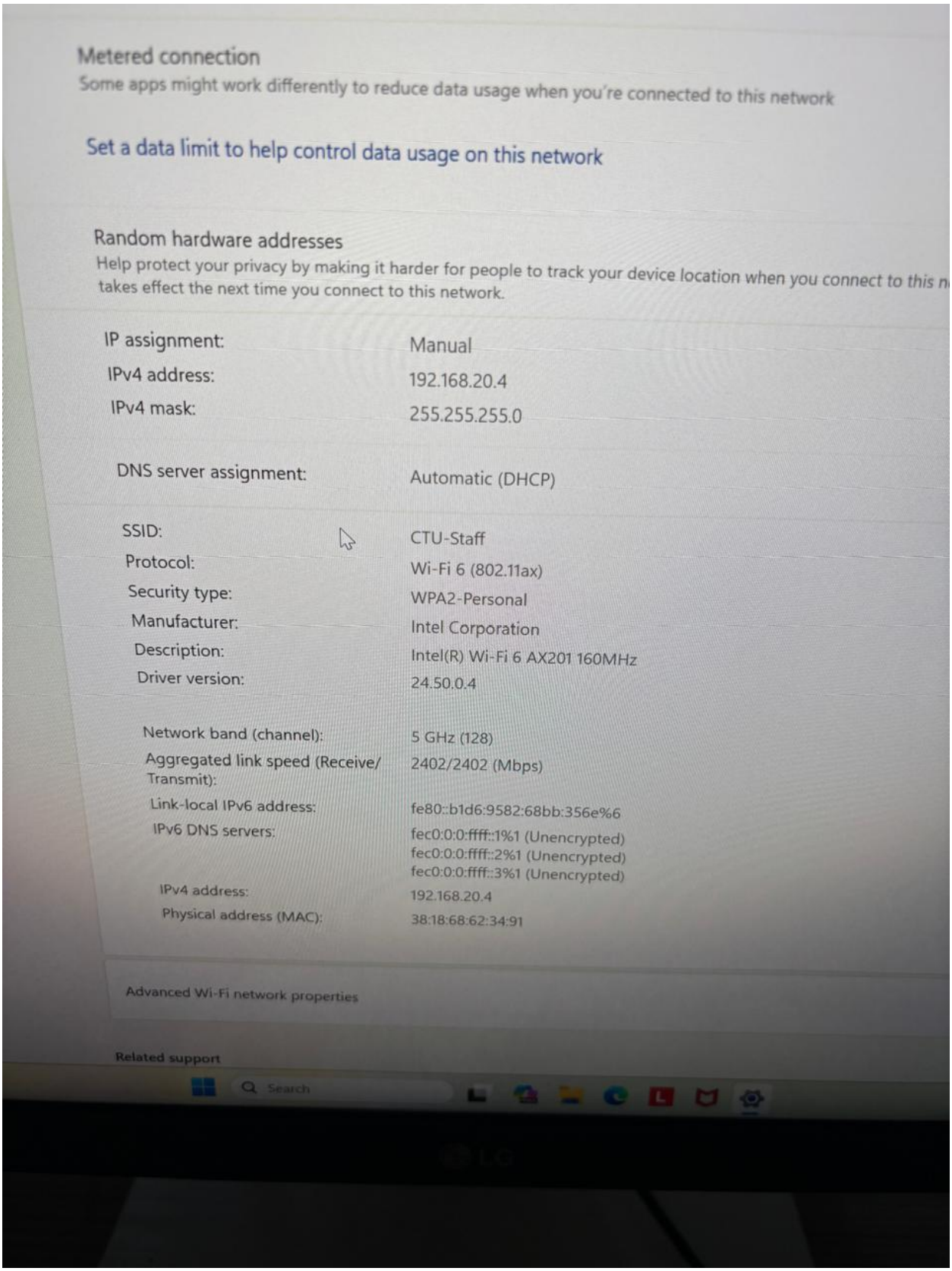

COM3 - PuTTY
Port-channel Ethernet Channel of interfaces
Portgroup Portgroup interface
Pos-channel POS Channel of interfaces
Tunnel Tunnel interface
Vif PGM Multicast Host interface
Virtual-Template Virtual Template interface
Virtual-TokenRing Virtual TokenRing
Vlan Catalyst Vlans
fcpa Fiber Channel
range interface range command
Switch(config)#interface g0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#eixt
% Invalid input detected at '^' marker.
Switch(config-if)#exit
Switch(config)#interface g0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/4
Switch(config-if)#switchport mode acce
00:31:35: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10,switchport mode access
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
00:32:43: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to upexi
Switch(config)#interface g0/8
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface 0/9
% Invalid input detected at '^' marker.
Switch(config)#interface g0/9
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface g0/10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#

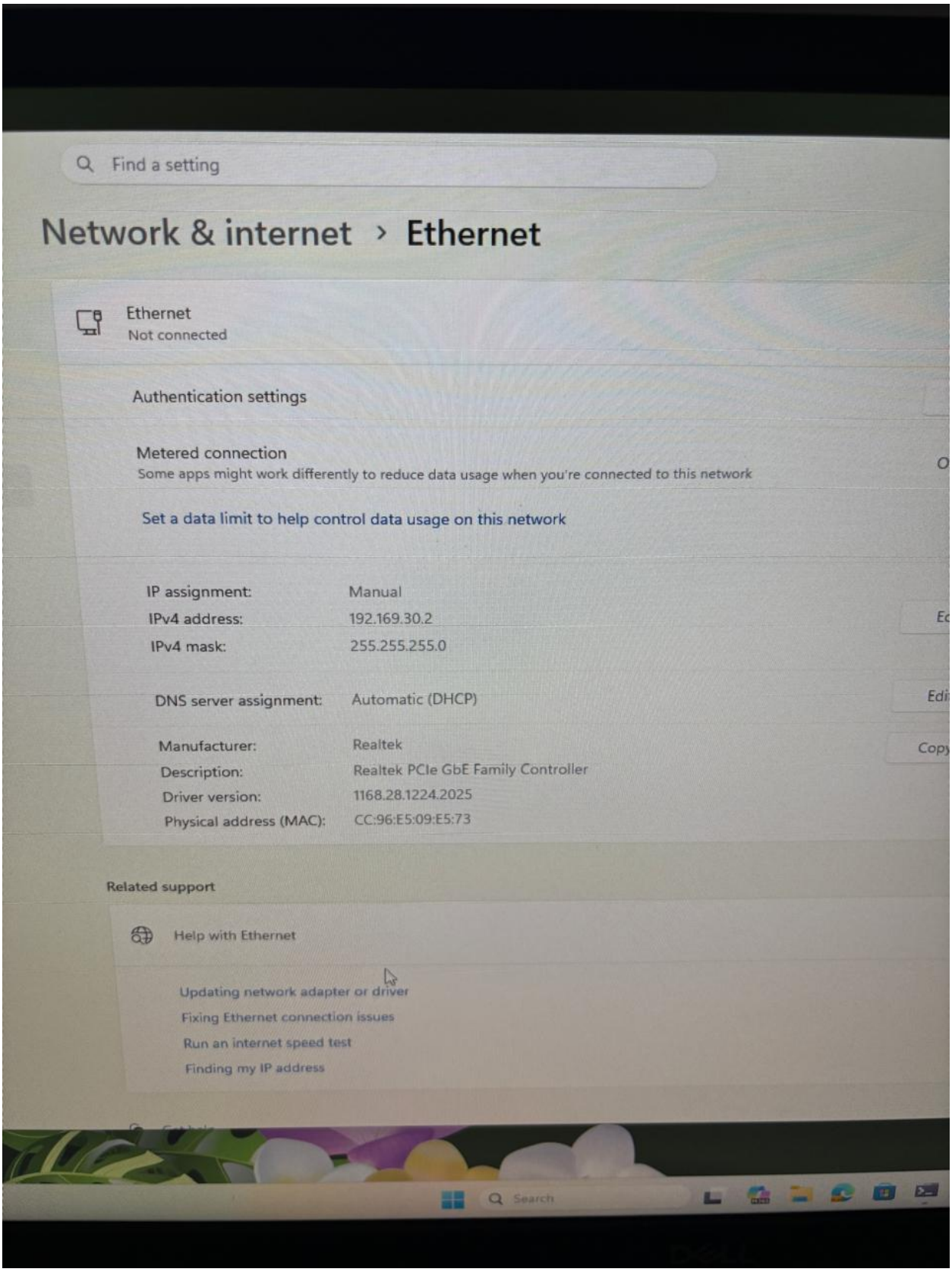

Switch(config-if)#switchport access vlan 10
Switch(config-if)#eixt
^
% Invalid input detected at '^' marker.
Switch(config-if)#exit
Switch(config)#interface g0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/4
Switch(config-if)#switchport mode acce
00:31:35: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10,switchport mode access
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface g0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
00:32:43: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to upexi
Switch(config)#interface g0/8
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface 0/9
^
% Invalid input detected at '^' marker.
Switch(config)#interface g0/9
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface g0/10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#
00:33:50: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to down
Search

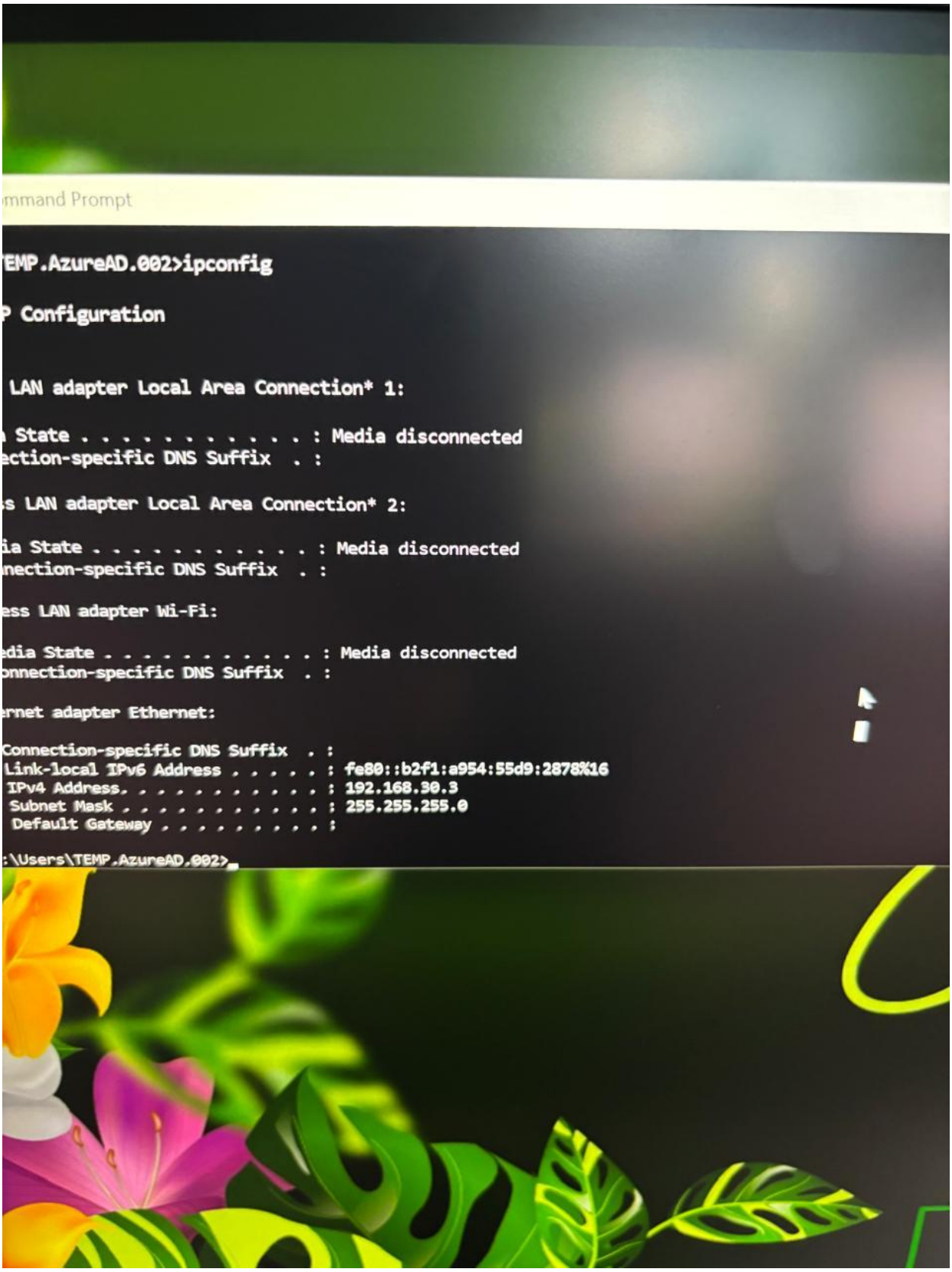

Switch(config)#interface g0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface g0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
00:32:43: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to upexi
Switch(config)#interface g0/8
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface 0/9
% Invalid input detected at '^' marker.
Switch(config)#interface g0/9
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface g0/10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#
00:33:50: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to down
Switch(config-if)#exit
Switch(config)#show vlan brief
% Invalid input detected at '^' marker.
Switch(config)#exit
Switch#show
00:34:02: %SYS-5-CONFIG_I: Configured from console by consolevlan
VLAN Name Status Ports
1 default active Gi0/11, Gi0/12, Gi0/13, Gi0/14
Gi0/15, Gi0/16, Gi0/17, Gi0/18
Gi0/19, Gi0/20, Gi0/21, Gi0/22
Gi0/23, Gi0/24, Gi0/25, Gi0/26
Gi0/27, Gi0/28
10 engeneering/management active Gi0/1, Gi0/2, Gi0/3, Gi0/4
20 sales/operations active Gi0/5, Gi0/6, Gi0/7, Gi0/8
30 server/admin active Gi0/9, Gi0/10
1002 fddi-default act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default act/unsup
1005 trnet-default act/unsup
VLAN Type SAID MTU Parent RingNo BridgeNo Stp BrdgMode Trans1 Trans2
1 enet 100001 1500 - - - - - 0 0
10 enet 100010 1500 - - - - - 0 0
20 enet 100020 1500 - - - - - 0 0
30 enet 100030 1500 - - - - - 0 0
1002 fddi 101002
--More--

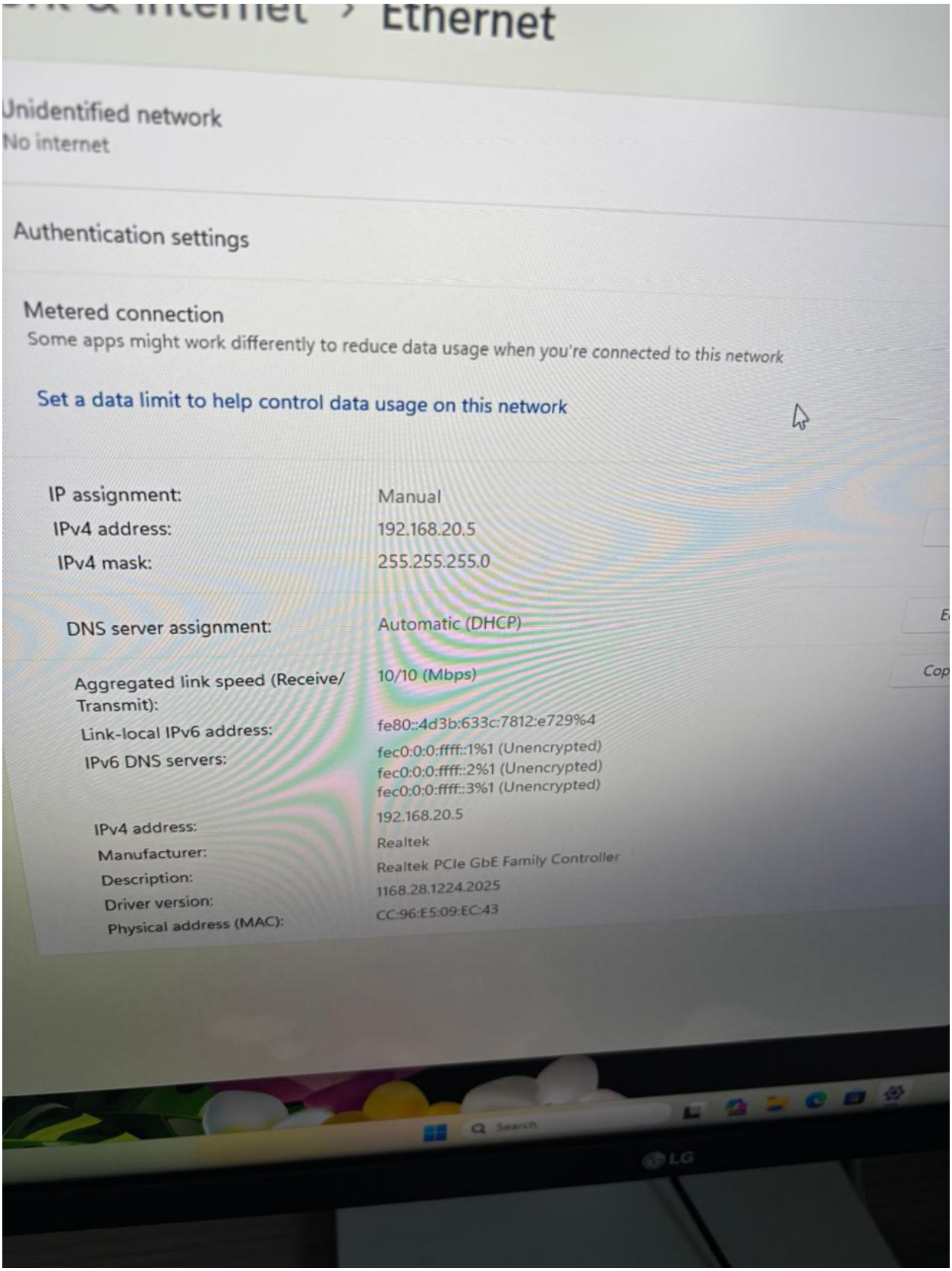

Switch(config)#exit
Switch(config)#interface g0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
00:32:43: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to upexi
Switch(config)#interface g0/8
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface 0/9
% Invalid input detected at '^' marker.
Switch(config)#interface g0/9
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface g0/10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#
00:33:50: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to down
Switch(config-if)#exit
Switch(config)#show vlan brief
% Invalid input detected at '^' marker.
Switch(config)#exit
Switch#show
00:34:02: %SYS-5-CONFIG_I: Configured from console by consolevlan
VLAN Name Status Ports
1 default active Gi0/11, Gi0/12, Gi0/13, Gi0/14
Gi0/15, Gi0/16, Gi0/17, Gi0/18
Gi0/19, Gi0/20, Gi0/21, Gi0/22
Gi0/23, Gi0/24, Gi0/25, Gi0/26
Gi0/27, Gi0/28
10 engeneering/management active Gi0/1, Gi0/2, Gi0/3, Gi0/4
20 sales/operations active Gi0/5, Gi0/6, Gi0/7, Gi0/8
30 server/admin active Gi0/9, Gi0/10
1002 fddi-default act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default act/unsup
1005 trnet-default act/unsup
VLAN Type SAID MTU Parent RingNo BridgeNo Stp BrdgMode Trans1 Trans2
1 enet 100001 1500 - - - - - 0 0
10 enet 100010 1500 - - - - - 0 0
20 enet 100020 1500 - - - - - 0 0
30 enet 100030 1500 - - - - - 0 0
1002 fddi 101002 1500 - - - - - 0 0
--More--
Search

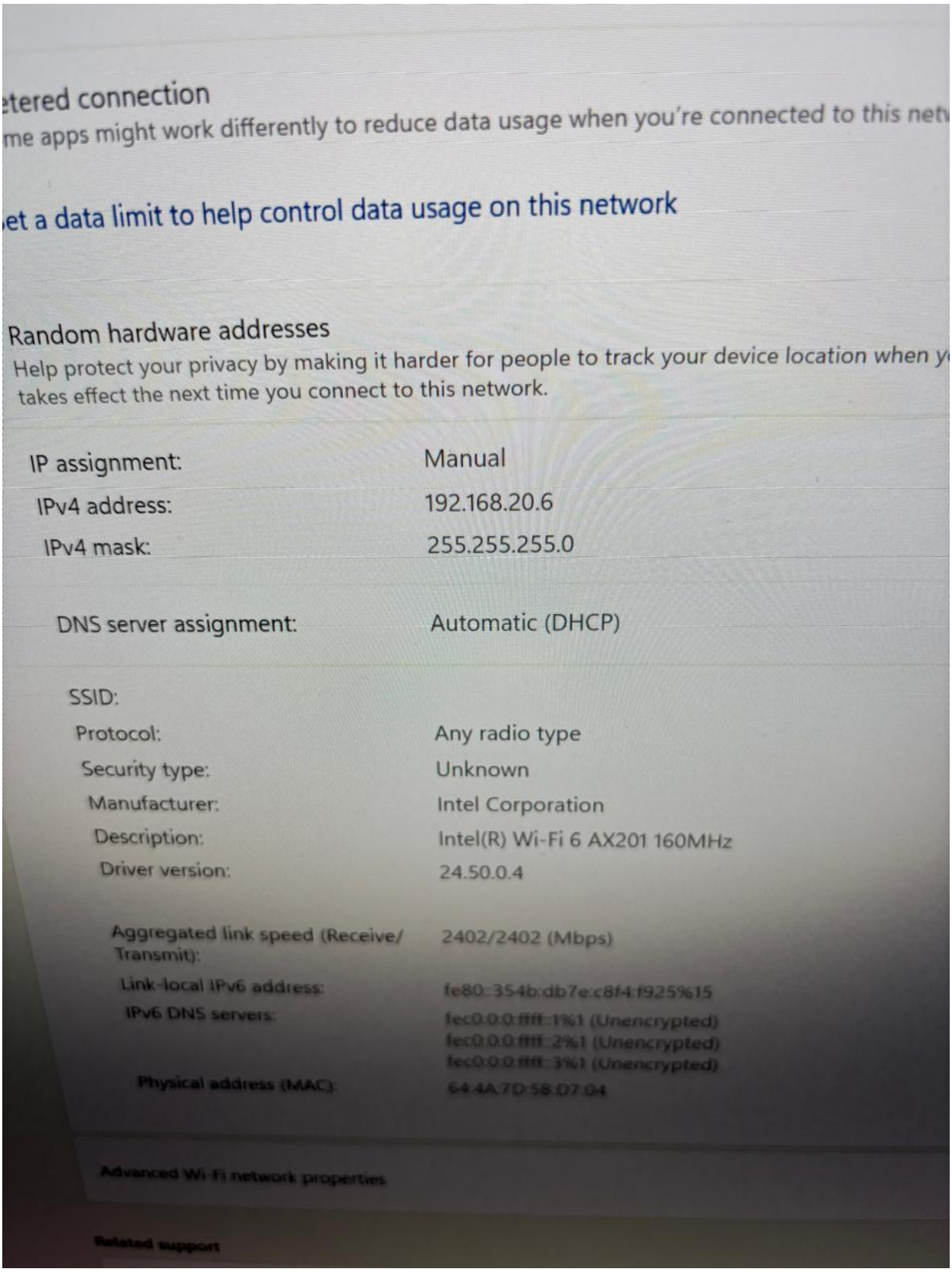

YOUR FUTURE IS BLOOMING.


Metered connection
Some apps might work differently to reduce data usage when you're connected to this network
Set a data limit to help control data usage on this network
Random hardware addresses
Help protect your privacy by making it harder for people to track your device location when you connect to this n
takes effect the next time you connect to this network.
IP assignment: Manual
IPv4 address: 192.168.20.4
IPv4 mask: 255.255.255.0
DNS server assignment: Automatic (DHCP)
SSID: CTU-Staff
Protocol: Wi-Fi 6 (802.11ax)
Security type: WPA2-Personal
Manufacturer: Intel Corporation
Description: Intel(R) Wi-Fi 6 AX201 160MHz
Driver version: 24.50.0.4
Network band (channel): 5 GHz (128)
Aggregated link speed (Receive/ Transmit): 2402/2402 (Mbps)
Link-local IPv6 address: fe80::b1d6:9582:68bb:356e%6
IPv6 DNS servers: fec0:0:0:ffff::1%1 (Unencrypted)
fec0:0:0:ffff::2%1 (Unencrypted)
fec0:0:0:ffff::3%1 (Unencrypted)
IPv4 address: 192.168.20.4
Physical address (MAC): 38:18:68:62:34:91
Advanced Wi-Fi network properties
Related support
Search


Find a setting
Network & internet › Ethernet
Ethernet
Not connected
Authentication settings
Metered connection
Some apps might work differently to reduce data usage when you're connected to this network
Set a data limit to help control data usage on this network
IP assignment: Manual
IPv4 address: 192.169.30.2
IPv4 mask: 255.255.255.0
DNS server assignment: Automatic (DHCP)
Manufacturer: Realtek
Description: Realtek PCIe GbE Family Controller
Driver version: 1168.28.1224.2025
Physical address (MAC): CC:96:E5:09:E5:73
Related support
Help with Ethernet
Updating network adapter or driver
Fixing Ethernet connection issues
Run an internet speed test
Finding my IP address
Search


mmand Prompt
EMP.AzureAD.002>ipconfig
P Configuration
LAN adapter Local Area Connection* 1:
State . . . . . : Media disconnected
ction-specific DNS Suffix . :
s LAN adapter Local Area Connection* 2:
ia State . . . . . : Media disconnected
nection-specific DNS Suffix . :
ess LAN adapter Wi-Fi:
edia State . . . . . : Media disconnected
onnection-specific DNS Suffix . :
rnet adapter Ethernet:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::b2f1:a954:55d9:2878%16
IPv4 Address. . . . . : 192.168.30.3
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
:\Users\TEMP.AzureAD.002>


Ethernet
Unidentified network
No internet
Authentication settings
Metered connection
Some apps might work differently to reduce data usage when you're connected to this network
Set a data limit to help control data usage on this network
IP assignment: Manual
IPv4 address: 192.168.20.5
IPv4 mask: 255.255.255.0
DNS server assignment: Automatic (DHCP)
Aggregated link speed (Receive/ Transmit): 10/10 (Mbps)
Link-local IPv6 address: fe80::4d3b:633c:7812:e729%4
IPv6 DNS servers: fec0:0:0:ffff::1%1 (Unencrypted)
fec0:0:0:ffff::2%1 (Unencrypted)
fec0:0:0:ffff::3%1 (Unencrypted)
IPv4 address: 192.168.20.5
Manufacturer: Realtek
Description: Realtek PCIe GbE Family Controller
Driver version: 1168.28.1224.2025
Physical address (MAC): CC:96:E5:09:EC:43
Search
LG



tered connection
me apps might work differently to reduce data usage when you're connected to this net

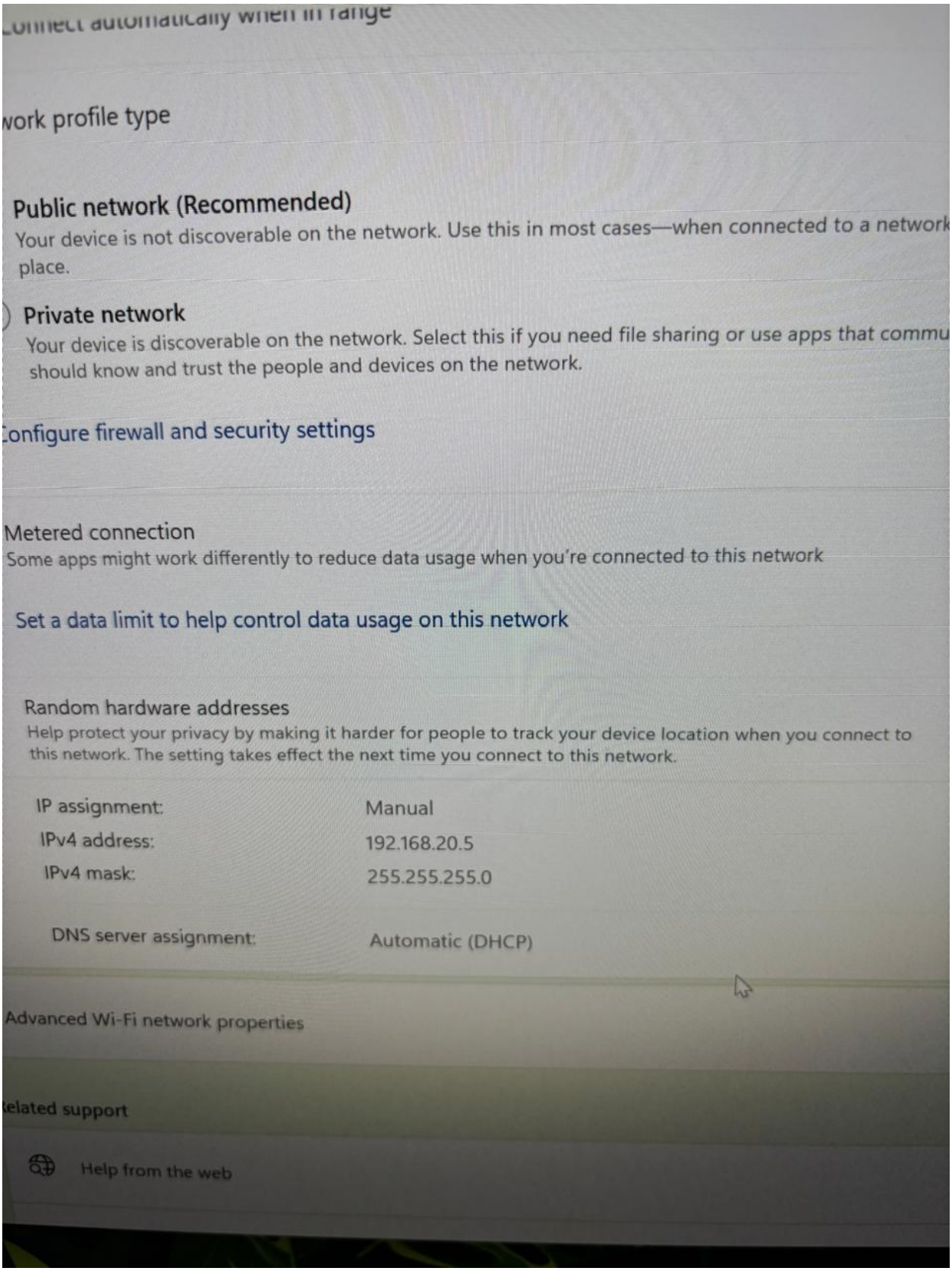
et a data limit to help control data usage on this network

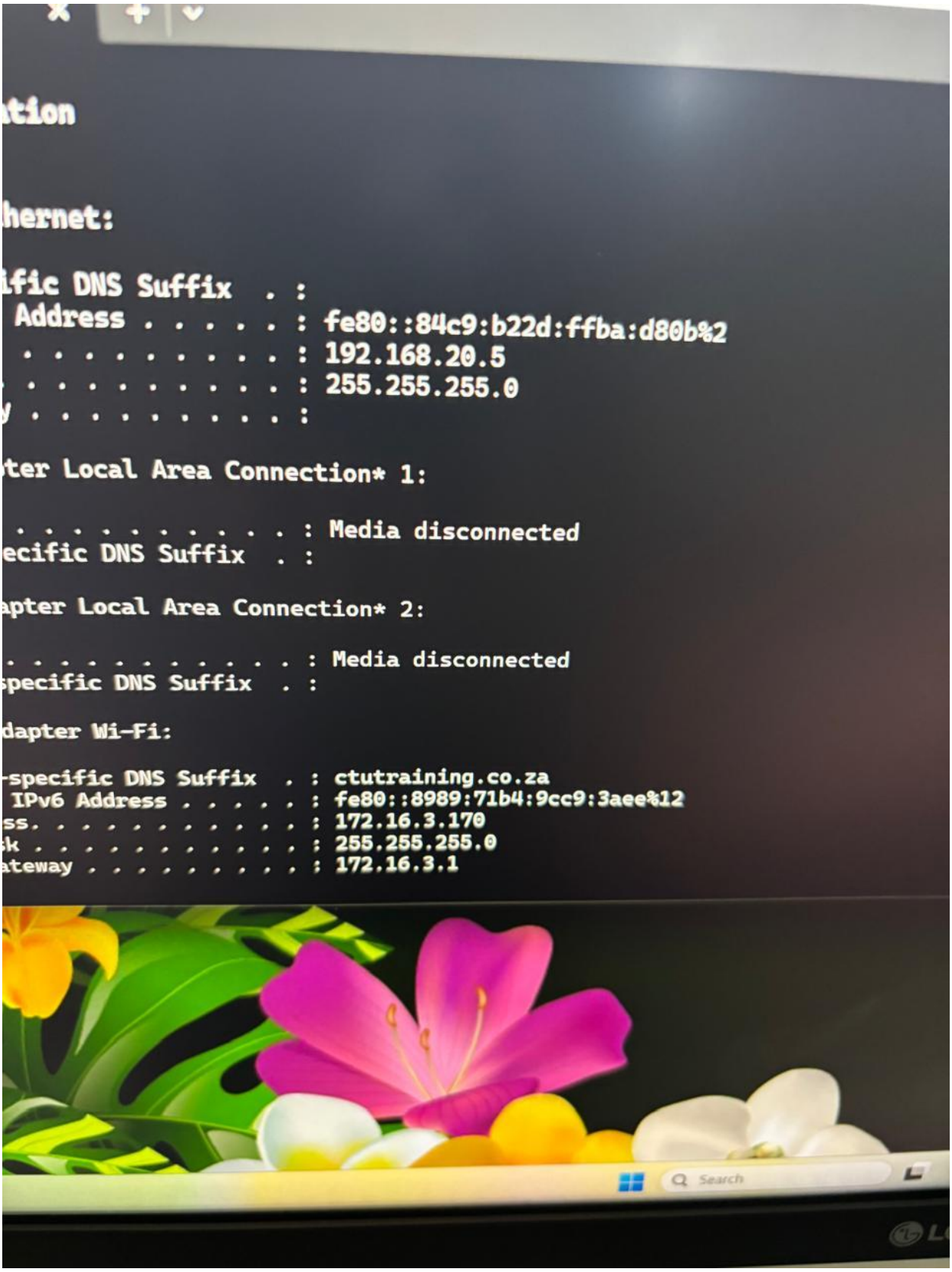
Random hardware addresses
Help protect your privacy by making it harder for people to track your device location when y
takes effect the next time you connect to this network.

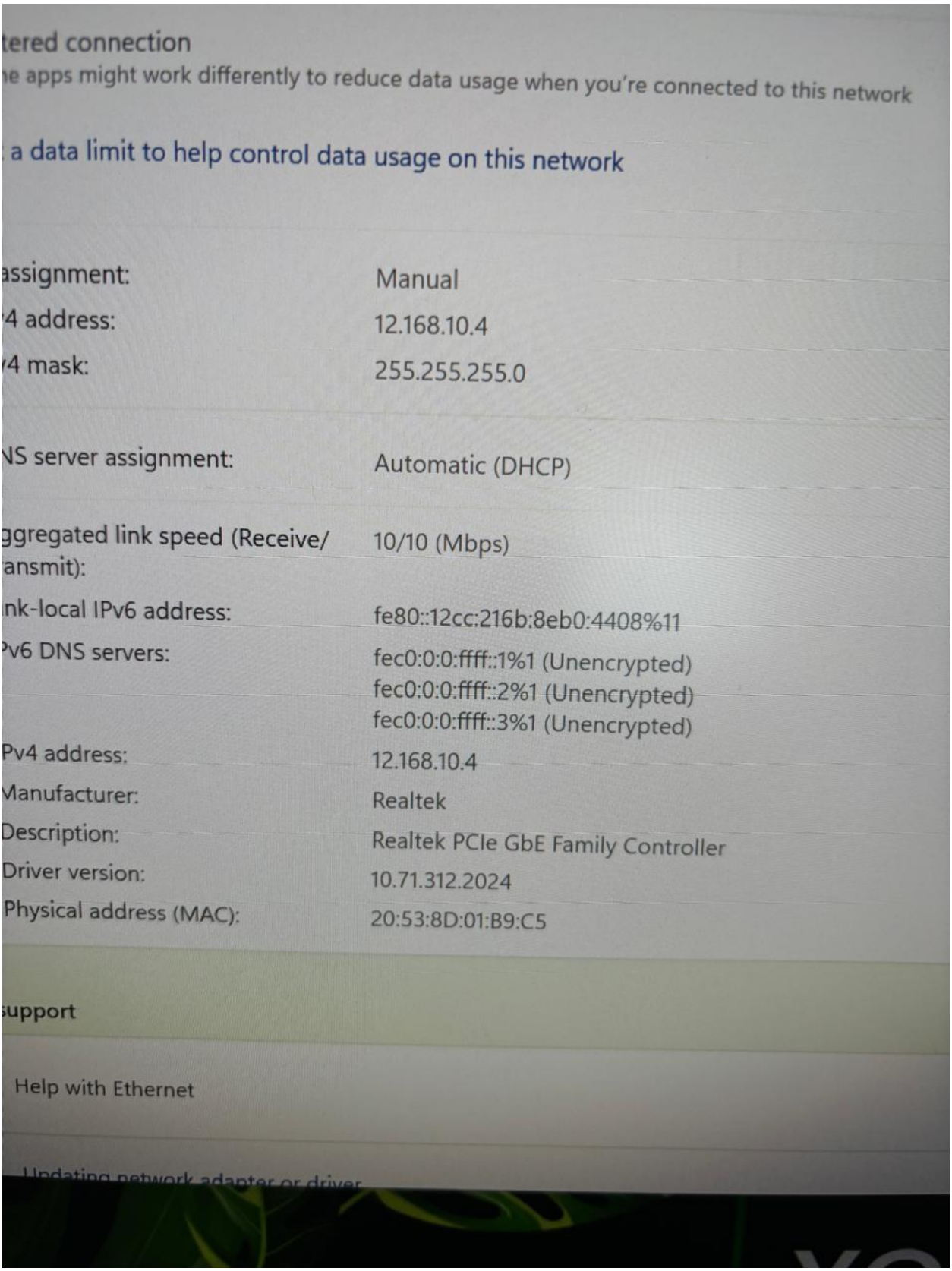
IP assignment:	Manual
IPv4 address:	192.168.20.6
IPv4 mask:	255.255.255.0
DNS server assignment:	Automatic (DHCP)
SSID:	
Protocol:	Any radio type
Security type:	Unknown
Manufacturer:	Intel Corporation
Description:	Intel(R) Wi-Fi 6 AX201 160MHz
Driver version:	24.50.0.4
Aggregated link speed (Receive/ Transmit):	2402/2402 (Mbps)
Link-local IPv6 address:	fe80::354b:db7e:c8f4:f925%15
IPv6 DNS servers:	fec0:0:0:ffff::1%1 (Unencrypted) fec0:0:0:ffff::2%1 (Unencrypted) fec0:0:0:ffff::3%1 (Unencrypted)
Physical address (MAC):	64:4A:7D:58:D7:04

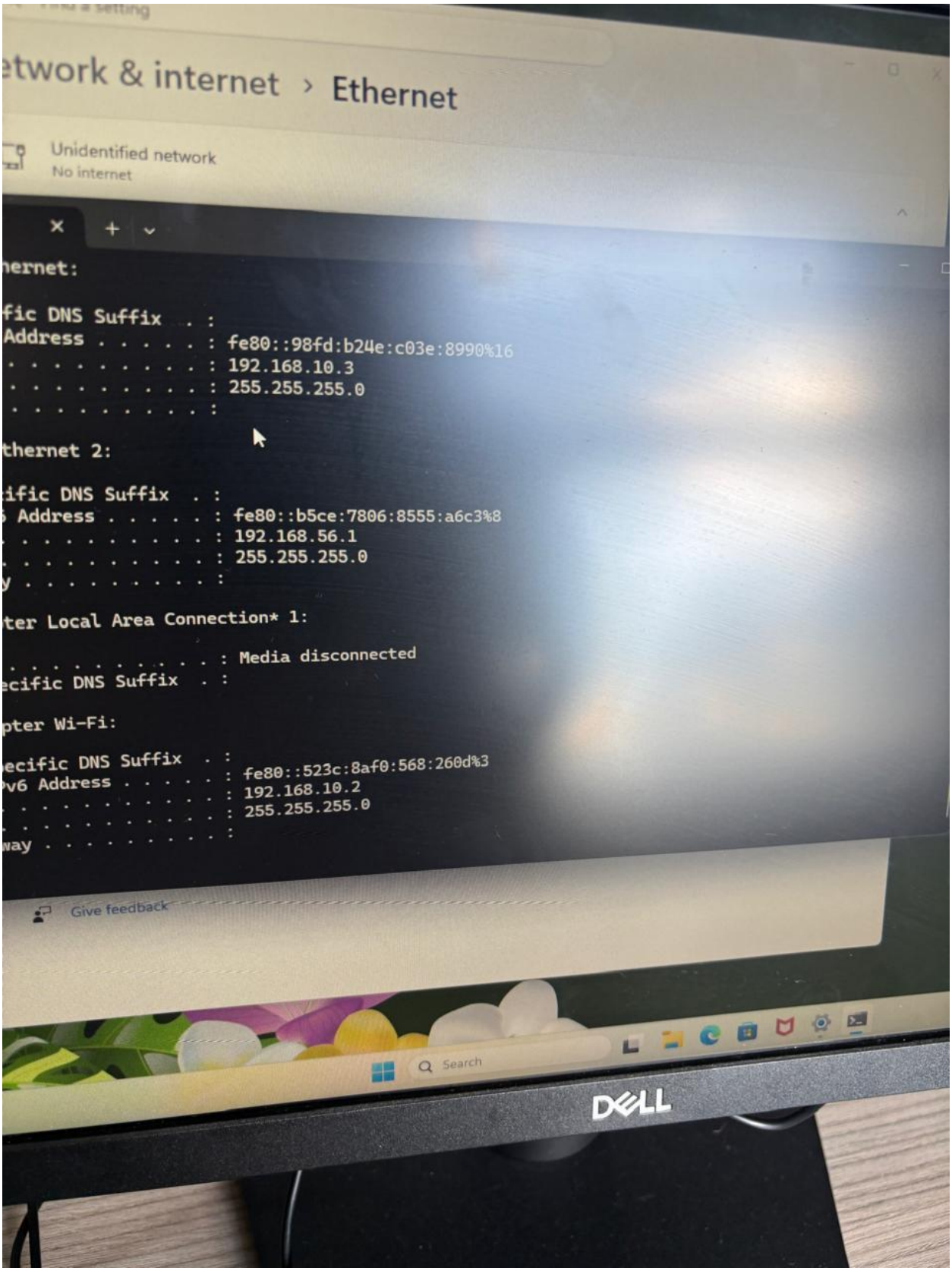
Advanced Wi-Fi network properties

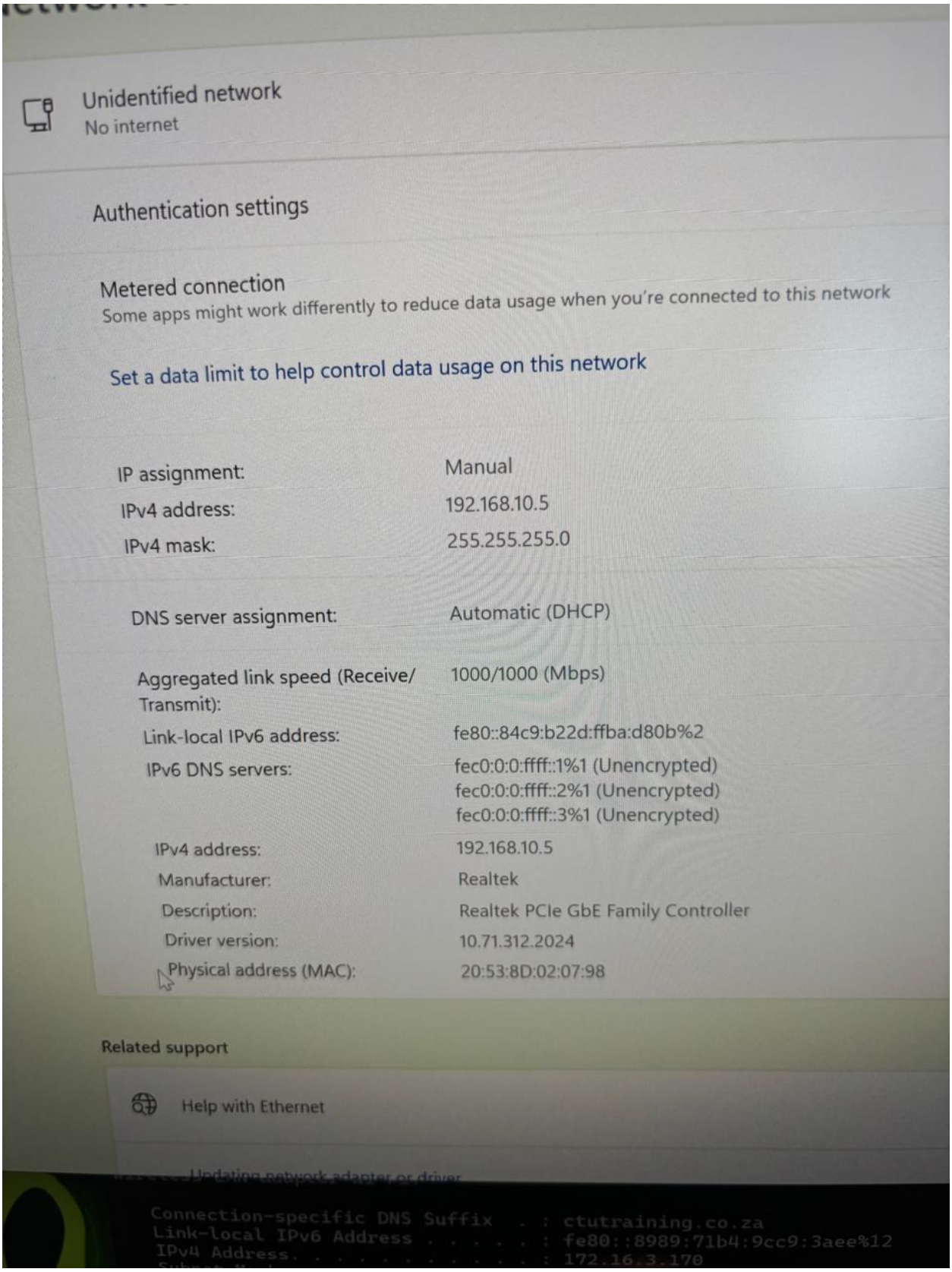
Related support

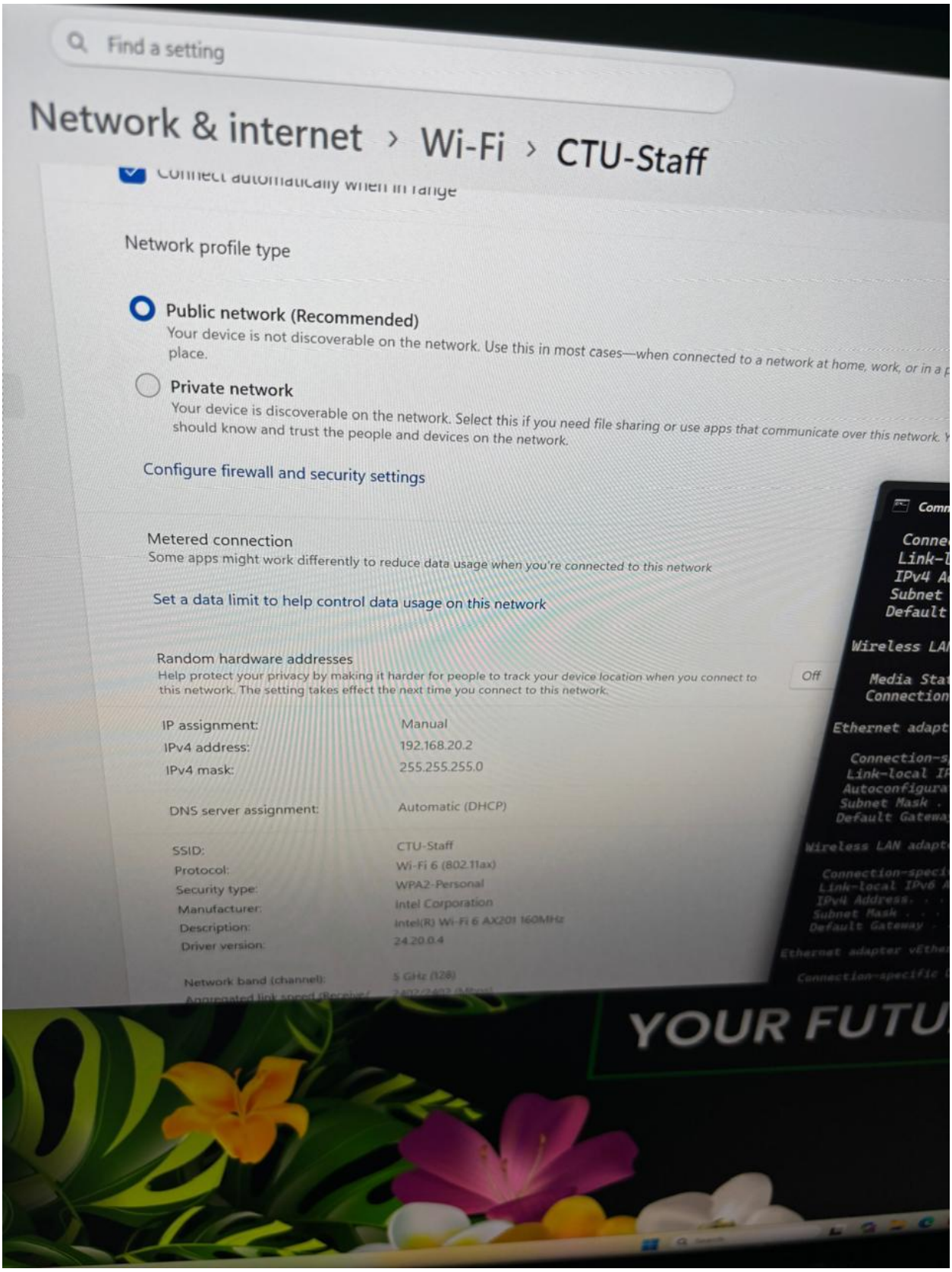

Connect automatically when in range
work profile type
Public network (Recommended)
Your device is not discoverable on the network. Use this in most cases—when connected to a network
place.
Private network
Your device is discoverable on the network. Select this if you need file sharing or use apps that commu
should know and trust the people and devices on the network.
Configure firewall and security settings
Metered connection
Some apps might work differently to reduce data usage when you're connected to this network
Set a data limit to help control data usage on this network
Random hardware addresses
Help protect your privacy by making it harder for people to track your device location when you connect to
this network. The setting takes effect the next time you connect to this network.
IP assignment: Manual
IPv4 address: 192.168.20.5
IPv4 mask: 255.255.255.0
DNS server assignment: Automatic (DHCP)
Advanced Wi-Fi network properties
Related support
Help from the web

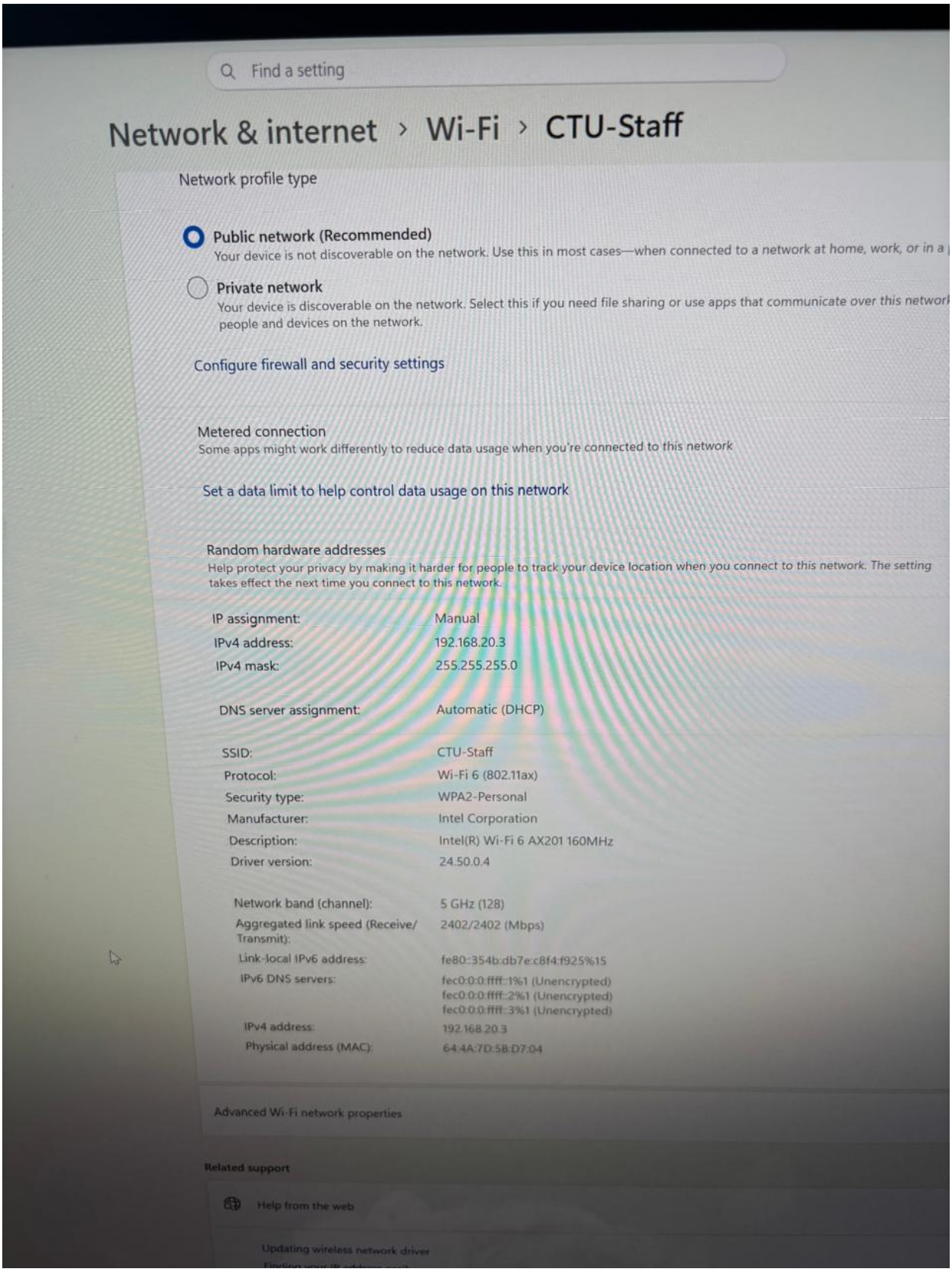

tion
hernet:
ific DNS Suffix . :
Address . . . . . : fe80::84c9:b22d:ffba:d80b%2
. . . . . : 192.168.20.5
. . . . . : 255.255.255.0
. . . . . :
ter Local Area Connection* 1:
. . . . . : Media disconnected
ecific DNS Suffix . :
apter Local Area Connection* 2:
. . . . . : Media disconnected
specific DNS Suffix . :
dapter Wi-Fi:
-specific DNS Suffix . : ctutraining.co.za
IPv6 Address . . . . . : fe80::8989:71b4:9cc9:3aee%12
ss. . . . . : 172.16.3.170
k . . . . . : 255.255.255.0
ateway . . . . . : 172.16.3.1
Search

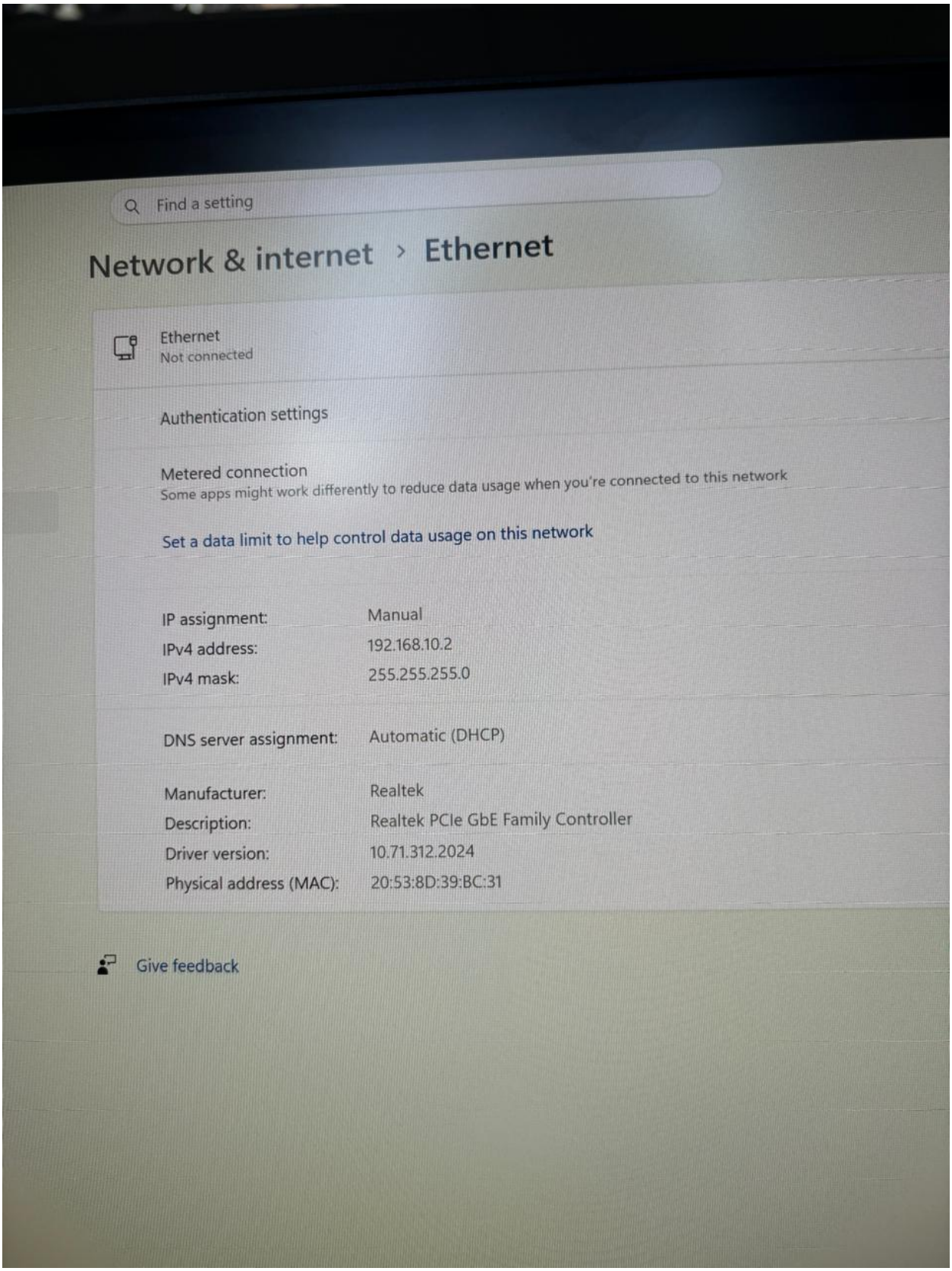

tered connection
ne apps might work differently to reduce data usage when you're connected to this network
a data limit to help control data usage on this network
assignment: Manual
4 address: 12.168.10.4
4 mask: 255.255.255.0
NS server assignment: Automatic (DHCP)
ggregated link speed (Receive/ ansmit): 10/10 (Mbps)
nk-local IPv6 address: fe80::12cc:216b:8eb0:4408%11
v6 DNS servers: fec0:0:0:ffff::1%1 (Unencrypted)
fec0:0:0:ffff::2%1 (Unencrypted)
fec0:0:0:ffff::3%1 (Unencrypted)
Pv4 address: 12.168.10.4
Manufacturer: Realtek
Description: Realtek PCIe GbE Family Controller
Driver version: 10.71.312.2024
Physical address (MAC): 20:53:8D:01:B9:C5
upport
Help with Ethernet
Updating network adapter or driver

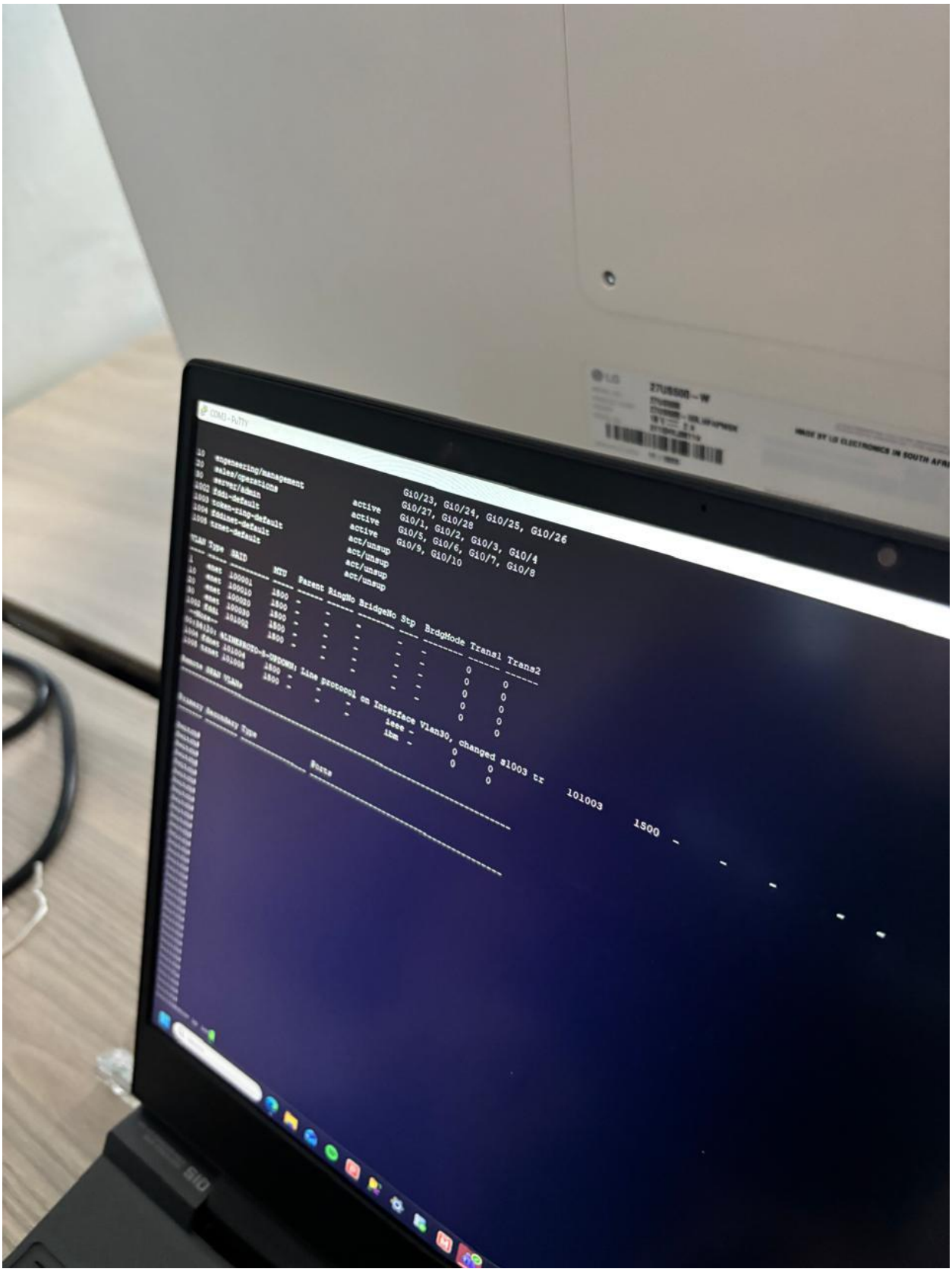

etwork & internet › Ethernet
Unidentified network
No internet
hernet:
fic DNS Suffix . :
Address . . . . . : fe80::98fd:b24e:c03e:8990%16
. . . . . : 192.168.10.3
. . . . . : 255.255.255.0
. . . . . :
thernet 2:
ific DNS Suffix . :
Address . . . . . : fe80::b5ce:7806:8555:a6c3%8
. . . . . : 192.168.56.1
. . . . . : 255.255.255.0
y . . . . . :
ter Local Area Connection* 1:
. . . . . : Media disconnected
ecific DNS Suffix . :
pter Wi-Fi:
ecific DNS Suffix . :
v6 Address . . . . . : fe80::523c:8af0:568:260d%3
. . . . . : 192.168.10.2
. . . . . : 255.255.255.0
way . . . . . :
Give feedback
Search
DELL

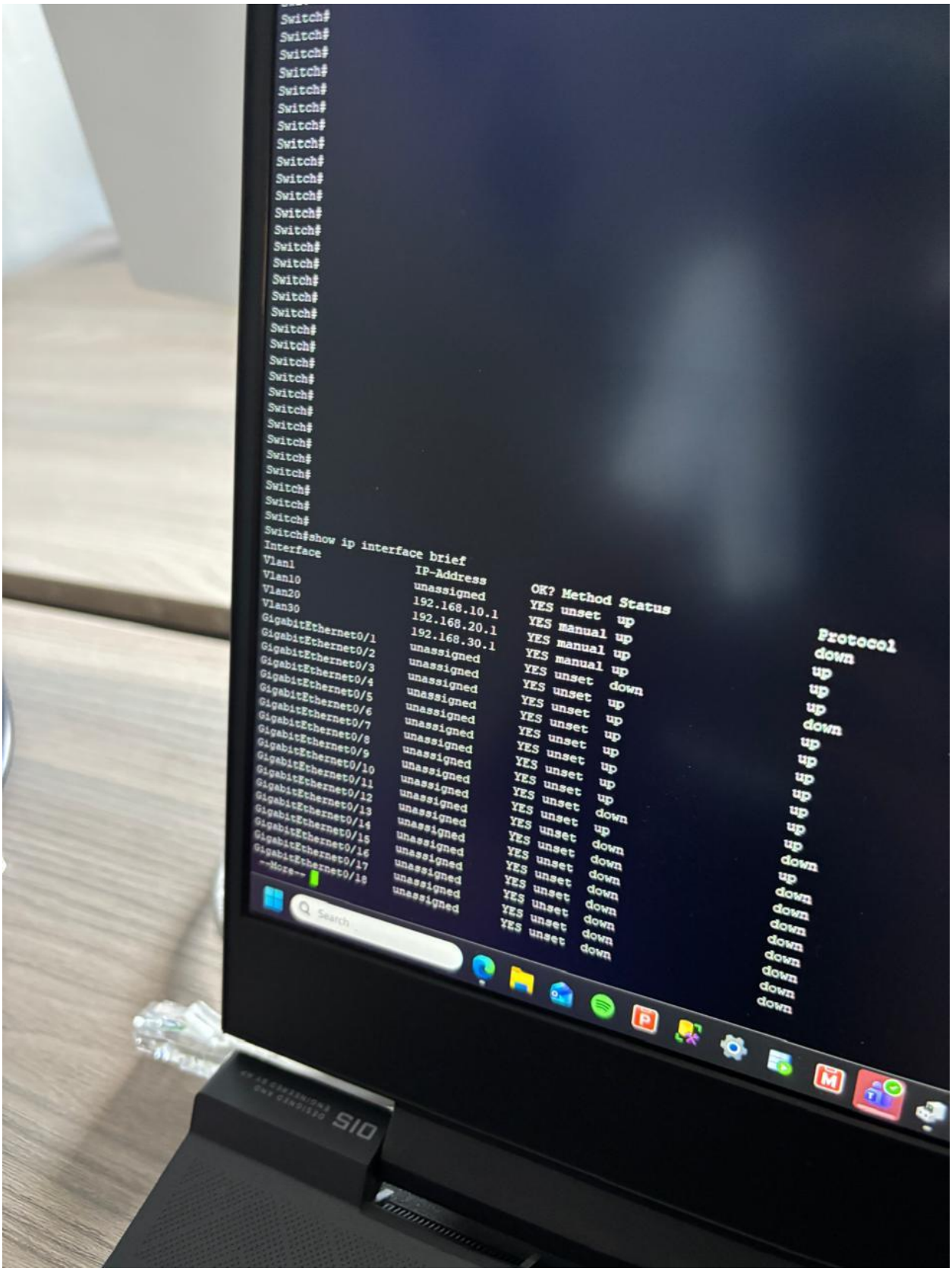

Unidentified network
No internet
Authentication settings
Metered connection
Some apps might work differently to reduce data usage when you're connected to this network
Set a data limit to help control data usage on this network
IP assignment: Manual
IPv4 address: 192.168.10.5
IPv4 mask: 255.255.255.0
DNS server assignment: Automatic (DHCP)
Aggregated link speed (Receive/ Transmit): 1000/1000 (Mbps)
Link-local IPv6 address: fe80::84c9:b22d:ffba:d80b%2
IPv6 DNS servers: fec0:0:0:ffff::1%1 (Unencrypted)
fec0:0:0:ffff::2%1 (Unencrypted)
fec0:0:0:ffff::3%1 (Unencrypted)
IPv4 address: 192.168.10.5
Manufacturer: Realtek
Description: Realtek PCIe GbE Family Controller
Driver version: 10.71.312.2024
Physical address (MAC): 20:53:8D:02:07:98
Related support
Help with Ethernet
Updating network adapter or driver
Connection-specific DNS Suffix . : ctutraining.co.za
Link-local IPv6 Address . . . . . : fe80::8989:71b4:9cc9:3aee%12
IPv4 Address. . . . . : 172.16.3.170


Find a setting
Network & internet › Wi-Fi › CTU-Staff
Network profile type
Public network (Recommended)
Your device is not discoverable on the network. Use this in most cases—when connected to a network at home, work, or in a p
place.
Private network
Your device is discoverable on the network. Select this if you need file sharing or use apps that communicate over this network. Y
should know and trust the people and devices on the network.
Configure firewall and security settings
Metered connection
Some apps might work differently to reduce data usage when you're connected to this network
Set a data limit to help control data usage on this network
Random hardware addresses
Help protect your privacy by making it harder for people to track your device location when you connect to this network. The setting takes effect the next time you connect to this network.
Off
IP assignment: Manual
IPv4 address: 192.168.20.2
IPv4 mask: 255.255.255.0
DNS server assignment: Automatic (DHCP)
SSID: CTU-Staff
Protocol: Wi-Fi 6 (802.11ax)
Security type: WPA2-Personal
Manufacturer: Intel Corporation
Description: Intel(R) Wi-Fi 6 AX201 160MHz
Driver version: 24.20.0.4
Network band (channel): 5 GHz (128)
YOUR FUTU


Find a setting
Network & internet › Wi-Fi › CTU-Staff
Network profile type
Public network (Recommended)
Your device is not discoverable on the network. Use this in most cases—when connected to a network at home, work, or in a
Private network
Your device is discoverable on the network. Select this if you need file sharing or use apps that communicate over this networ
people and devices on the network.
Configure firewall and security settings
Metered connection
Some apps might work differently to reduce data usage when you're connected to this network
Set a data limit to help control data usage on this network
Random hardware addresses
Help protect your privacy by making it harder for people to track your device location when you connect to this network. The setting
takes effect the next time you connect to this network.
IP assignment: Manual
IPv4 address: 192.168.20.3
IPv4 mask: 255.255.255.0
DNS server assignment: Automatic (DHCP)
SSID: CTU-Staff
Protocol: Wi-Fi 6 (802.11ax)
Security type: WPA2-Personal
Manufacturer: Intel Corporation
Description: Intel(R) Wi-Fi 6 AX201 160MHz
Driver version: 24.50.0.4
Network band (channel): 5 GHz (128)
Aggregated link speed (Receive/ Transmit): 2402/2402 (Mbps)
Link-local IPv6 address: fe80::354b:db7e:c8f4:f925%15
IPv6 DNS servers: fec0:0:0:ffff::1%1 (Unencrypted)
fec0:0:0:ffff::2%1 (Unencrypted)
fec0:0:0:ffff::3%1 (Unencrypted)
IPv4 address: 192.168.20.3
Physical address (MAC): 64:4A:7D:5B:D7:04
Advanced Wi-Fi network properties
Related support
Help from the web
Updating wireless network driver


Find a setting
Network & internet › Ethernet
Ethernet
Not connected
Authentication settings
Metered connection
Some apps might work differently to reduce data usage when you're connected to this network
Set a data limit to help control data usage on this network
IP assignment: Manual
IPv4 address: 192.168.10.2
IPv4 mask: 255.255.255.0
DNS server assignment: Automatic (DHCP)
Manufacturer: Realtek
Description: Realtek PCIe GbE Family Controller
Driver version: 10.71.312.2024
Physical address (MAC): 20:53:8D:39:BC:31
Give feedback


engineering/management
sales/operations
secret/admin
fddi-default
token-ring-default
fddinet-default
trnet-default
active
active
active
act/unsup
act/unsup
act/unsup
act/unsup
Gi0/23, Gi0/24, Gi0/25, Gi0/26
Gi0/27, Gi0/28
Gi0/1, Gi0/2, Gi0/3, Gi0/4
Gi0/5, Gi0/6, Gi0/7, Gi0/8
Gi0/9, Gi0/10
VLAN Type SAID MTU Parent RingNo BridgeNo Stp BrdgMode Trans1 Trans2
1 enet 100001 1500 - - - - - 0 0
20 enet 100020 1500 - - - - - 0 0
30 enet 100030 1500 - - - - - 0 0
1002 fddi 101002 1500 - - - - - 0 0
1003 tr 101003 1500 - - - - - 0 0
1004 fdnet 101004 1500 - - - - - 0 0
1005 trnet 101005 1500 - - - - - 0 0
Remote SPAN VLANs
Primary Secondary Type Ports
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up
101003 1500 - - - ibm - 0 0 0


Switch#
Switch#
Switch#

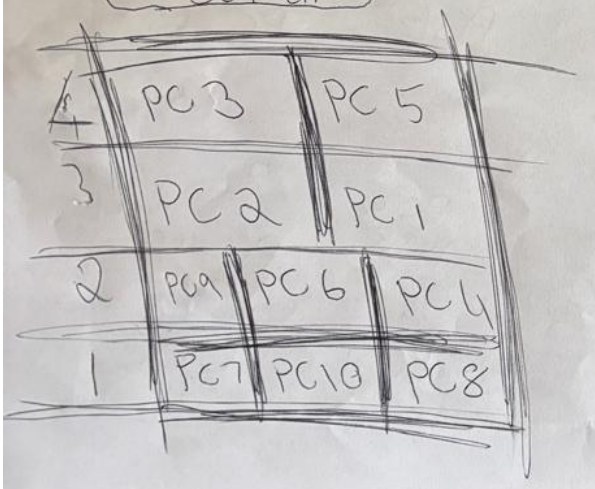

D-Link

~~Mdm@ctu~~car

mdm@ctucareer.co.za

CTU@2026

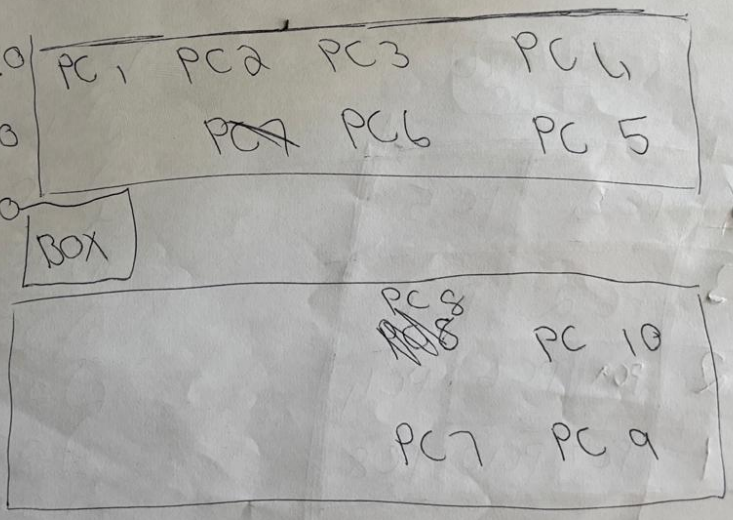
PATCH PANEL SET UP

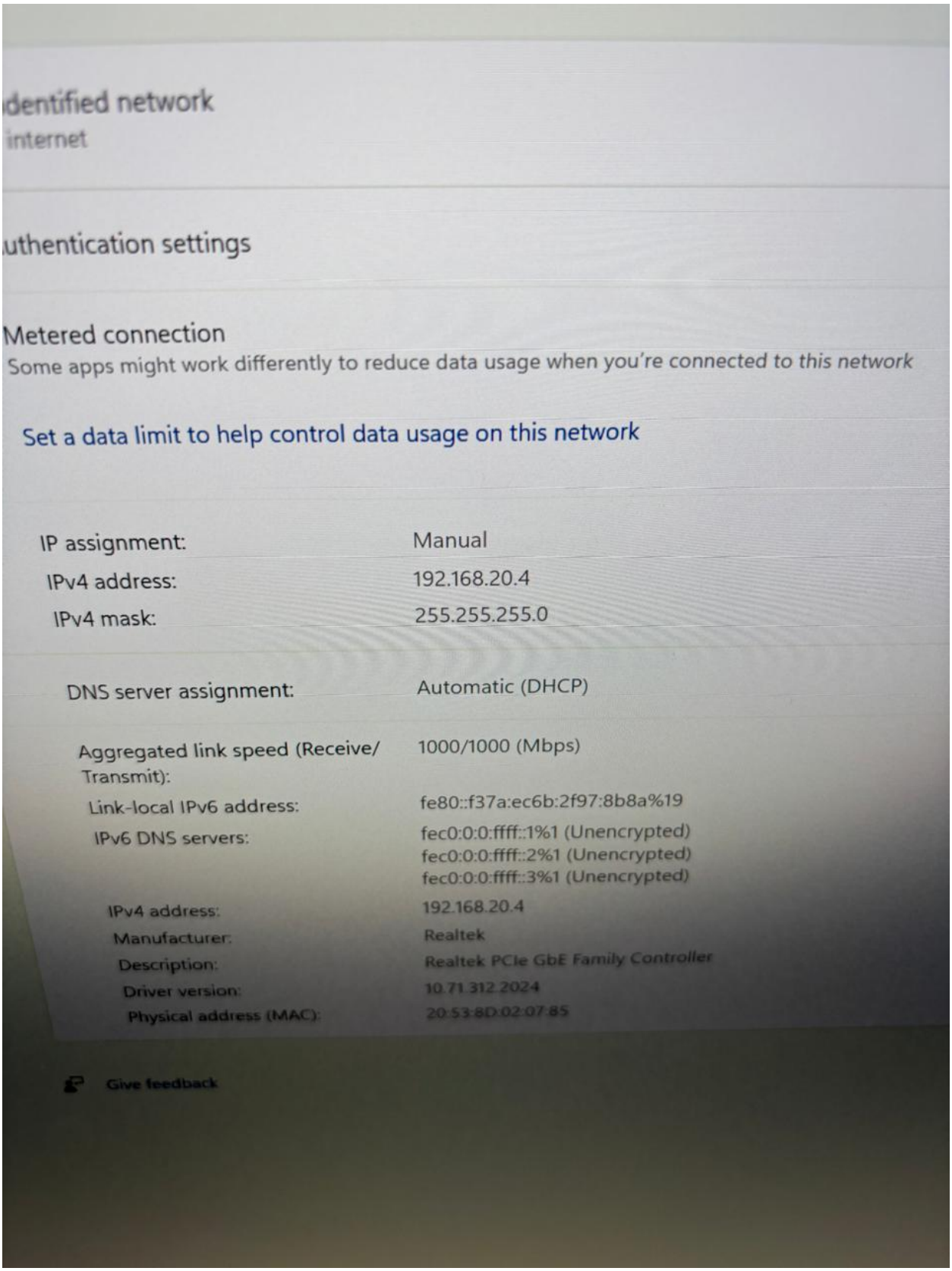


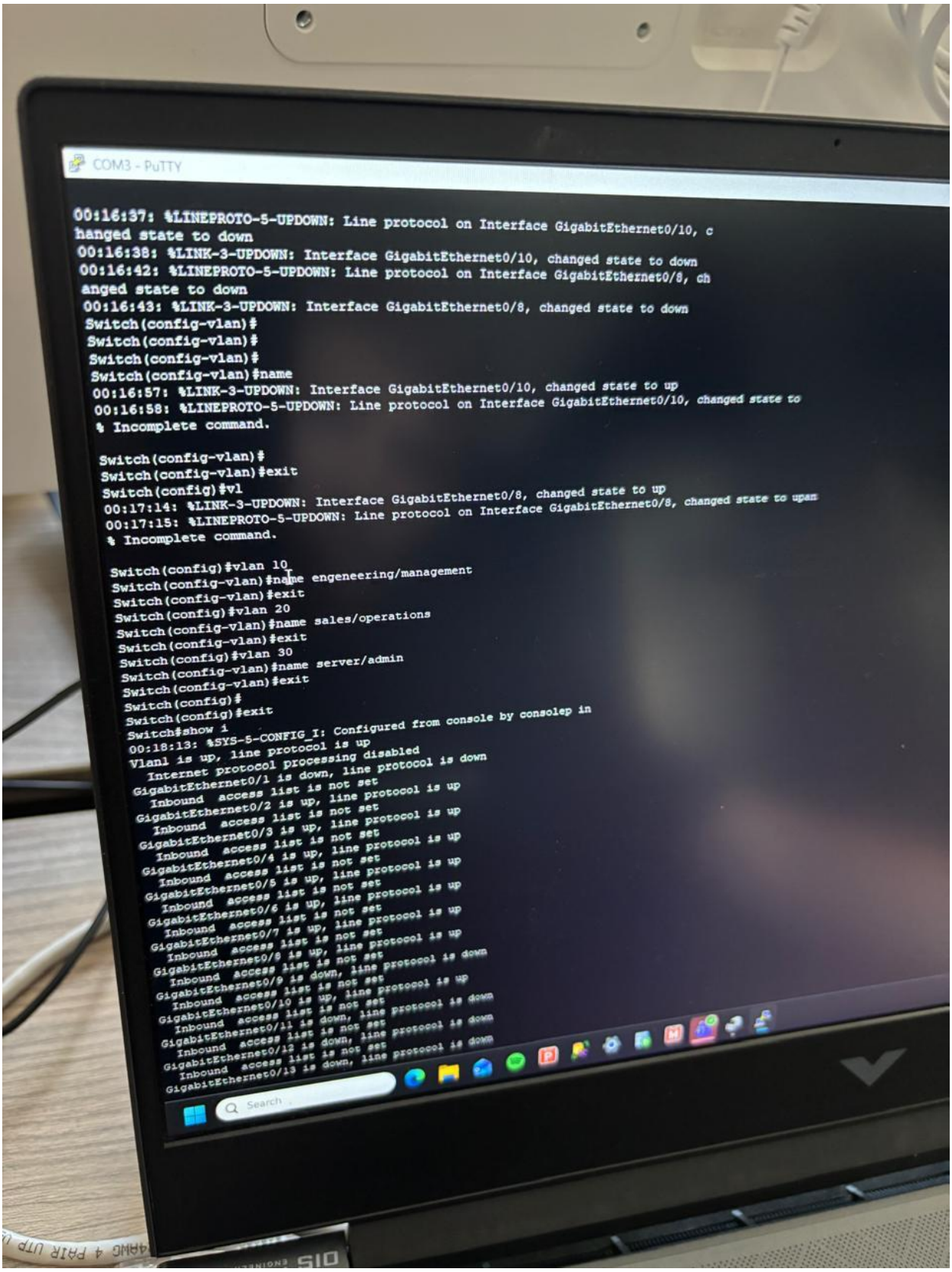
1) 10.2
2) 10.3
3) 10.4
4) 10.5
5) 20.2
6) 20.3
7) 20.4
8) 20.5
9) 30.2
10) 30.3

Vlan 10 (engeneering/management) PC 1-4
Vlan 20 (Sales/Operations) PC 5-8
Vlan 30 (Server/Admin) PC 9-10

PC 1-4: 192.168.10.0
PC 5-8: 192.168.20.0
PC 9-10: 192.168.30.0

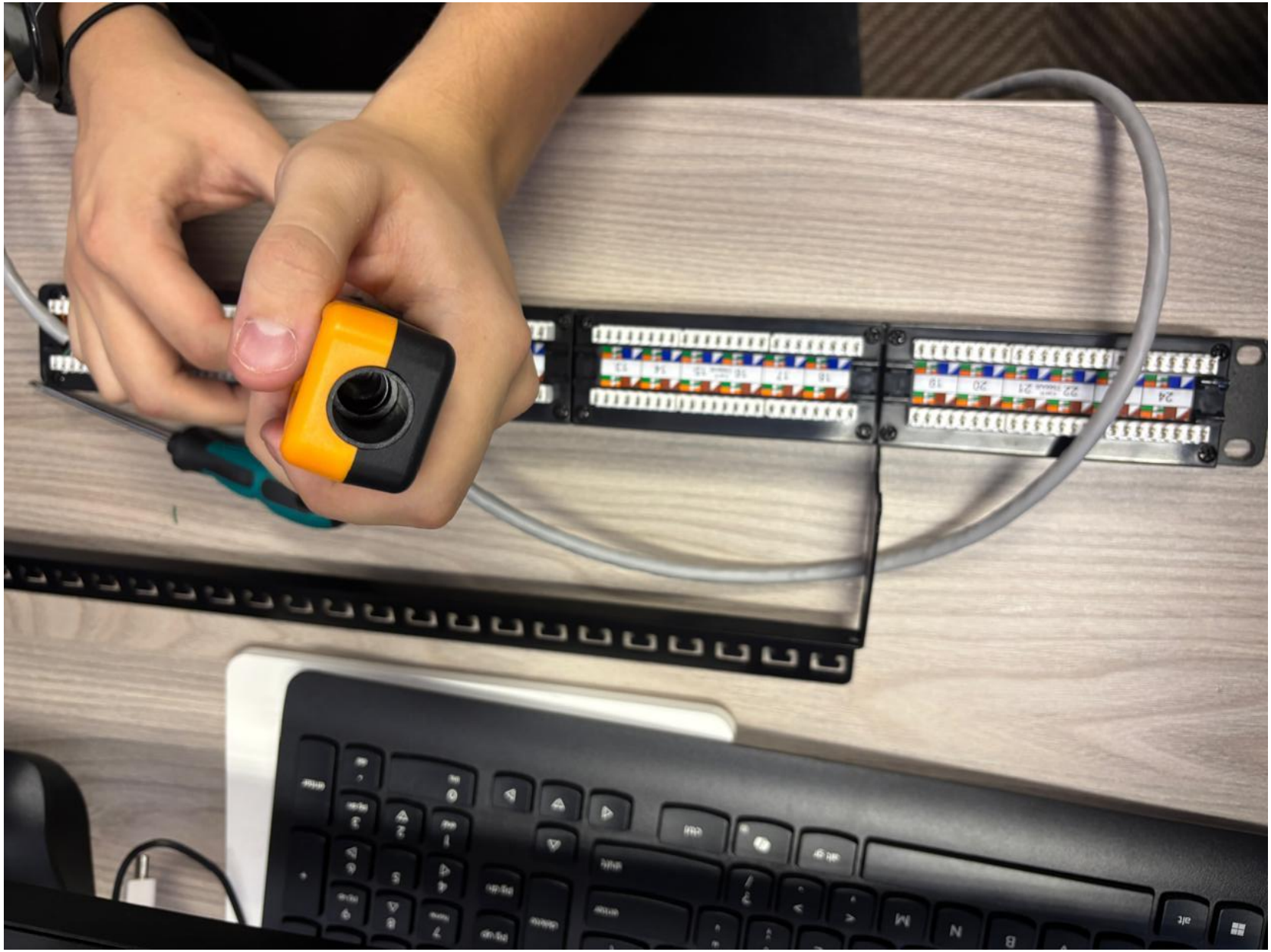


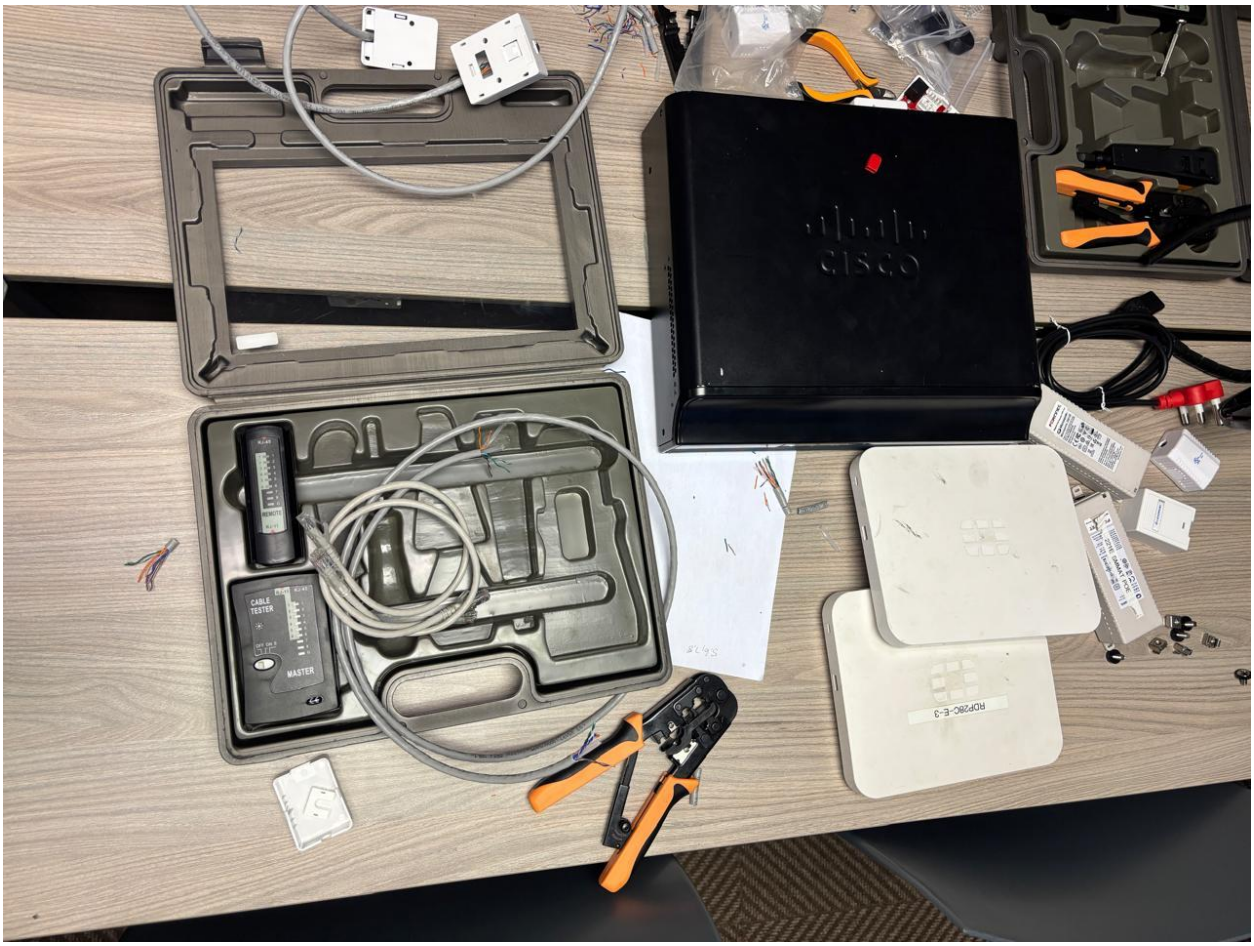

dentified network
internet
uthentication settings
Metered connection
Some apps might work differently to reduce data usage when you're connected to this network
Set a data limit to help control data usage on this network
IP assignment: Manual
IPv4 address: 192.168.20.4
IPv4 mask: 255.255.255.0
DNS server assignment: Automatic (DHCP)
Aggregated link speed (Receive/ Transmit): 1000/1000 (Mbps)
Link-local IPv6 address: fe80::f37a:ec6b:2f97:8b8a%19
IPv6 DNS servers: fec0:0:0:ffff::1%1 (Unencrypted)
fec0:0:0:ffff::2%1 (Unencrypted)
fec0:0:0:ffff::3%1 (Unencrypted)
IPv4 address: 192.168.20.4
Manufacturer: Realtek
Description: Realtek PCIe GbE Family Controller
Driver version: 10.71.312.2024
Physical address (MAC): 20-53-8D-02-07-85
Give feedback


COM3 - PuTTY
00:16:37: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/10, c
hanged state to down
00:16:38: %LINK-3-UPDOWN: Interface GigabitEthernet0/10, changed state to down
00:16:42: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/8, ch
anged state to down
00:16:43: %LINK-3-UPDOWN: Interface GigabitEthernet0/8, changed state to down
Switch(config-vlan)#
Switch(config-vlan)#
Switch(config-vlan)#
Switch(config-vlan)#name
00:16:57: %LINK-3-UPDOWN: Interface GigabitEthernet0/10, changed state to up
00:16:58: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/10, changed state to
% Incomplete command.
Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#vl
00:17:14: %LINK-3-UPDOWN: Interface GigabitEthernet0/8, changed state to up
00:17:15: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/8, changed state to upan
% Incomplete command.
Switch(config)#vlan 10
Switch(config-vlan)#name engeneering/management
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name sales/operations
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name server/admin
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#exit
Switch#show i
00:18:13: %SYS-5-CONFIG_I: Configured from console by consolep in
Vlan1 is up, line protocol is up
Internet protocol processing disabled
GigabitEthernet0/1 is down, line protocol is down
Inbound access list is not set
GigabitEthernet0/2 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/3 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/4 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/5 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/6 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/7 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/8 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/9 is down, line protocol is down
Inbound access list is not set
GigabitEthernet0/10 is up, line protocol is up
Inbound access list is not set
GigabitEthernet0/11 is down, line protocol is down
Inbound access list is not set
GigabitEthernet0/12 is down, line protocol is down
Inbound access list is not set
GigabitEthernet0/13 is down, line protocol is down
Search


CONSOLE






cisco
CABLE TESTER
MASTER

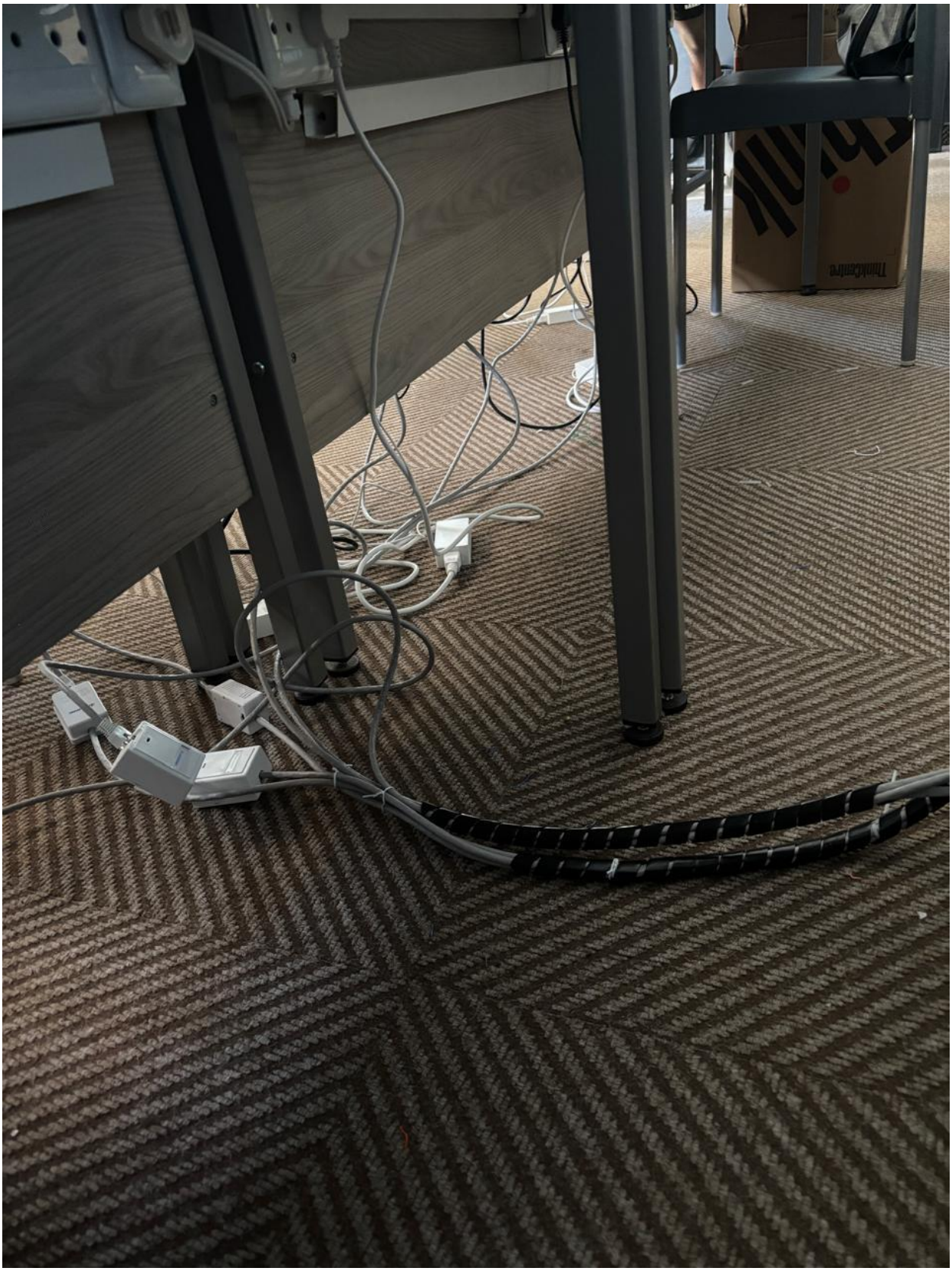

cisco
cisco Cisco 4300 Series
cisco Cisco 4300 Series
Cat 6
FORTINET


Connect to Configure
RTINET

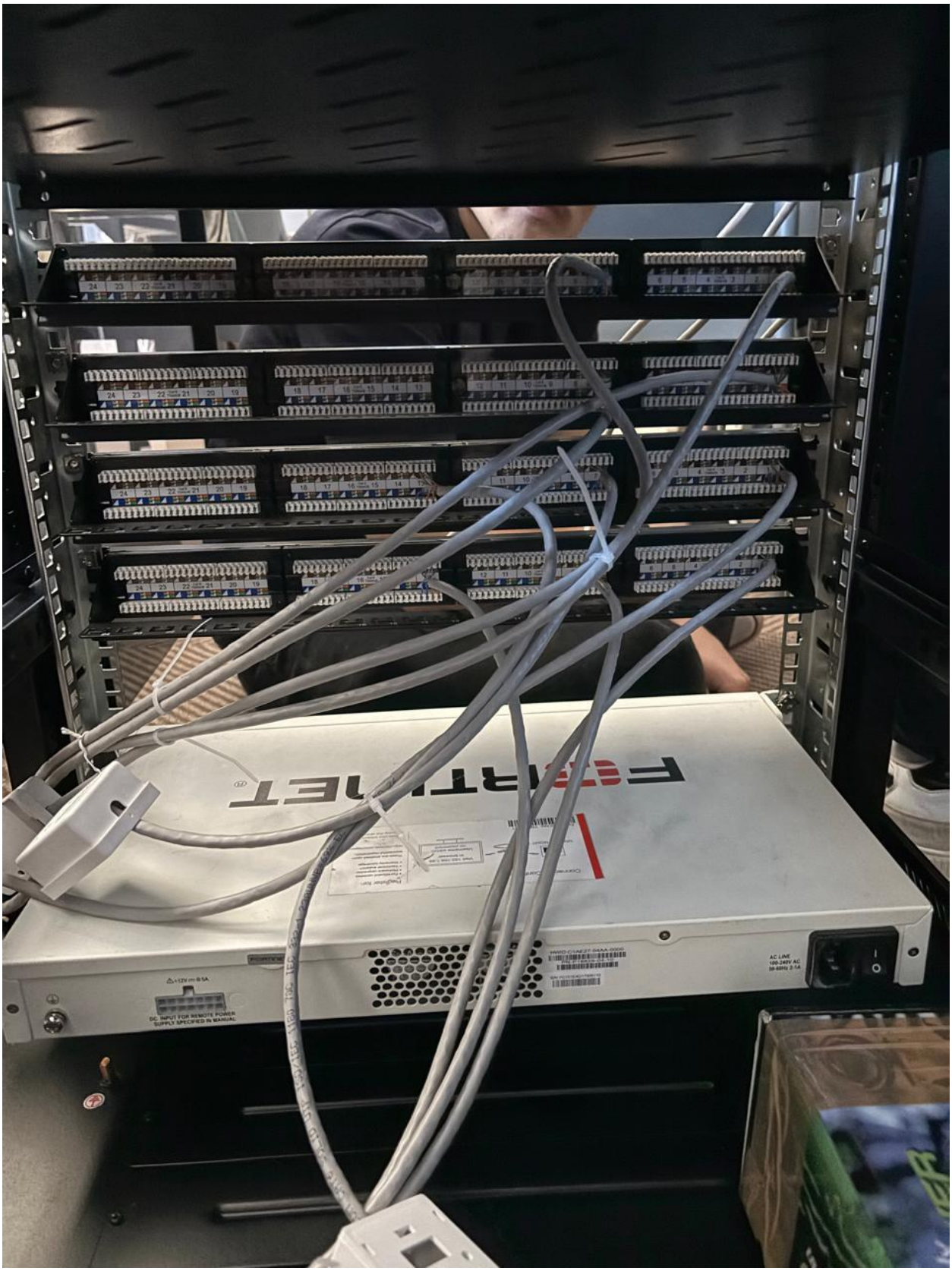

CISCO
Cisco 4300 Series
CISCO
Cisco 4300 Series
Cat 6
Cat 6
Cat 6
Cat 6
FORTINET
FORTINET
FortiGate 101E

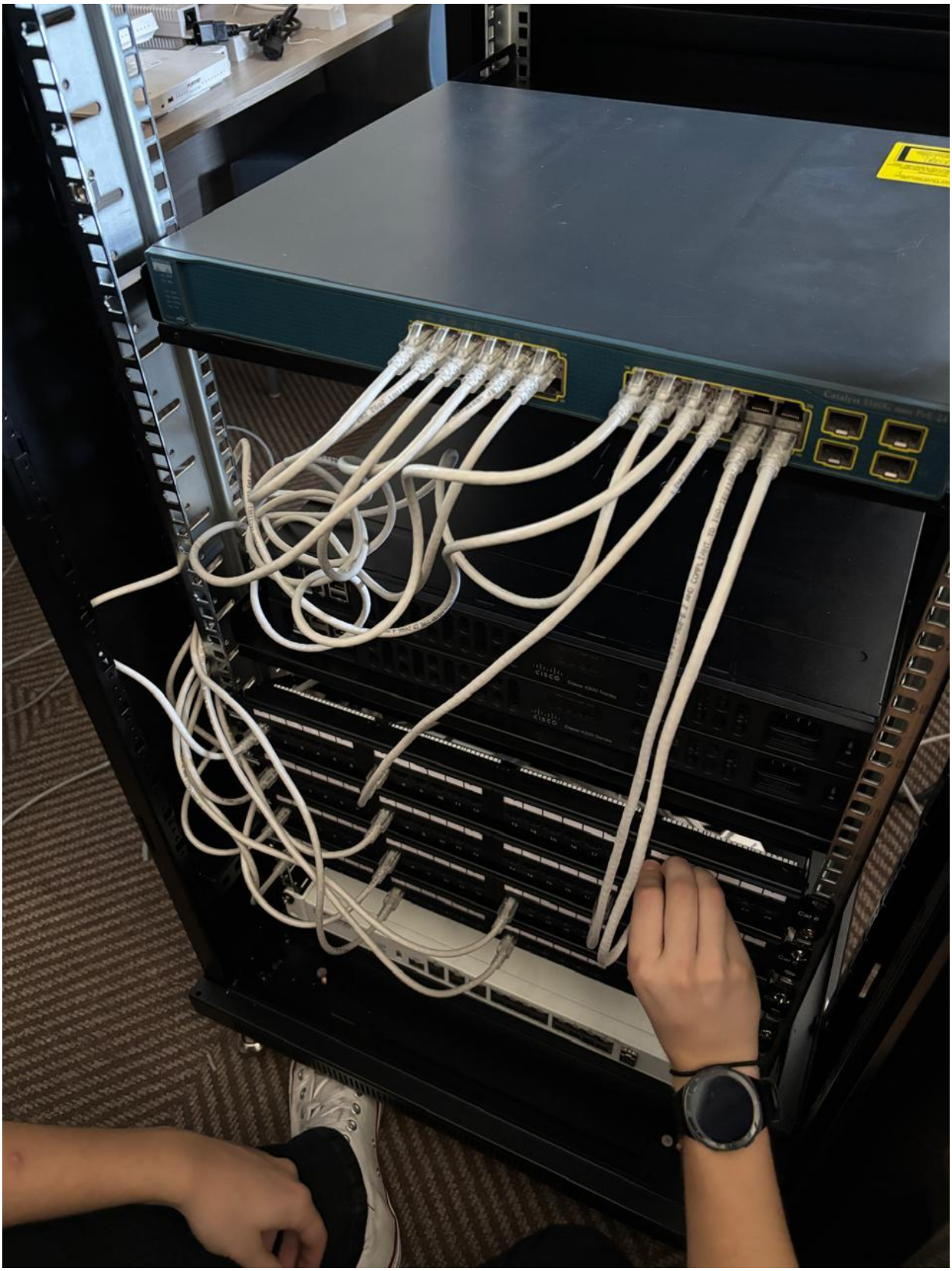

FORTINET
GREEN POWER



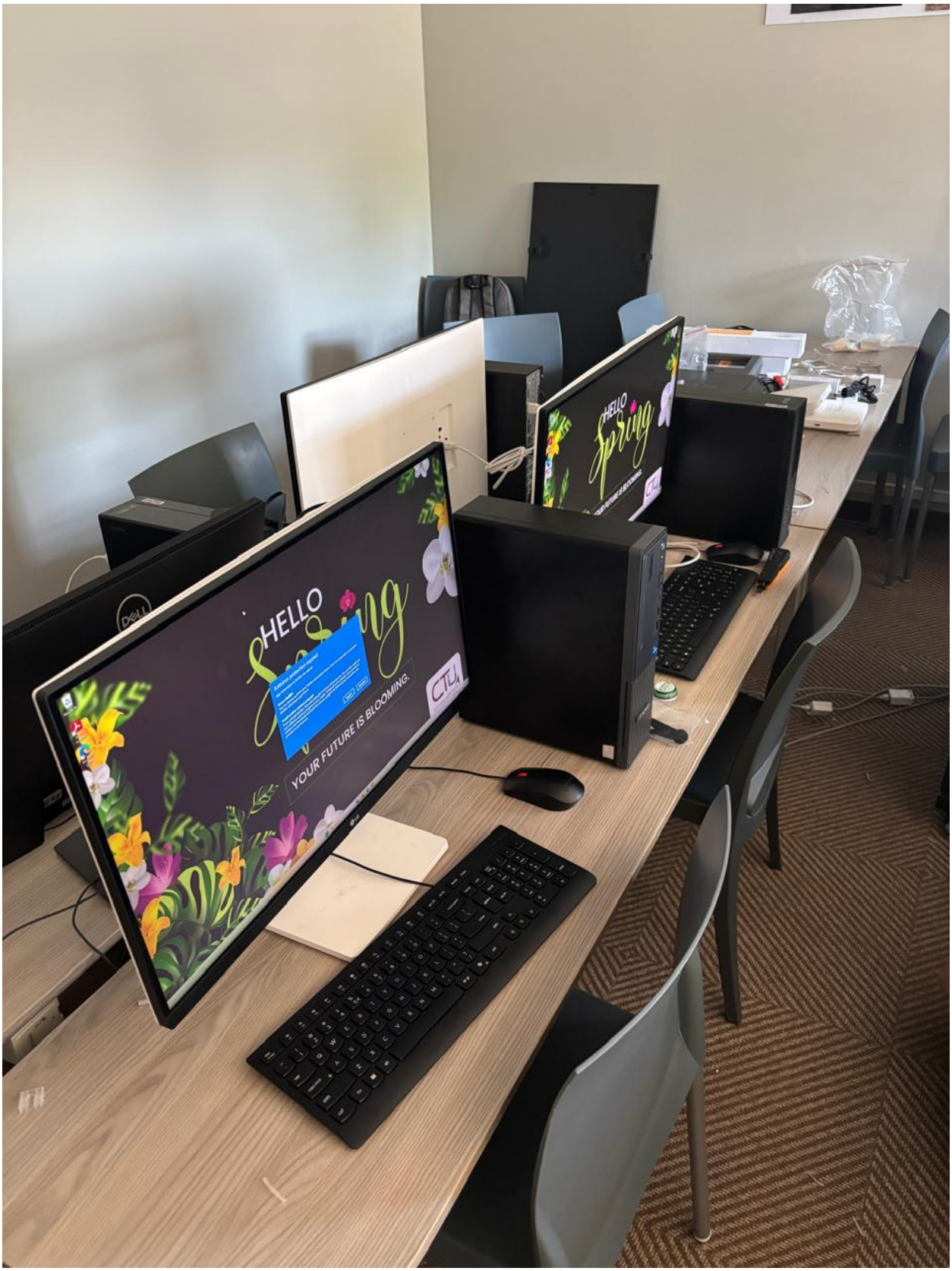



D-Link
DGS-1024D
FORTINET
FORTINET
FortiGate 101E


FORTINET
DC INPUT FOR REMOTE POWER SUPPLY SPECIFIED IN MANUAL
AC LINE
100-240V AC


cisco
cisco


hello spring
YOUR FUTURE IS BLOOMING.


HELLO
YOUR FUTURE IS BLOOMING.
CTU






DELL






umbro
DELL
RE IS BLOOMING.
MDM

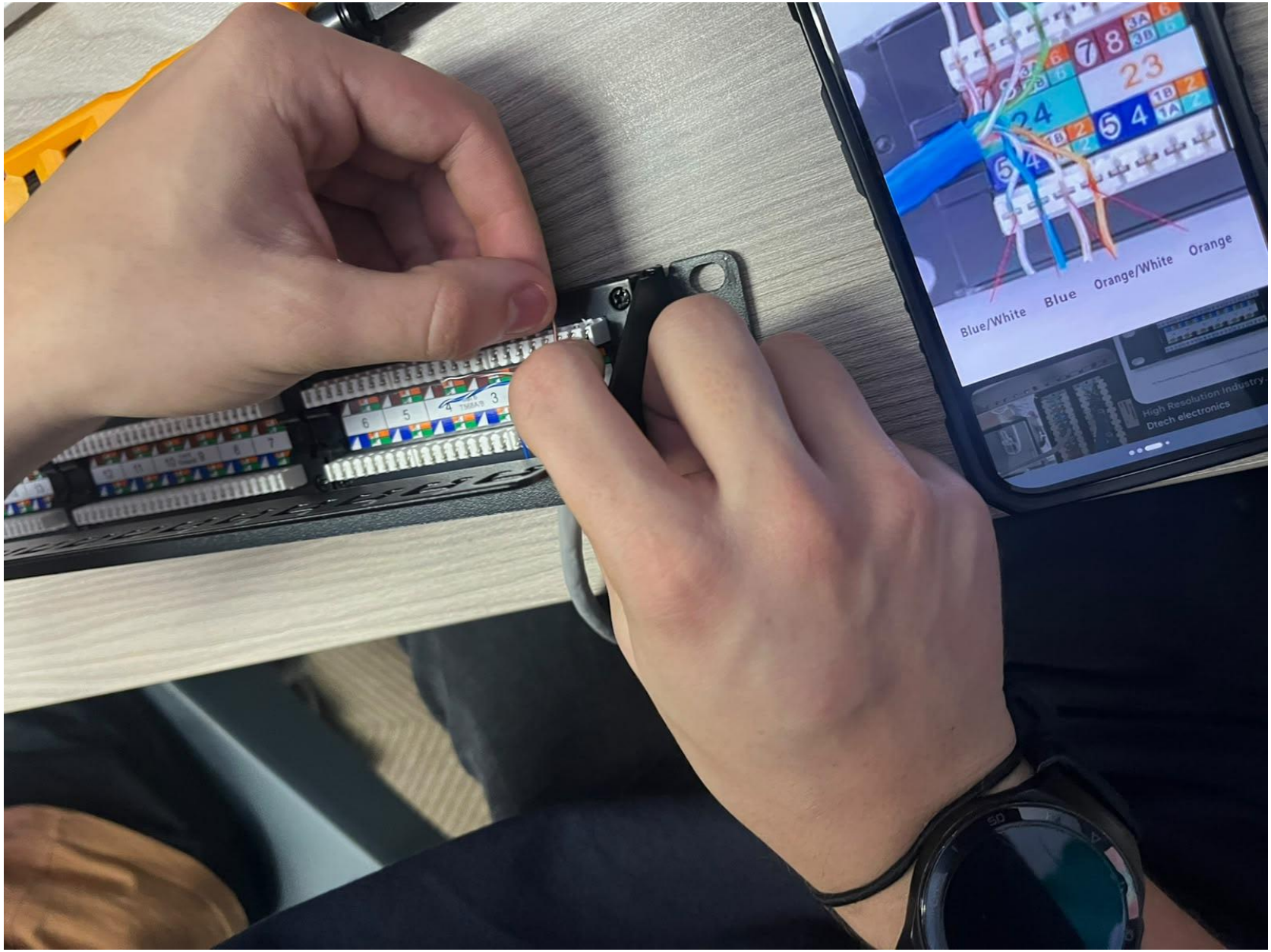


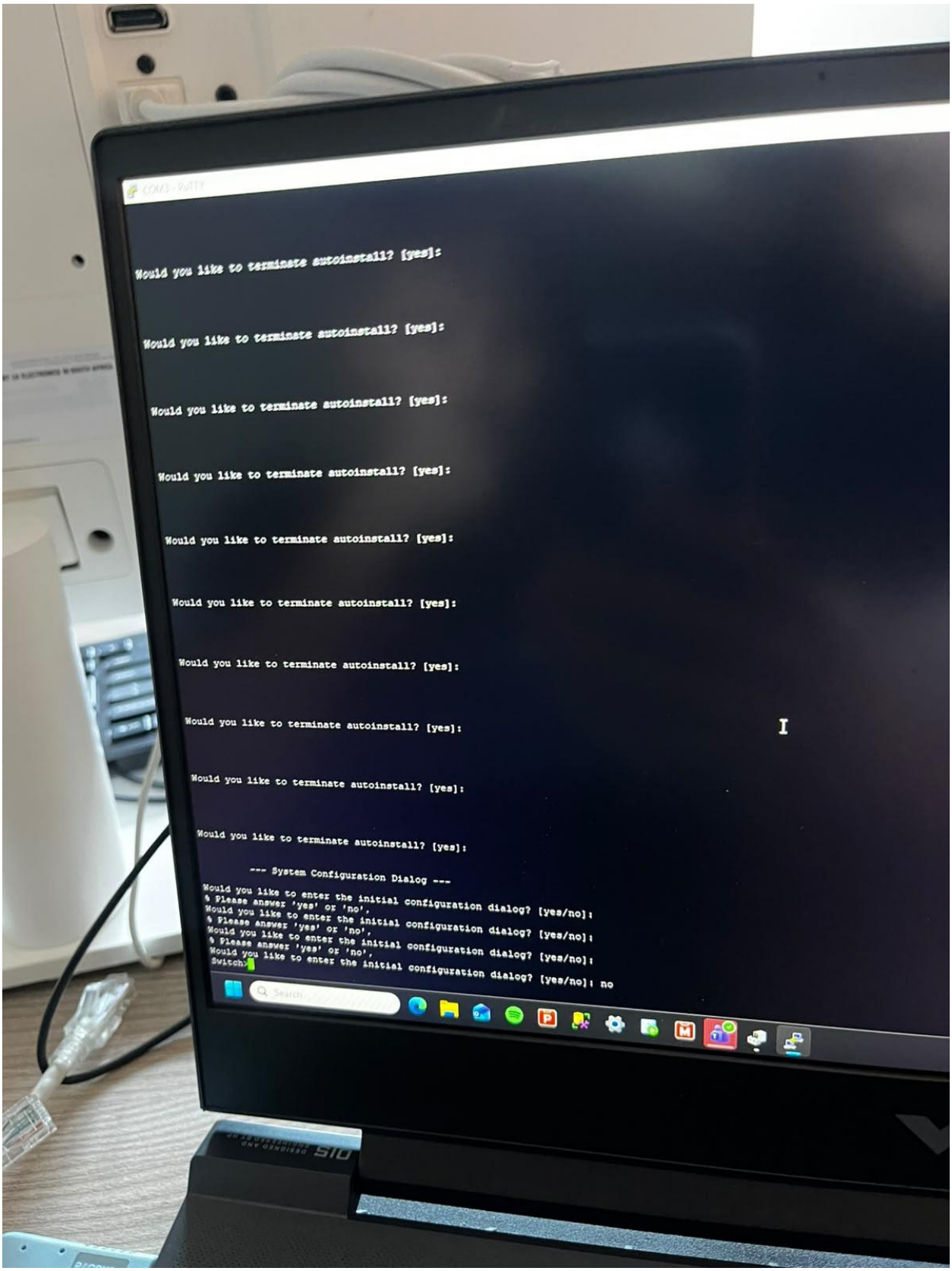
10.2a

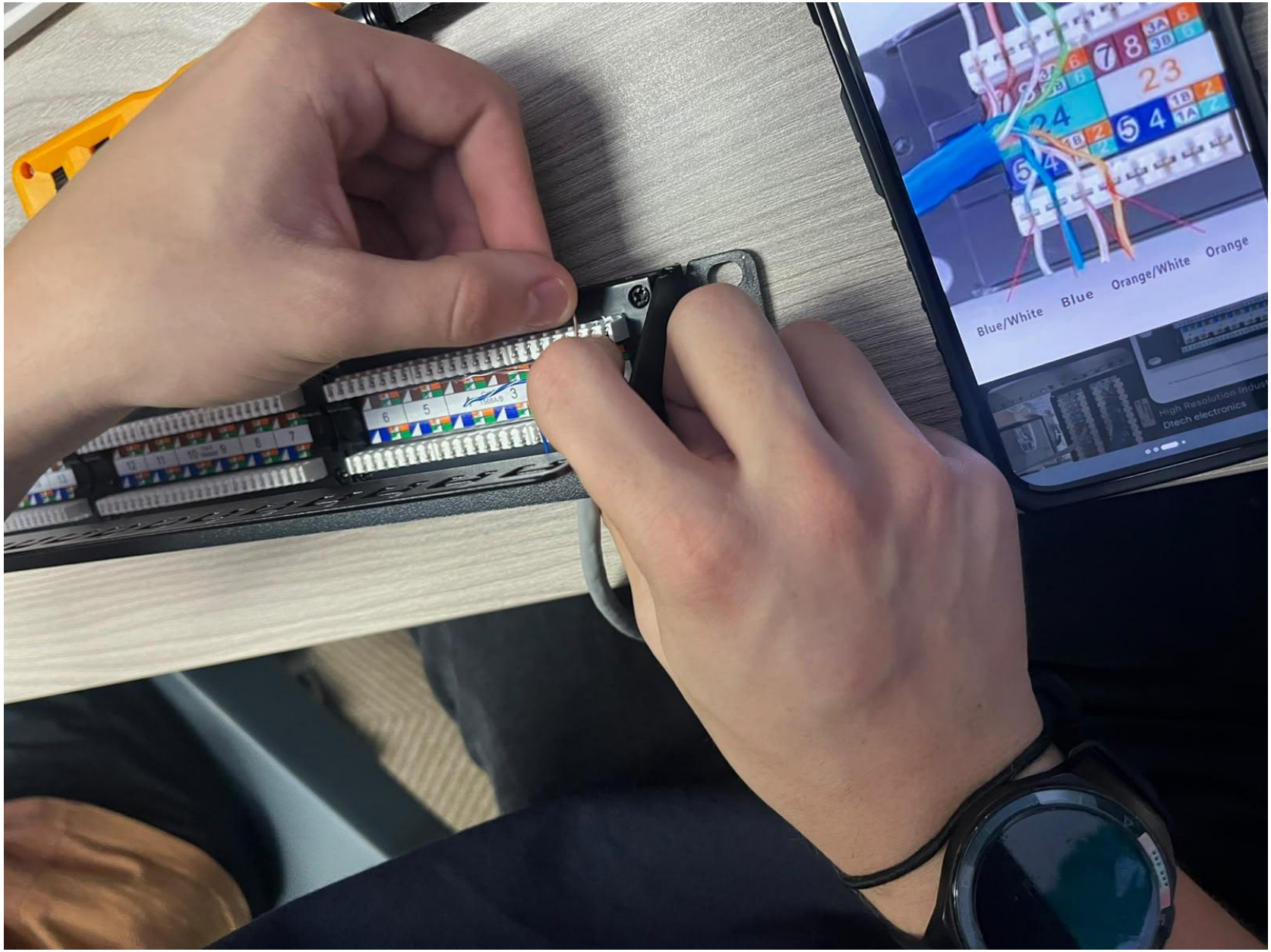
1) 10.2
2) 10.3
3) 10.4
4) 10.5
5) 20.2
6) 20.3
7) 20.4
8) 20.5
9) 30.2
10) 30.3

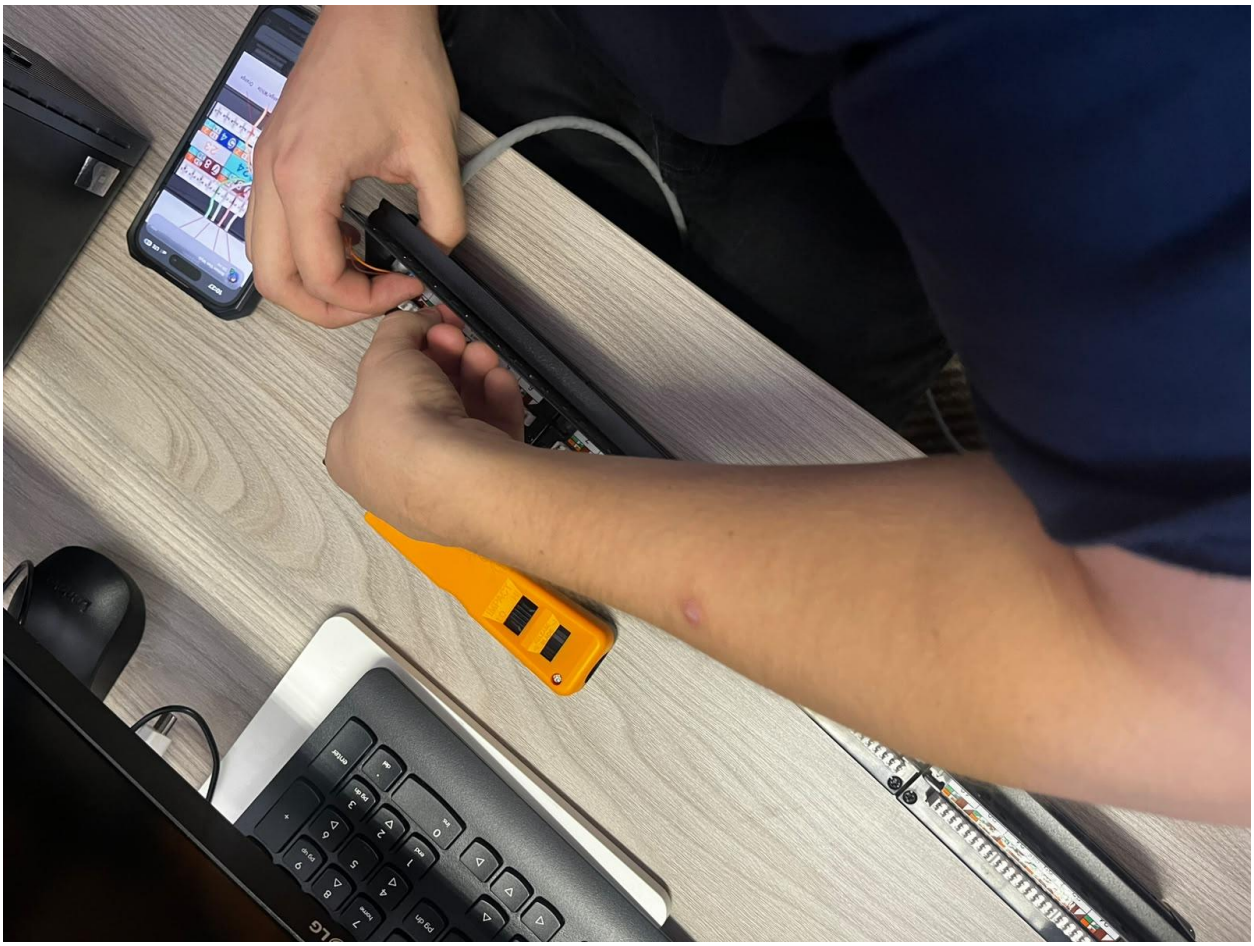




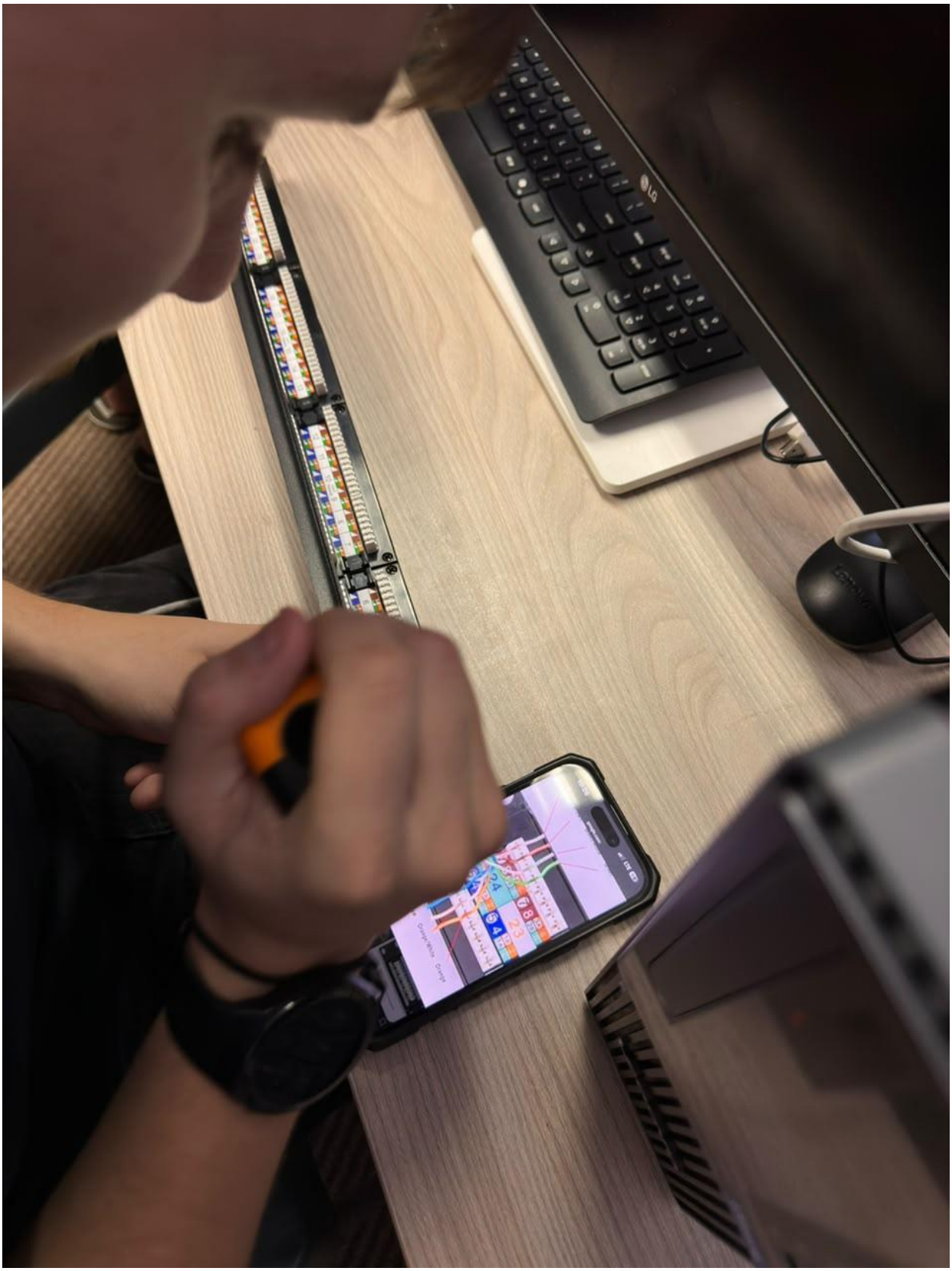

Blue/White
Blue
Orange/White
Orange
High Resolution Industry
Dtech electronics


Would you like to terminate autoinstall? [yes]:
Would you like to terminate autoinstall? [yes]:
Would you like to terminate autoinstall? [yes]:
Would you like to terminate autoinstall? [yes]:
Would you like to terminate autoinstall? [yes]:
Would you like to terminate autoinstall? [yes]:
Would you like to terminate autoinstall? [yes]:
Would you like to terminate autoinstall? [yes]:
Would you like to terminate autoinstall? [yes]:
Would you like to terminate autoinstall? [yes]:
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]:
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]:
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]:
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]: no
Switch>


Blue/White
Blue
Orange/White
Orange
High Resolution Indust
Dtech electronics

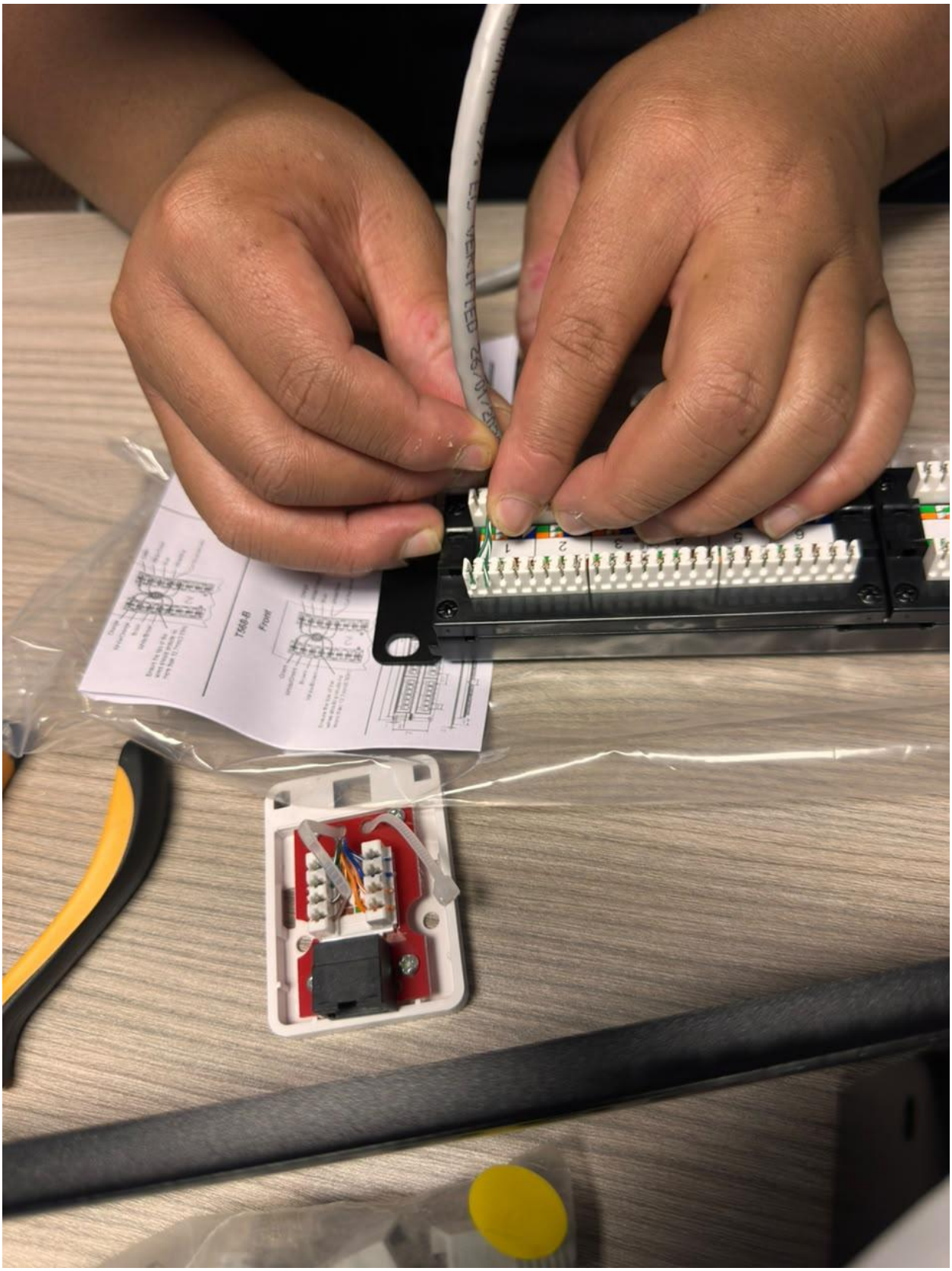


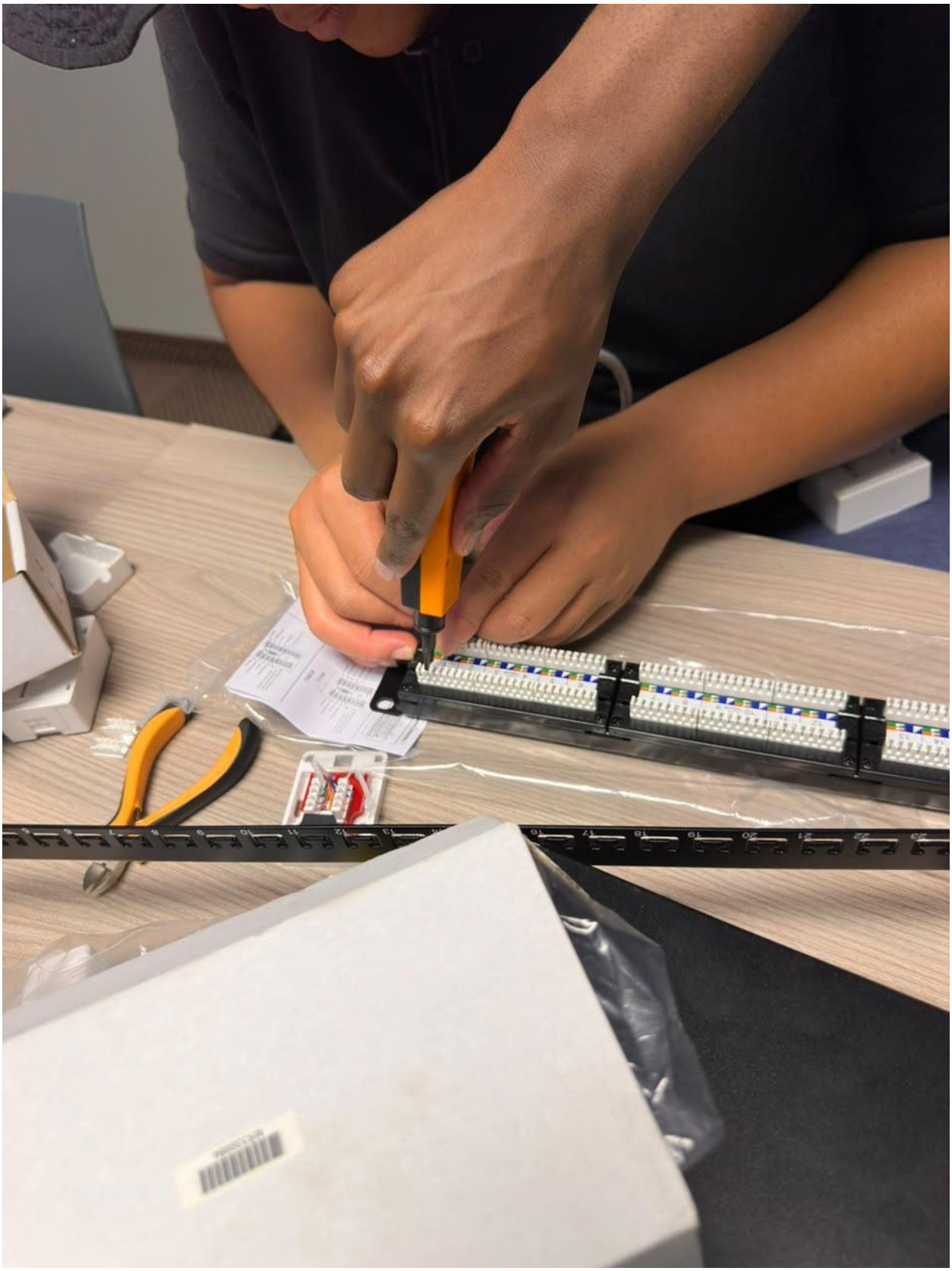

cisco

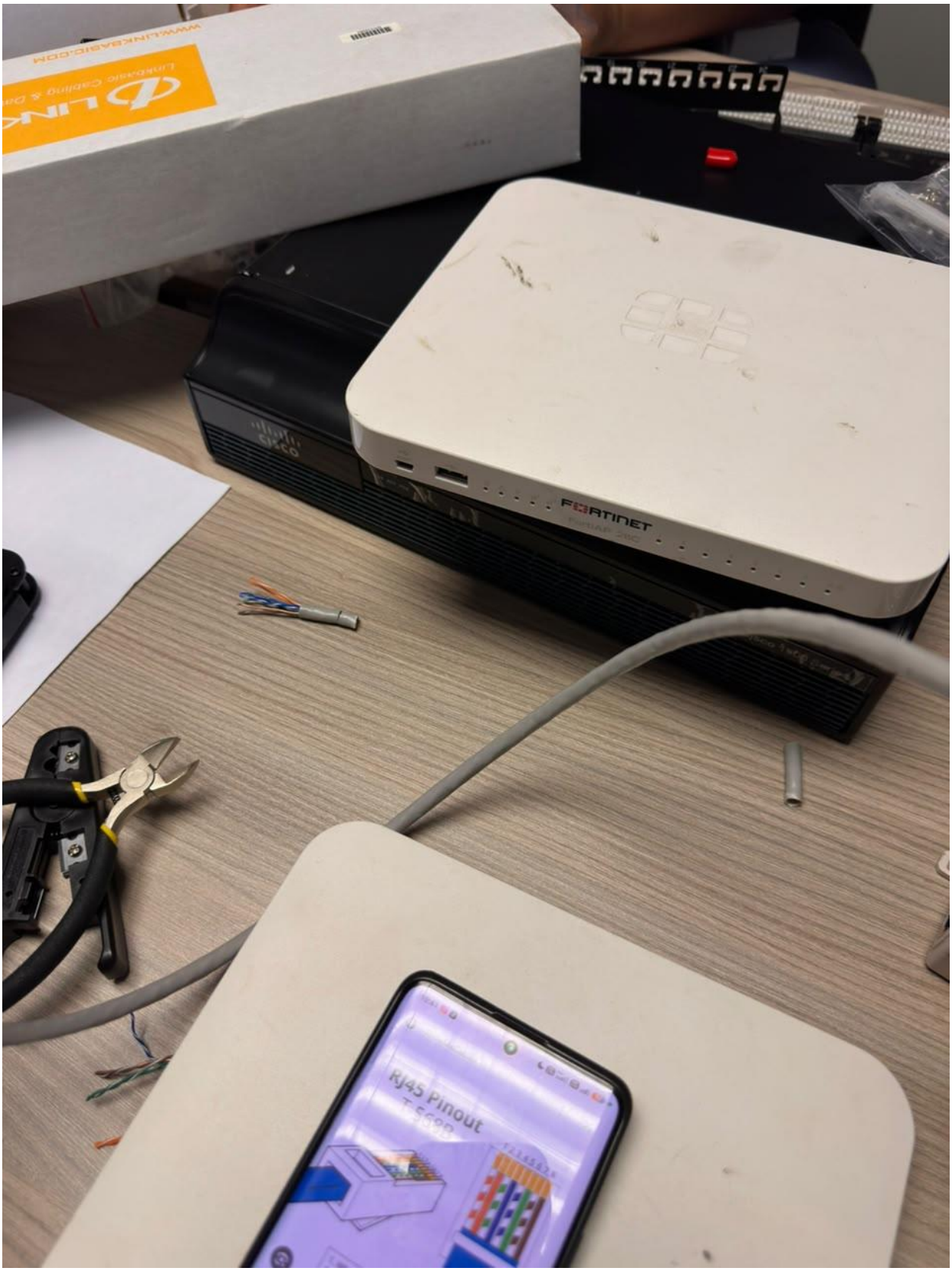



D-Link
FORTINET

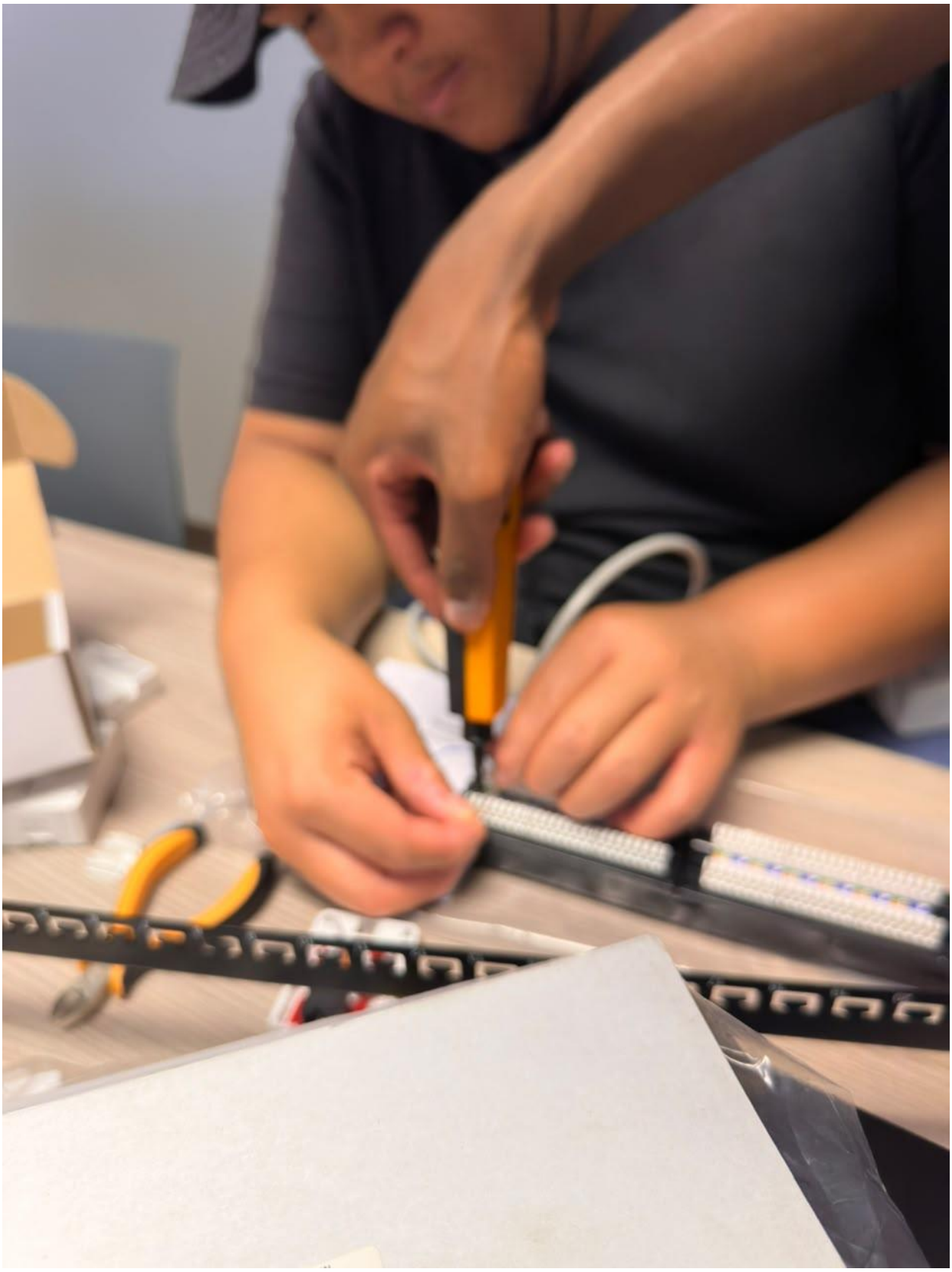

cisco

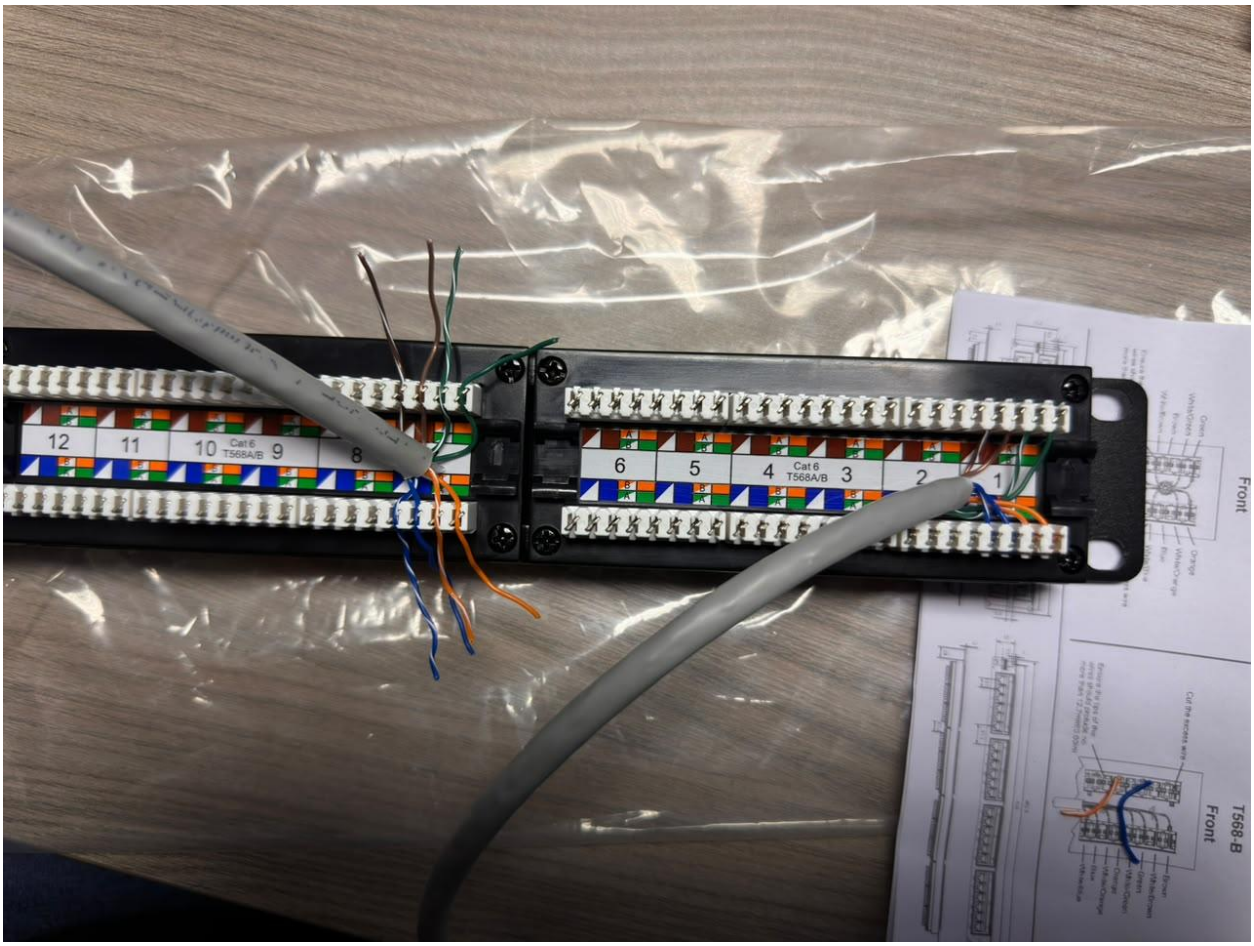

T568-B
Front

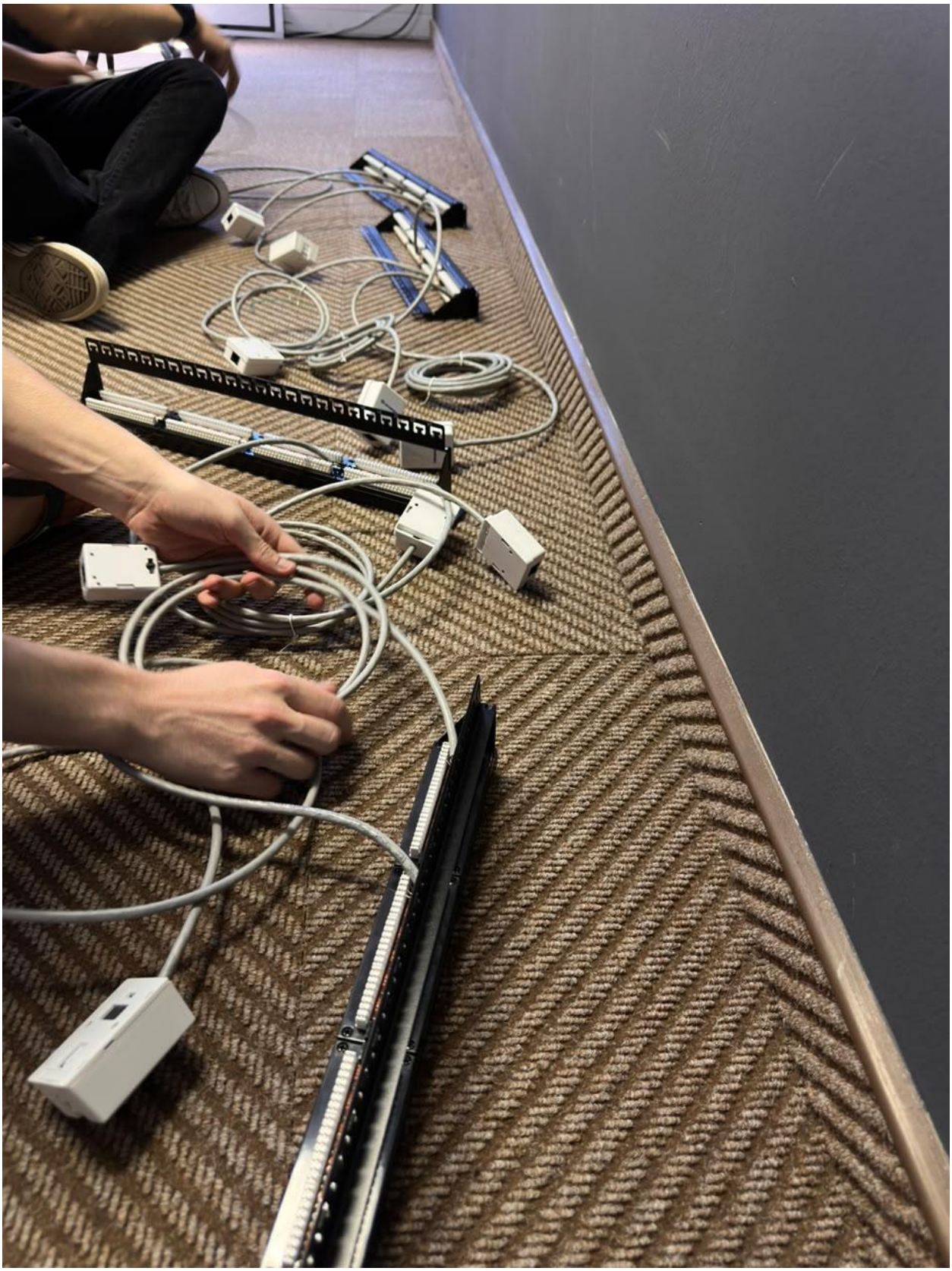



FORTINET
RJ45 Pinout

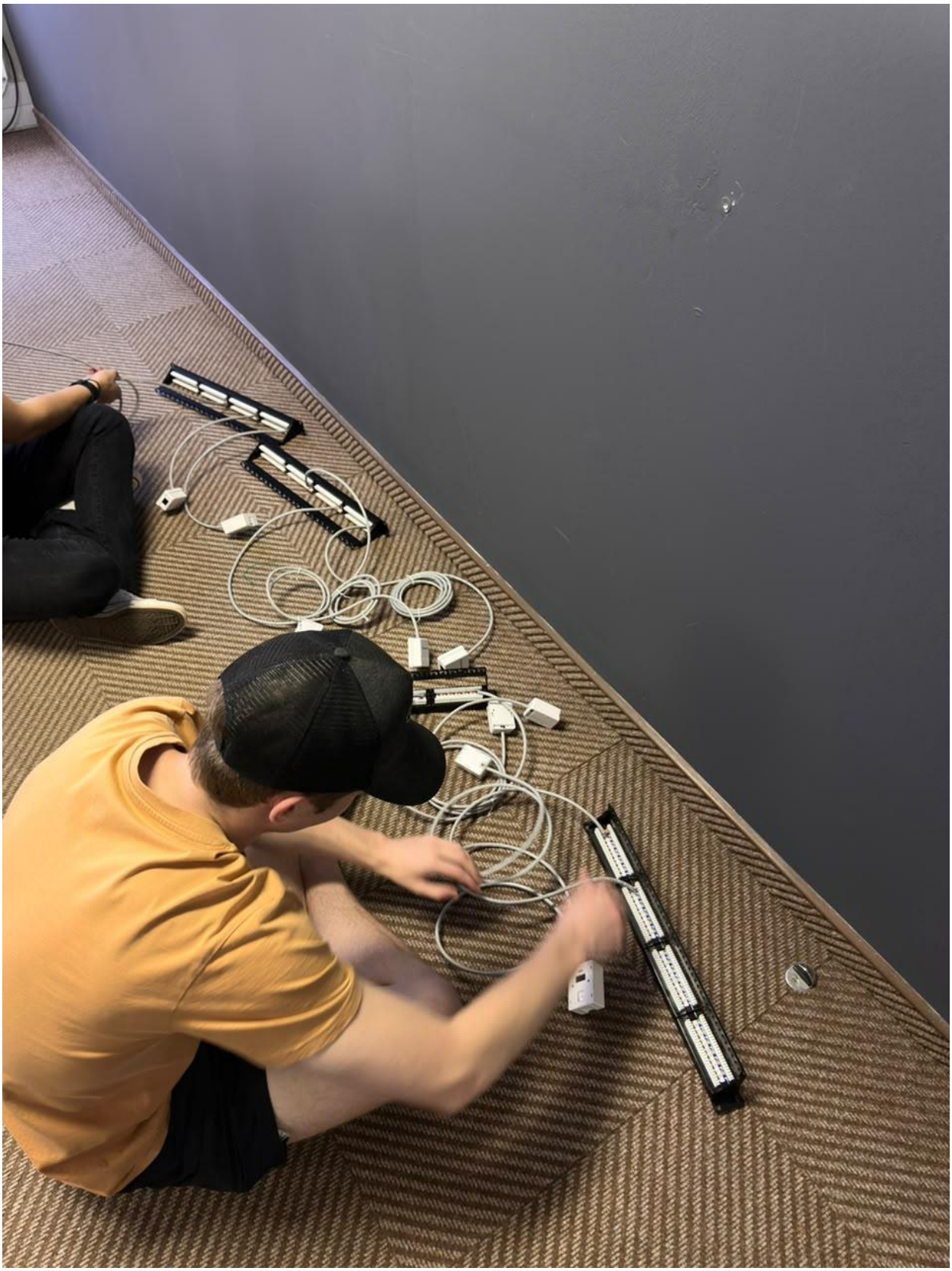





12
11
10
Cat 6
T568A/B
9
8
6
5
4
Cat 6
T568A/B
3
2
1
Front
T568-B
Front

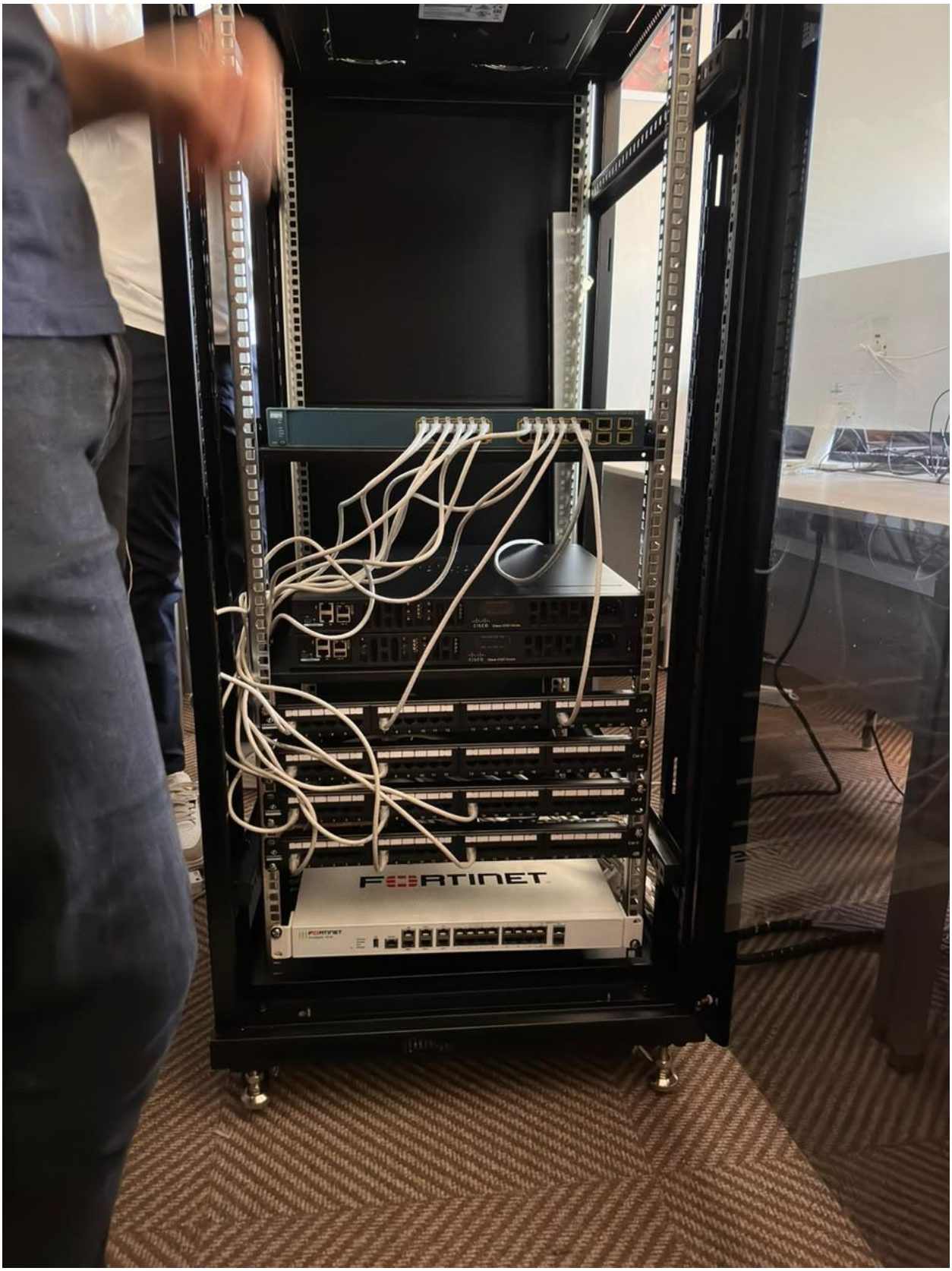



LINKBASIC®
Linkbasic Cabling & Data Center Solutions
WWW.LINKBASIC.COM
ISO9001-2015

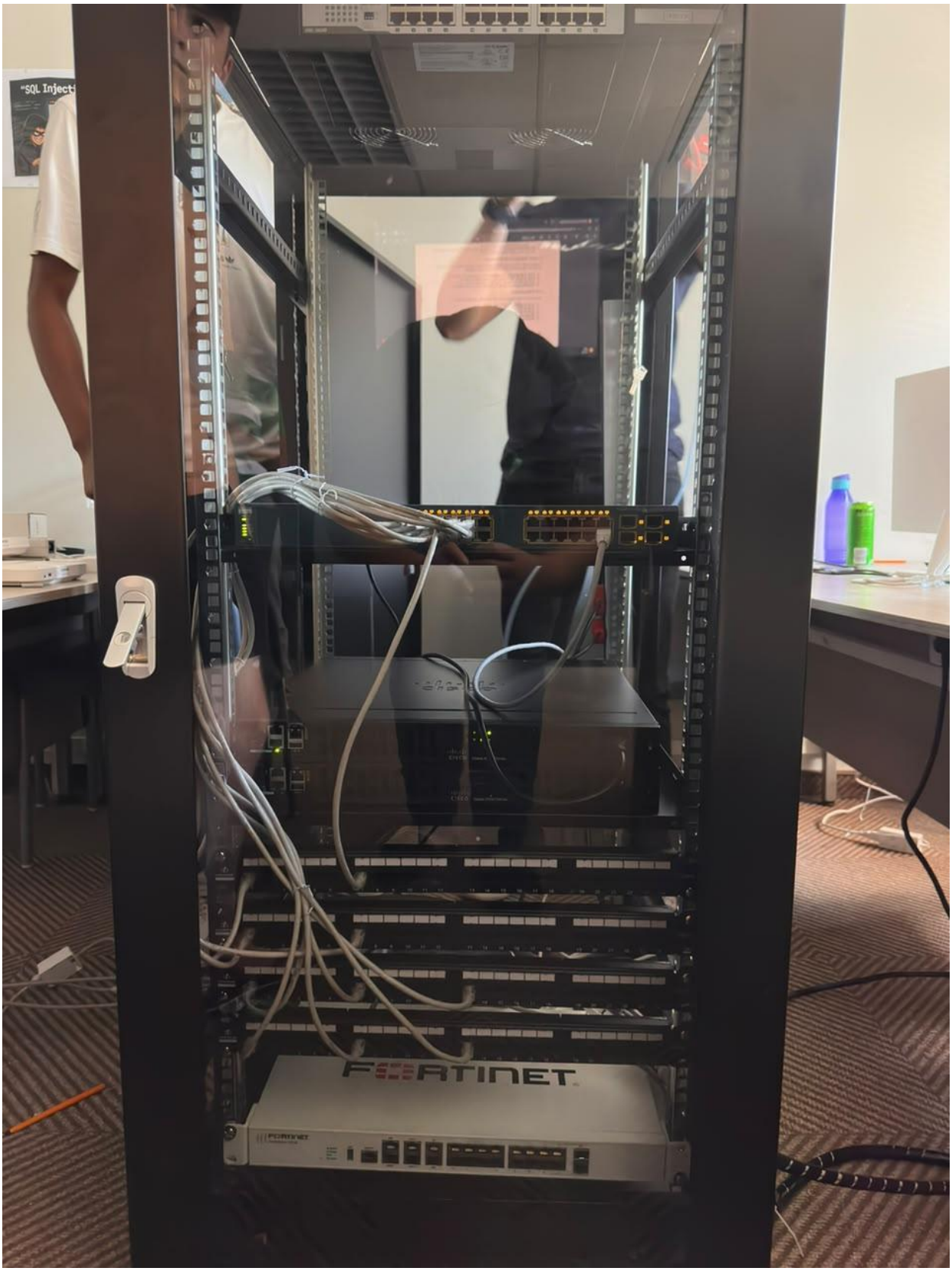



D-Link

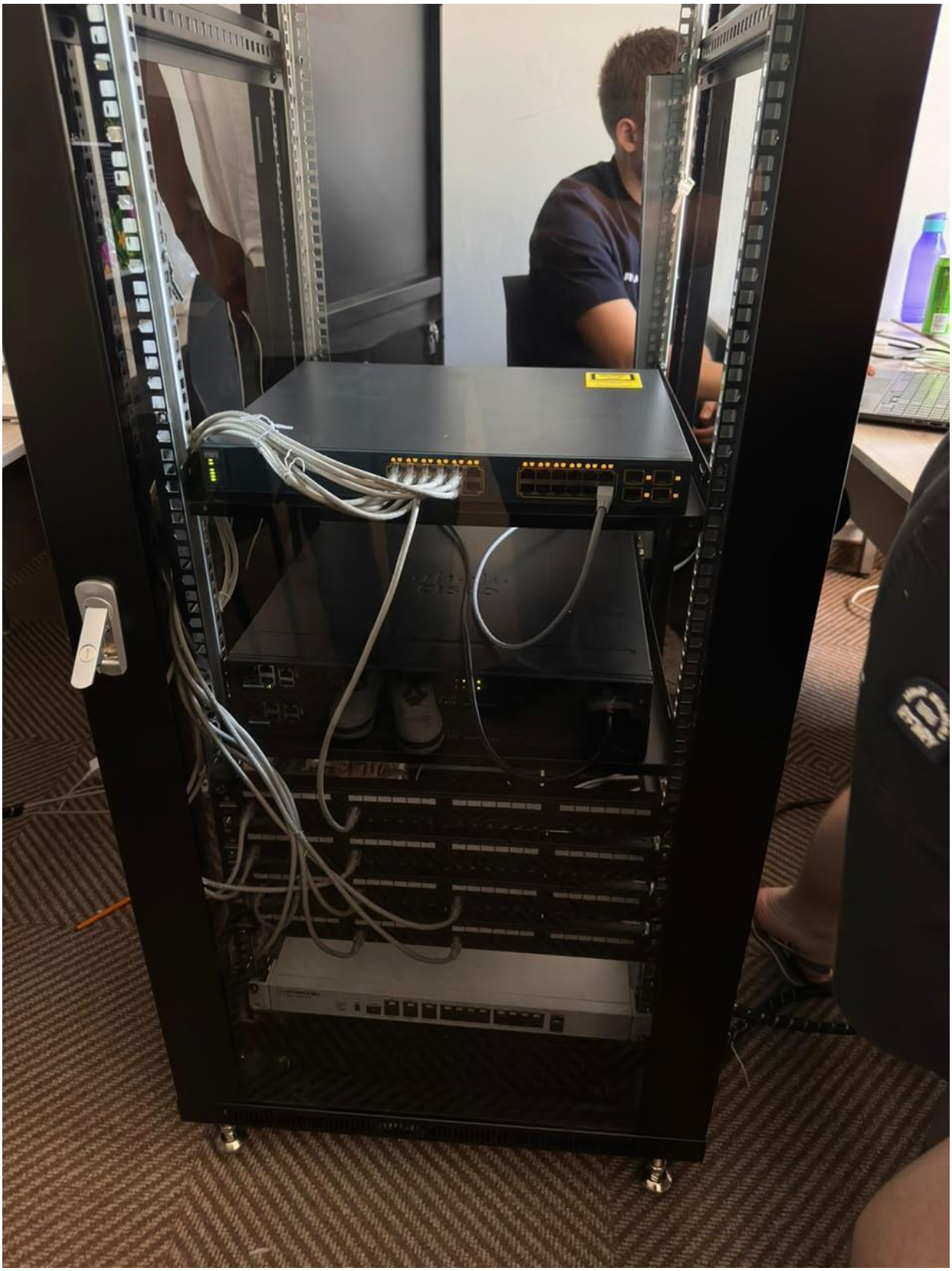

D-Link


FORTINET


cisco
Cisco 4300 Series
cisco
Cisco 4300 Series


FORTINET



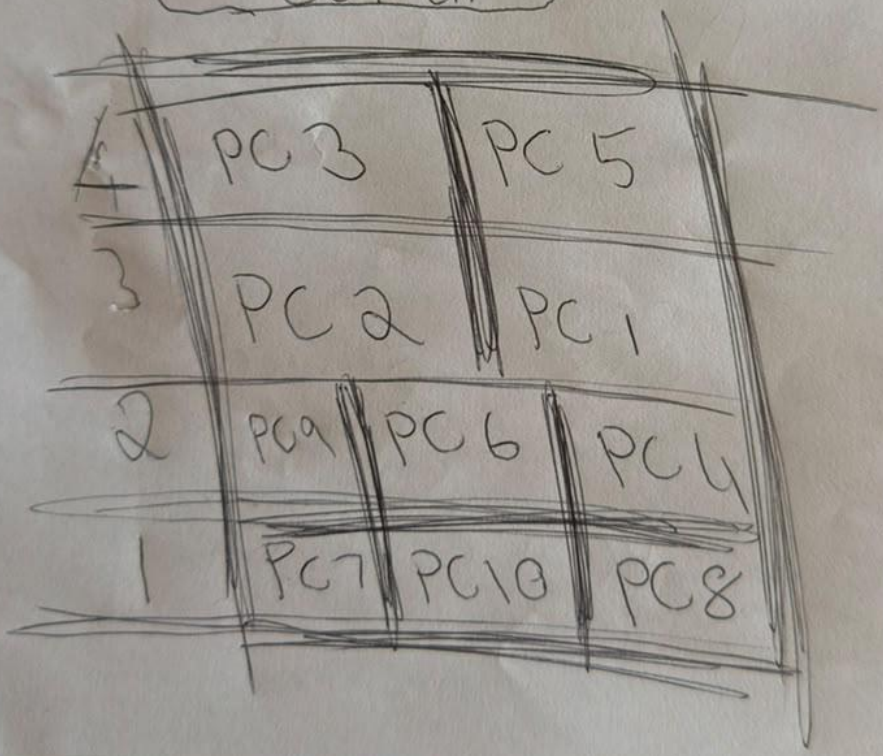


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PATCH PANEL SET UP

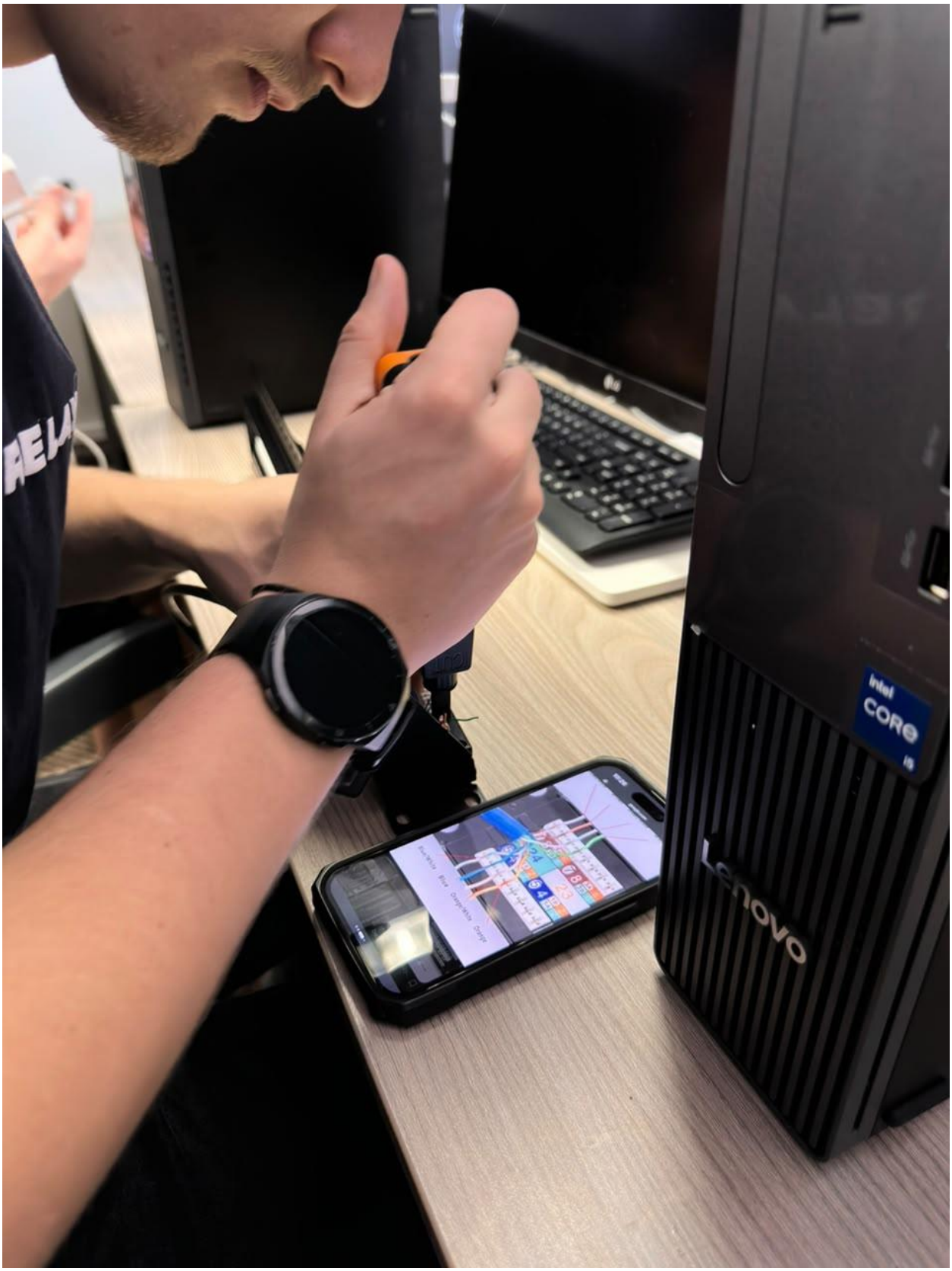





SQL Injection Attack




CABLE TESTER
MASTER


intel
CORE
lenovo